Buyer-Initiated Auctions: Nominations in Oil & Gas

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Abstract

Markets served by auctions can fail when too few bidders participate. Pre-auction nomination is a common but understudied remedy in oil and gas lease sales: bidders pay a nonrefundable fee and commit to purchase at a minimum price to propose tracts for sale, and only nominated tracts are auctioned. I study nomination when bidders' values are positively correlated and learned only at a cost. Correlation makes a nomination informative: it signals to uninformed bidders that their own values are likely high, drawing them in and thickening the market. But the same signal can make waiting optimal, so fewer bidders learn early and fewer tracts are nominated. To sign the net effect, I characterize a two-stage nomination-auction game, derive equilibrium learning and nomination strategies, and structurally estimate learning costs and value dependence using North Dakota's state oil and gas auctions. Counterfactuals show that nomination works mainly as a *screening* device. Relative to a no-nomination benchmark, it raises revenue by about \$3,090 (32%) per sold auction, and by 46% overall once its higher sale rate is included. Roughly 80% of this comes from screening on *demand*, which guarantees a committed bidder, and the rest from screening on *value*. The market-thickening signal matters only for the lowest-value tracts.

1 Introduction

Auctions serve as a primary mechanism for allocating resources, from private goods to government assets. When designing these mechanisms, sellers face a fundamental challenge:

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which items should be offered for sale? This question becomes more acute when the seller incurs non-trivial costs from running auctions in thin markets. The problem is illustrated by recent U.S. government attempts to auction oil and gas leases in Alaska's Arctic National Wildlife Refuge, where the attempted Bureau of Land Management (BLM) auction in January 2025 received zero bids after offering 400,000 acres worth of tracts.[1] When markets are too thin to the point where demand does not exist, traditional auction design tools, such as reserve prices and entry fees, are insufficient.

This paper examines an institutional feature often adopted in mineral rights markets to address this problem: pre-auction nominations. Rather than the seller deciding which items to auction, interested buyers nominate specific items they wish to see offered, typically by paying a small fee and committing to purchase the item at a minimum price if no other bidder appears. Only items receiving sufficient nominations proceed to auction.[2]

This mechanism has been implemented in various U.S. state government mineral rights leases. For example, U.S. Government Accountability Office [2021, p. 18] documents that between 2009 and 2019 about 87 million acres were nominated for onshore oil and gas leasing under the BLM nomination process. Before area-wide leasing was adopted on the Outer Continental Shelf in 1983, a similar nomination system determined which tracts the federal government offered [Hendricks and Porter, 2014].

Despite this prevalence, little is known about how nomination affects seller revenue relative to the standard auction models, where the seller chooses what to sell without explicit input from buyers. Buyers' nomination incentives determine the auction trigger rate (how often a tract is nominated) while buyers' entry incentives, once an auction is triggered, determine market thickness (how many bidders show up to compete). When bidders' values are correlated but unknown ex ante, entailing a learning cost, these two channels move in opposite directions.

Nomination becomes a public signal which pulls in bidders who would otherwise never participate. However, it also pushes some bidders to wait and defer the learning cost on

[1] For such BLM-run oil and gas lease auctions, environmental reviews required under the National Environmental Policy Act must be completed before the auction, regardless of whether or not those lands are ultimately leased. Across twelve reviews conducted between June 2020 and January 2021 the cost ranged from $17,000 to $178,000 collectively per lease auction, and BLM Utah reported an average of about $2,100 per parcel across its two most recent auctions [U.S. Government Accountability Office, 2021, pp. 28–29, n.42].

[2] In U.S. federal onshore oil and gas leasing, this same mechanism is officially termed an *expression of interest* (EOI): a member of the public may nominate a parcel for a lease sale by submitting an EOI, and BLM may also select parcels directly [U.S. Government Accountability Office, 2021]. I use *nomination* throughout to match the terminology of my empirical setting (North Dakota state leasing). Section 1.1 situates this mechanism relative to related but distinct notions of "expressions of interest" and "indications of interest" in the finance literature (IPO bookbuilding, M&A sale processes).

potentially low-value tracts until someone else has nominated, suppressing how often tracts get nominated. Therefore, whether nominated auctions outperform those initiated solely by the seller is an empirical question. If they do, the natural question is how to design the mechanism to make the best use of both channels: how many nominations should be required to trigger an auction, what fee should be charged, and whether a fee or a reserve price better serves the seller's goals.

This paper builds a structural model of buyers' nomination incentives and uses the estimated model to identify the design choices that maximize seller revenue. I use data from North Dakota (ND) mineral rights auctions conducted between 2017 and 2018 for estimation, including digitized nomination records obtained through Freedom of Information Act (FOIA) requests. These auctions provide an ideal setting for several reasons. First, the nomination process is well-documented. Bidders must nominate tracts 2–3 months before the auction, paying a $10 fee per nomination and committing to purchase the tract at the reserve price if no other bidder appears. ND commits to a single-nomination trigger-rule, where tracts are auctioned if and only if there was at least one nomination. Second, the stakes are substantial. 878 tracts generated 2,740 bids with winning bids averaging $170 per acre (standard deviation of $829) over a total of 116,750.2 acres, totaling about $12.2 million in revenue. Third, and most importantly, the data reveal two patterns that challenge standard auction theory predictions.

I begin by documenting a stark discrepancy in the number of nominators and number of the non-nominator bidders. While some bidders nominate hundreds of tracts, others nominate far fewer (or nothing) and yet bid aggressively on tracts nominated by others. For instance, one bidder, who nominated 397 tracts during the two-year period of interest, placed bids on 22 tracts nominated by a different bidder. When these non-nominators do bid, they bid substantial amounts indistinguishable from nominators' bids.

The total number of nominations received is remarkably low compared to the numerous tracts available for nomination in each season. Despite the availability of approximately 28,000 tracts totaling up to 2.2 million in net acreage, typically only 100–200 tracts receive nominations per season.[3]

In addition to these data patterns, several features of the oil and gas market motivate my modelling choices. First, bidders' valuations are unlikely to be independent: firms draw on the same public information about a tract (size, royalty rate, location, and the production history of nearby wells) and the same prevailing oil and gas prices, so their valuations move together, even though what that shared information is *worth* is private to each firm and

[3]Table 1 reports the number of available tracts and nomination counts for each of the eight auction seasons.

depends on its drilling technology, machinery, and lease portfolio. I therefore model values as correlated *private* values in the sense of Milgrom and Weber [1982] rather than as common values; Section 4 develops this structure and its post-fracking justification in full.

Second, precisely because these public characteristics do not determine a firm's payoff on their own, discovering one's own value is costly: translating them into a firm-specific dollar figure requires due diligence, the *learning cost* (Section 2). This is a cost of nomination well beyond the $10 fee itself, and is a candidate explanation for why so few tracts are nominated.

These facts point to a fundamental tension inherent in nomination systems. From the seller's perspective, nomination serves as a demand-screening mechanism that ensures auctions proceed only when a sufficient number of buyers exhibit interest. From bidders' perspectives, it is a quality-screening device: because values are correlated, a nomination signals that another firm judged the tract worth pursuing, and hence that a bidder's own value is likely high as well. This creates incentives for bidders to wait and learn later on others' nominations, discouraging early learning and reducing how many auctions are initiated. [4] The dual role is what generates the competing effects described above.

I develop a two-stage nomination-auction game where bidders know their cost of discovering their values (learning cost) upon arriving in Stage 1 (pre-auction nomination). They must decide whether to learn early and nominate or instead wait and incur the learning cost only on tracts others have nominated. Crucially, bidders do not know their values until they incur this learning cost.

Stage 2 (auction) is triggered if at least one nomination is received. If the auction proceeds, the non-nominators from Stage 1 (later-learners) and any additional bidders who arrive in Stage 2 explore and decide whether to participate. The exact number of nominations received per tract is not public, just like in ND's setting. Nominated tracts are sold in a second-price sealed-bid auction.

The model generates two insights. First, there exists a learning cost threshold below which bidders learn immediately to determine whether to nominate and above which they wait and learn later. This cutoff balances the benefit of ensuring the auction occurs against the cost of nominating, including learning cost, nomination fee, and revealing information to non-nominating entrants.

Second, positive dependence reduces nomination incentives through two distinct channels. Observing a high own value signals that competitors likely have high values too, compressing the expected surplus from winning. Separately, dependence is what makes a nomination informative to bidders who have not yet learned their values, which is the source of the

[4] Phone interviews suggest this is a deliberate strategic choice; see the non-nominating bidder quoted in Section 3.

late-learning incentive. The estimates in Section 6 indicate the first channel dominates.

Taking the model to data requires recovering the distribution of private values and the strength of their dependence from observed bids. One empirical challenge arises from ND's auction format. ND uses an ascending-price auction with proxy bidding, which is essentially a button auction played by the computer on a bidder's behalf. Since the winner pays one dollar more than the second-highest bid, only the losing bids are observed. While standard approaches exist for handling censored auction data under independent private values, these methods do not apply to positively correlated private values. This paper develops a copula-based estimation technique to address this challenge, extending the framework of Li et al. [2002] using David and Nagaraja [2004]'s theoretical formula for cumulative distribution function of order-statistic under exchangeable copula dependence. Among the copula families considered, including Clayton and Gumbel, the Gaussian copula fits best, with estimated correlation parameter $\hat{\rho} = 0.861 \pm 0.096$, substantially different from independence.

The estimated structural parameters reveal that the binding barrier to nomination is the scarcity of early, pre-nomination arrivals. The pre-period arrival rate is estimated at ≈ 0.005 potential nominators per available tract per season, orders of magnitude below the post-nomination arrival rate. The mean exploration cost is modest relative to auction stakes (approximately \$1,161, against an average total bid of \$11,803). The estimated equilibrium cost cutoffs are non-binding in the high-value tracts, where essentially all arrivals would explore, but do screen participation in the low-value, low-value tracts. Even where cost imposes no screen, however, the trigger rate is only $\approx 0.5\%$, so it is the low pre-period arrival intensity that generates the low observed nomination rate and a non-nominator to nominator ratio of approximately 2.3:1 (see Table 2 and Table 16).

Relative to a no-nomination benchmark, the single-nomination rule raises revenue by about \$3,090 per sold auction. That is a 32% gain conditional on a sale, and 46% once the higher sale rate is included. Nomination works mainly as a *screening* device: about 80% of the gain comes from screening on demand, which guarantees that a committed bidder participates, and about 20% from screening on value, which selects higher-value tracts. The correlation-driven thickening signal is negligible except for the lowest-value tracts (Section 7.1).

I then examine the externality that non-nominating entrants impose on nominators. Although these entrants compete away part of a nominator's surplus, their participation also raises the seller's expected revenue and lets the seller post a substantially lower reserve price. Netting the two channels, the average nominator is barely harmed, and is in fact better off than under a hypothetical nominators-only mechanism on all but the most valuable tracts.

1.1 Contributions

This paper makes several contributions. First, I provide the first structural analysis of pre-auction nomination mechanisms, quantifying their dual role as demand screening (from the sellers' perspective) and quality screening (from the buyers' perspective). The closest formal model is Gorbenko and Malenko [2024], who study pre-auction "indications of interest" in corporate finance settings. In their framework, signaling interest reveals an asset's value, which in common-value environments discourages other buyers from nominating. My setting works in a correlated private values framework where observing a nomination updates a rival's decision about whether to enter. This quality-screening-for-entry effect is absent from Gorbenko and Malenko [2024], who focus on asset revaluation due to the common-value setting. I also show that single-nomination trigger-rule is the unique trigger-rule that prevents complete coordination failure, a viability floor not discussed in prior work.

This dual screening role also connects nomination to two literatures on pre-auction signaling that are typically treated separately. The first studies non-binding *indicative bidding* as a device for screening bidders under costly entry [Ye, 2007, Quint and Hendricks, 2018], with experimental support in Kagel et al. [2008]: a preliminary interest signal helps sellers select higher-value bidders for a costly second stage, though Ye [2007] shows fully separating equilibria generally fail to exist when the signal is non-binding.

Nomination differs from this literature along two dimensions. First, the nonrefundable $10 fee is small but strictly binding. Second, and more distinctively, nomination carries a commitment where the nominator must complete the purchase at the minimum price when they face no competing bidders, guaranteeing a nonnegative payoff that non-nominating entrants, who enter only conditional on the auction already having been triggered by someone else, can never obtain.

The structural estimates (Section 6) show that the binding constraint on market thickness is the scarcity of early, pre-nomination arrivals rather than the height of any single cost. Counterfactuals confirm that the $10 fee and the purchase commitment are non-binding and removing the fee raises the trigger rate by only 1.6%. The learning cost, by contrast, is not costless: it screens participation in the lower-value tracts, where the early learning share falls from 100% in the high-value bins to 3.5% in the low-value bins. Nomination therefore remains a costly signal in the Spence sense for low-value tracts, where it bundles a fee, an exploration cost, and a purchase commitment. This places nomination closer to the costly-entry tradition of Levin and Smith [1994] and Bulow and Klemperer [2009] than to either non-binding indicative bidding or a competitively-bid entry-rights auction. What the estimates add is that the quantitative friction limiting the *overall* nomination rate is the thinness of the early arrival process rather than the height of any one cost, while exploration

cost shapes *which* tracts are nominated.

A related literature documents an analogous, non-binding pre-public screening phase in M&A sale processes [Hansen, 2001, Boone and Mulherin, 2007], where a cheap early signal feeds the later, binding sale stage. My nominator-welfare counterfactual (Section 7.2) shows how that information shapes pricing in the nomination setting: because a nomination draws rival bidders into the auction, competition substitutes for a high floor, and the reserve-optimizing seller holds the reserve far below the level it would set facing nominators alone ($65 versus $143 per acre). This is the entry-inducement logic of Bulow and Klemperer [2009], whereby attracting additional bidders is worth more to the seller than optimizing the reserve against a fixed set of participants.

I also extend affiliated private value (APV) empirical methodologies, developed for settings where all bids are observed [Li et al., 2002, Athey and Haile, 2002], to a setting with both a censored highest bid and positively dependent values. When the full bid profile is not observed, values are not nonparametrically identified under standard affiliation assumptions [e.g., Athey and Haile, 2002]; some parametric or semiparametric structure is needed to estimate the dependence parameter and the marginal value distribution from the observed lower order statistics alone. Under independence, Espín-Sánchez et al. [2025] show that the number of bidders and the marginal value distribution remain nonparametrically identified from any two order statistics, even when the highest bid is unobserved.Their framework also extends this identification to dependent bidders for max-stable correlation structures (e.g., Gumbel-type dependence). Positive dependence estimated from natural resource auction data need not take this max-stable form, and no general nonparametric identification result is available once both censoring and non-max-stable dependence (e.g., Gaussian or Archimedean copulas) are present simultaneously. I address this gap by pairing a parametric copula for the dependence structure with a flexible, spline-based approximation to the marginal. Identification of the dependence parameter is driven by the copula family selected and therefore explicitly parametric, while the marginal itself is estimated nonparametrically.

The findings also connect to the literature on information design in auctions. Bergemann and Pesendorfer [2007], Board [2009], and Ganuza and Penalva [2010] study how much information the *seller* should disclose prior to an auction to maximize revenue. My setting inverts this question: it is *buyers* who choose what information to reveal through nomination, and the seller designs the nomination threshold and nomination fee to elicit costly demand revelation. The late-learning externality I document creates a public goods problem in information production that does not arise in standard seller-disclosure frameworks, because the cost of generating information is borne by the buyers, not the seller.

Delayed-learning-for-information motive appears at the exploration stage of this industry

as well. Hodgson [2024] estimates a dynamic model of UK offshore drilling in which firms delay costly exploration of wells to observe rivals' outcomes. There, the information spilling over is a signal about a commonly-valued unknown (the location of oil), so firms learn from realized well results of rivals. In my setting, what a bidder infers from a rival's nomination is instead something about *their own* private value, transmitted through the positive dependence of values rather than through any revision of a common estimate.

The paper proceeds as follows. Section 2 describes the institutional setting in detail, including the two-stage process of nomination followed by an ascending-price auction, and the proxy bidding system that implements button auction-like outcomes. Section 3 discusses the data structure, including the digitization of nomination records and the construction of bidder-level participation variables. Section 4 formalizes the game structure with a timeline showing information flows and decision points across the two stages, and develops the theoretical model, characterizing equilibrium nomination strategies and deriving the expected surplus functions under affiliated values. Section 5 outlines the estimation strategy, with particular attention to the two-step copula-based method for recovering the value distribution from censored bids under positive dependence. Section 6 presents the estimated structural parameters from the main model. Section 7 conducts the counterfactual analysis: it first decomposes the value of nomination against a no-nomination benchmark into market-thickening and screening (on demand and on value) components, and then studies the externality that non-nominating entrants impose on nominators and how the seller's reserve-price choice partly offsets it. Section 8 concludes.

2 Institutional Details

The ND state government holds quarterly auctions for oil and gas lease sales through the Department of Trust Lands. Each cycle, referred to as a *season* throughout this paper, begins almost immediately after the previous auction concludes. At this time, the state announces the date of the next auction and calls for nominations from interested parties along with the deadline for nominations. The list of tracts open for leasing is made public in their website. Tracts need at least one nomination to be considered for auction. The size of the tract can vary, ranging from 0.01 acres to a maximum of 160 acres as specified by state regulations. The nomination process requires a nonrefundable $10 fee per tract. Nominators can determine the minimum opening bid. If not specified, the opening bid is $2.

Following the nomination period, the state conducts ascending price auctions to individually sell nominated tracts. Each tract opens for bidding individually, with auctions closing sequentially at two-minute intervals. Bids are received at per-acre level. If no one bids, the

nominator becomes the automatic winner and has to pay the minimum opening bid they specified.

Upon winning a lease auction, the successful bidder must pay several upfront costs. For example, a $2 per-acre bid (the minimum possible) on 40 net acres incurs the following costs: the bonus payment ($80), a ND administrative fee ($10), a five-year prepaid rental fee ($1 per net acre annually, totaling $200), and a 2% buyer premium ($1.60) combine for a total initial cost of $291.60. Once a well is drilled and begins producing, the leaseholder pays a royalty rate, which is the percentage of oil or gas revenue paid to the mineral rights owner, of either 3/16ths or 1/6th, depending on lease terms.[5]

The winning bidders obtain exclusive rights to explore, drill, and produce oil and gas from their tracts for a maximum lease period of five years. Extensions beyond this period are only permitted if the tract is being actively developed and producing.

Bidders have access to what is called *proxy bidding*, an online system that bids on behalf of the user until the highest competing bid exceeds the value the user has set as their *proxy bid*. In other words, as long as the competing bidders are bidding lower than the proxy bid value, the proxy bidder will keep its winning status with a bid incrementally higher ($1) than the highest competing bid. If the competing bid is equal to the proxy bid, say at $10, but was placed after the proxy bid was placed, the proxy bidder will show up as the winner at $10 and the bidder who bid $10 but placed after the proxy bidder will lose with their bid recorded as $9.

Under the presence of proxy bidding system and the private value assumption, the optimal bidding strategy of the bidders is to place a proxy bid at their private value. For this reason, most winning bids are a dollar higher than the second highest bid. In this sense, the winning bid is *censored*. However, oftentimes bidders will not use this proxy bid system. When this happens, a distinct *jump* is observed in the data. For example, instead of placing a proxy bid of $10 after the previous bidder placed a proxy bid of $5, the next highest recorded bid will be $10 instead of a bid incrementally higher than $5, which in this case is $6. Such instances are rare and informative in that they reveal the winner's true bid directly. However, I do not explicitly use these jump-bid auctions and they enter the estimation on equal footing with all other auctions. In addition, I abstract away from the fact that the losing bidder's bid may be lower than than what is recorded due to this system.

While the broad geological formation underlying a given region is public knowledge, translating that shared information into a firm-specific dollar value for a particular tract is not costless. For instance, identifying whether a given tract sits within such a reservoir

[5]Royalty rate differs by county, and most counties are one/sixth (1/6) except for counties like Billings, Divide, Dunn, Golden Valley, McKenzie, Mountrail and Williams.

– a localized pocket within the broader formation where the rock is naturally fractured or under sufficiently high pressure to make extraction economical – and how much it would cost a specific firm to develop it requires firm-specific analysis. Phone interviews with market participants conducted for this project confirm that this evaluation step is a real and heterogeneous cost across firms: larger firms typically maintain in-house land men and geologists (or even their "bosses" with years of experience are capable of such evaluation on the spot from "looking at the map") to evaluate prospective tracts, while smaller firms rely on third-party consultants whose availability, and hence cost, is uncertain from one tract to the next. This heterogeneity in the cost of acquiring firm-specific information motivates modeling this cost, which I refer to as the *learning cost*, as privately observed and drawn independently across bidders, rather than as a fixed or publicly known cost of participation.

3 Data

The main data for estimation is auction level-bid data and nomination information for each auction for years 2017–2018. This is constructed by combining 1. auction-level bid data for all bidders scrapped from an online platform that conducts online auction and 2. nomination data. Ever since the Mineral Leasing Act was amended by Congress to allow for internet-based lease sales as part of the National Defense Authorization Act (NDAA) for Fiscal Year 2015, many state-government run mineral rights sales auctions adopted internet-based auction through the bidding platform EnergyNet. Past bidding history from open auctions (English auctions) conducted on this platform is available to public. Additional information about each auction, such as royalty rate, size in acres if the tract, identity of nominator and start and end times of the auction that was publicized during auction is available as well. Detailed information on nominations received per tract was acquired through FOIA request. Bidder identity during auction is anonymized using bidder id numbers. The winning bidder's identity is revealed. Using the bidder id and winning bidder's names, bidders who have won at least once during the given season can be traced within the season. Bidders who never won stay anonymous. This means firm-level characteristics (size, drilling technology, or existing lease portfolio) are observed for only a small subset of bidder-tract pairs (winners within a season). Consequently, I do not model them explicitly as bidder-level heterogeneity.

Supplementary data is needed to understand which lands were available for nomination for each auction. This information is reconstruct using 1. comprehensive list of tracts available for leasing and 2. past auction history from years 2012–2016. Potential nominators can check current lease status and available tracts for nomination through the public records of ND Department of Trust Lands. This includes detailed geographical information according

to the World Geodetic System 1984 (WGS 84) coordinate system for both currently leased and available for lease tracts. It also includes information on most current past lease associated with the tract and whether the lease was extended beyond it maximum five-year lease period.[6] Past auction history for both offline (before NDAA was passed) and online auctions is acquired from ND Department of Trust lands public data. Combining these two data, I reconstruct the list of tracts that would have been available for nomination by the time a potential nominator arrived to nominate. Figure 8 in Appendix A maps the geographic distribution of available tracts alongside the locations of nearby producing wells, illustrating the variation in well proximity across tracts that motivates including distance to the nearest well as a covariate in the bid-homogenization regression.

There are 878 number of auctions in total, covering 2,740 bids. Below, I report the summary statistics from the entire data unless specified otherwise. The number of auctions used for estimation is 728 covering 1,875 bids. I keep only the auctions where there is more than one competitor and auctions where the size is greater than 0.01 acres. In addition, I use auctions where the winning bidder is not observed for estimation.

Figure 1 shows the number of bidders and nominators per tract, showing a stark discrepancy in the number of the two. While number of bidders per auction ranges from one to 11 with the median being three, those who nominated and triggered the auction is at most three with majority of auctions being triggered by a single nominator. This shows how much delayed-learning takes place and most bidders take advantage of someone else's nomination, despite the monetary cost of triggering an auction being very low ($10 per nomination).

Figure 2 documents this late-learning pattern at the individual bidder level. Because bidder identity is unobserved unless a bidder wins and their winning status is announced by the seller, the figure is restricted to bidders who won at least once during the auction season. Within a season each bidder carries a unique identifier, making their participation across tracts traceable conditional on at least one win. Each point is a nominator-auction pair for this identified sample: the horizontal axis counts all tracts the bidder participated in during that auction, and the vertical axis counts the tracts they actually nominated. Points on the 45-degree dashed line represent bidders who nominated every tract they participated in; points below the line represent late-learning. The vast majority of observations in green circles lie along the horizontal axis with almost zero nominations, despite participating in dozens or even over a hundred tracts. By contrast, the small cluster of orange triangles near the 45-degree line represents the handful of nominators who actually trigger the auction. Even among nominators, some still defer learning until later on a substantial share of tracts

[6] To the best of my knowledge using the most current lease data, most lease terminate after the five year period unless a new well was drilled.

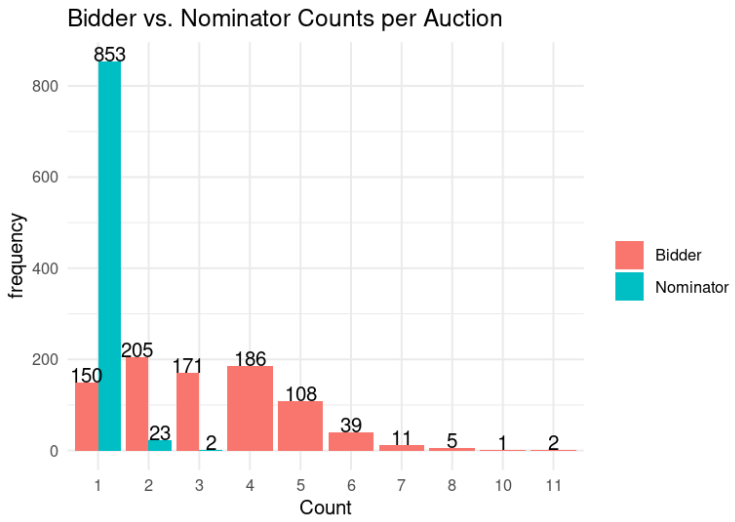


Figure 1: Histogram of per Auction Bidder vs. Nominator

Notes: Histogram comparing the number of bidders versus nominators per auction. The stark discrepancy shows extensive delayed-learning: while auctions receive 1–11 bidders (median of 3), the vast majority are triggered by a single nominator, with only a few receiving 2–3 nominations.

they enter. With Figure 1, these two figures together establish that late-learning is pervasive.

Beyond the raw magnitude of late-learning documented in Figures 1 and 2, phone interviews with market participants suggest that non-nomination is a deliberate strategic choice. One non-nominating bidder reasoned that it is "unlikely there's anything valuable if no one else nominated it" and describe nominating as "a waste of time" when they can instead "rely on other nominators" to trigger the auction on their behalf. This paper directly models this delayed learning mechanism: nomination requires paying the learning cost upfront and the $10 nomination fee, while late-learning lets a bidder defer that cost until the auction is confirmed to occur.

I do not model tract quality explicitly, as not all tracts have realized oil and gas production data available. Instead, I use bids as a revealed-preference measure of tract value: each bid reflects the bidder's perceived value of holding the exclusive right to explore, drill, and extract from that tract over the five-year lease period, incorporating their private assessment of geological prospects, extraction costs, and future commodity prices. The bid homogenization procedure in Section B.1 controls for observable tract characteristics, including geographic location, isolating the private value component.

Table 1 reports the same number but broken down by season. The second row indicates the number of auctions that were triggered, meaning these are the number of tracts that received at least one nomination. The third column reports the number of tracts that would have been open for nomination, calculated using the tract list data and auction history

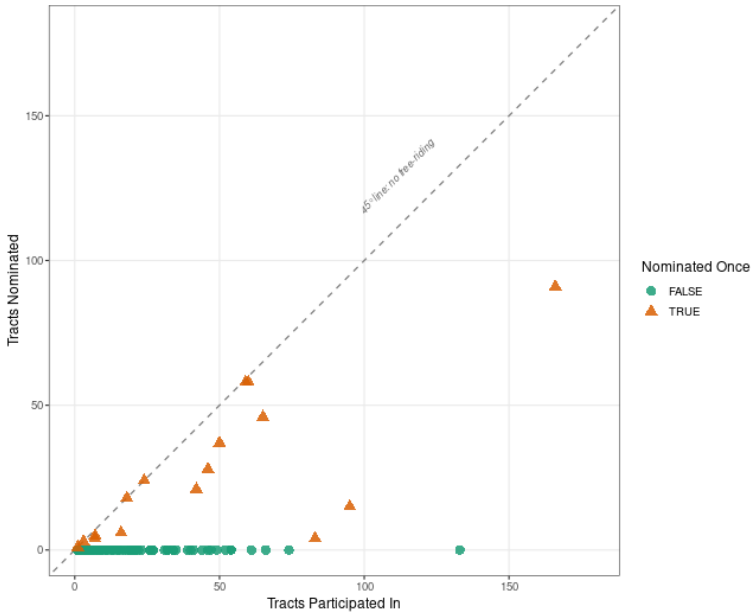


Figure 2: Tracts Participated vs. Tracts Nominated, by Nominator-Auction Pair

Notes: Each point is a bidder-auction pair restricted to bidders whose identities are known. The horizontal axis is the total number of tracts the bidder participated in (bid on); the vertical axis is the number of tracts they nominated.

data from five years before. The table also reports the number of parcels that received two and three nominations, obtained through FOIA request. Total number of bids received for that season is reported in the last column. During 2017–2018, a total of 878 auctions were nominating, which is only about 2.3% of the total available tracts back then.

Table 1: Summary of Available Tracts, Nomination Counts and Bids Received

Season	**# Auctions**	**Tracts Available**	**Nom. ×2**	**Nom. ×3**	**# Bids Received**
2017-Q1	173	27,867	21	2	361
2017-Q2	234	27,885	0	0	661
2017-Q3	166	27,885	0	0	504
2017-Q4	67	27,965	0	0	270
2018-Q1	52	28,062	2	0	240
2018-Q2	58	28,064	0	0	191
2018-Q3	29	28,408	0	0	148
2018-Q4	99	28,461	0	0	365
Total	**878**		**23**	**2**	**2,740**

Notes: Table summarizes the number of auctions and nominations by season during 2017–2018. The 'Tracts Available' column shows the total number of tracts that could have been nominated. The 'Nom.×2' and 'Nom.×3' columns indicate the number of tracts that received multiple nominations.

Summary statistics on bid values, parcel sizes, royalty rates, nearest-well characteristics, and starting bids are reported in Appendix A (Tables 3–10).

4 Model

I model nomination and auction as two separate stages. Potential bidders, or "buyers", arrive sequentially and must incur a cost to learn their private values. They face strategic decisions about when to learn their values and nominate. The auction only proceeds if sufficient nominations are received, creating coordination and late-learning incentives.

4.1 Game Setup

I model the sale of a single tract t to a population of potential bidders who arrive over time; the total number that ever arrive, N_t, is random and determined endogenously by the Poisson arrival processes described below. Bidders are symmetric, and conditional on N bidders arriving, their private values $\mathrm{V}_t = (V_{1t}, \ldots, V_{Nt})$ are exchangeable and jointly distributed according to H_V on a common support $[0, \infty)^N$.[7] The marginal CDF $F_V(v_{it})$ is then defined as $H_V(\infty, \ldots, v_i, \ldots, \infty)$.

I allow for positive dependence across values so that for $V_{it}, V_{jt}\,(i \neq j)$, $F_V(v_{jt} \mid v'_{it}) \leq F_V(v_{jt} \mid v_{it})$ for all $v'_{it} > v_{it}$. This captures the reality that firms observing similar information (production from nearby wells or market conditions) form positively correlated value estimates. [8]

This modeling choice is consistent with a broader shift in the recent empirical auction literature away from common-value frameworks for onshore oil and gas lease sales [Covert and Sweeney, 2023].[9] Improvements in horizontal drilling and hydraulic fracturing have made oil extraction far less dependent on locating a single, uncertain geologic reservoir. Instead, modern techniques target a broad source rock rather than a narrow, hard-to-locate point deposit, reducing the production uncertainty that historically justified a common-value treatment of these auctions [Kong, 2020]. Consequently, differences in bid values are driven primarily by bidder-specific factors (development strategy, drilling technology, portfolio fit, and cost of capital) rather than by private information about a shared underlying quantity, which is precisely the private-values structure this paper models. At the same time, allowing

[7] H_V denotes this consistent family of exchangeable distributions, indexed implicitly by the realized number of arrivals.

[8] This is the affiliated private values (APV) property of Milgrom and Weber [1982]; see Appendix D.1 for the formal definition and for verification that the copula adopted in Section 5 satisfies it.

[9] See, e.g., Hendricks and Porter [1988] for the classic common-value treatment of offshore auctions.

for zero correlation across bidders would be too strong of an assumption in this setting: bidders drawing on the same regional geological and market information are still likely to reach correlated, if not identical, conclusions about a tract's value.[10] The correlated private values structure takes a middle path appropriate to this setting where bidder values remain genuinely private and require costly exploration to learn, but are permitted to be positively correlated through a common factor reflecting shared uncertainty about the region's geology and prevailing oil and gas prices.

No bidder observes any realization of V_t ex ante, including her own, and must incur a learning cost ($c_{it} \sim F_C$, i.i.d.) to learn her value, i.e. paying c_{it} reveals v_{it} to buyer i alone. This learning cost is privately observed and independent of V_t.

There are three policy instruments exogenously given and common across tracts: the nomination threshold $K \in \{0\} \cup \mathbb{N}$, the per-nomination fee $\phi \geq 0$, and the reserve price $r \geq 0$. These are public.

I describe the model below for a single tract, suppressing tract-level heterogeneity; Section 4.5 extends the model to heterogeneous tracts and introduces the observable tract characteristics X_t.

The sale proceeds in two stages: the nomination (Stage 1) and the auction (Stage 2). In Stage 1, the seller receives nominations for t. If total received nominations reach the nomination threshold ($M_t \geq K$), t is sold in second-price auction in Stage 2. The realized nomination count M_t is not revealed. Bidders arrive according to two independent, homogeneous Poisson processes, one for each stage. I use λ_{pre} and λ_{post} to represent the arrial rates of Stage 1 and Stage 2, respectively.[11]

An important timing distinction clarifies how positive dependence in H_V operates. A bidder who has not yet paid c_{it} to learn can update her beliefs about her own yet-to-be-learned value upon observing nomination signals from others. Let $n(v) \in \{0, 1\}$ denote a bidder's (as yet unspecified) nomination rule, mapping a realized value v to a nomination decision, and let $\mathcal{E}_m \equiv \{n(V_1) = 1, \ldots, n(V_m) = 1\}$ be the event that exactly m other bidders have nominated. By Bayes' rule and the joint distribution H_V, being exchangeable and symmetric across bidders, the posterior density of bidder i's own value conditional on

[10] Krasnokutskaya and Seim [2011] make a related point in the context of highway procurement auctions, arguing that correlation in bids can arise either from heterogeneity that is observable to bidders but not to the econometrician, or from genuine correlation of private costs, and that distinguishing between the two is generally not possible without strong auxiliary assumptions.

[11] I do not have data on how long each stages were. Hence, I normalize each stage to be of one unit of time, so the two arrival rates represents the expected number of arrivals

$\mathcal{E}_m$ is

$$f_{V_i|\mathcal{E}_m}(v) \;=\; \frac{\displaystyle\int_{\mathbb{R}_+^m} h_{m+1}(v,\, u_1, \ldots, u_m) \prod_{j=1}^{m} \mathbb{I}\{n(u_j) = 1\}\, du_1 \cdots du_m}{\Pr(\mathcal{E}_m)}, \tag{1}$$

where h_{m+1} is the $(m{+}1)$-dimensional marginal density of H_V and $\Pr(\mathcal{E}_m) = \mathbb{E}\Big[\prod_{j=1}^{m} \mathbb{I}\{n(V_j) = 1\}\Big]$ is the normalizing constant. Section 4.4 shows that the equilibrium nomination rule takes the threshold form $n(v) = \mathbb{I}\{v \geq v^*\}$, in which case $\mathcal{E}_m$ reduces to $\{V_1 \geq v^*, \ldots, V_m \geq v^*\}$ and the integral above is taken over $(v^*, \infty)^m$; this is the form used throughout the rest of the paper. Under positive dependence of H_V and any weakly increasing nomination rule $n(\cdot)$, the posterior stochastically dominates the prior:

$$F_{V_i|\mathcal{E}_m}(v) \;\leq\; F_V(v) \qquad \text{for all } v, \tag{2}$$

because knowing that others nominated raises the probability that V_i is also high.[12]

4.2 Strategy Set

I assume the following:

Assumption A1. *Learning is required before nominating or bidding.*

Under Assumption A1, I restrict the strategy set of buyers accordingly: l_i, or learning timing, for bidders who arrived in Stage 1; n_i, or nomination decision for bidders who learned their values; d_i, or learning decision for bidders who arrived in Stage 2.

Potential bidder i arriving during Stage 1 chooses a learning timing $l_i \in \{1, 2, 3\}$. **Stage 1 (early) learner** ($l_i = 1$) pays learning cost c_{it} immediately and learns private value $v_{it} \sim F_V(\cdot)$. Nomination decision $n_i = n(v_{it}) \in \{0 \text{ (don't nominate)}, 1 \text{ (nominate)}\}$ is made based on v_{it}.

Stage 2 (late) learner ($l_i = 2$) waits without learning v_{it} and pays the exploration cost c_{it} to learn in Stage 2 only if the auction proceeds. **Exit** ($l_i = 3$) means leaving without learning their private value. Let $\pi_{l=1}$ and $\pi_{l=2}$ be the ex-ante probability of a randomly arriving bidder choosing $l = 1$ and $l = 2$ respectively.

If Stage 2 (the auction stage) is triggered ($M_t \geq K$), more potential bidders arrive according to Poisson(λ_{post}). Upon observing c_{it}, they decide whether to enter ($d_i \equiv d(c_{it}) \in \{1 \text{ (Learn)}, 0 \text{ (Exit)}\}$. $d_i = 1$ bidders observe their values. Let $\pi_{d=1}$ be the ex-ante probability

[12] This follows directly from positive correlation of H_V and holds for any copula satisfying that property, not only the Gaussian copula adopted in estimation; see Appendix D.1.

of a randomly arriving bidder in Stage 2 choosing $d = 1$. Bidders who chose $l_i = 2$ in Stage 1 also learn their value by paying the learning cost c_{it}. All potential bidders who have learned their value ($l \in \{1, 2\}$ from Stage 1 and $d = 1$ from Stage 2) enter the auction and bid $b_i(v_i)$ if the value exceeds reserve price ($v_{it} \geq r$) and exit otherwise (Figure 3).

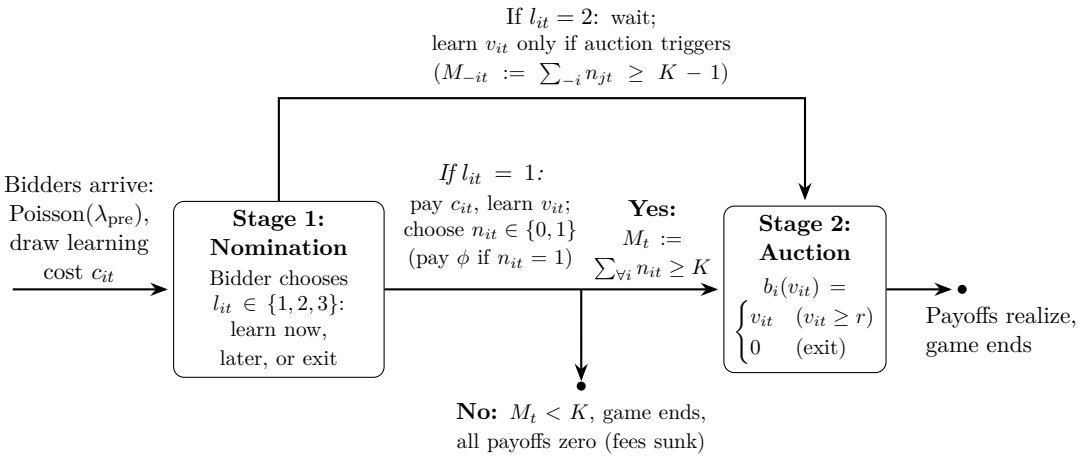


Figure 3: Timing of the model.

4.3 Payoffs

4.3.1 Stage 2: Expected Auction Surplus

A bidder with realized value v who enters the auction bids $b(v) = v$ (weakly dominant strategy under second-price rules) and wins if and only if her value exceeds both the reserve r and the highest competing value. Given the joint value distribution H_V, the expected surplus given j total competitors is

$$\tilde{\Pi}(v;\, j,\, H_V) \;=\; \int_r^v (v - y)\, f_{Y|v}(y \mid v,\, j,\, H_V)\, dy, \tag{3}$$

where $Y \equiv \max\{V_1, \ldots, V_j\}$ is the highest competing value and $f_{Y|v}$ is its conditional density given own value v, accounting for positive dependence through H_V (derived in Appendix B.4). With $j = 0$, $\tilde{\Pi}(v; 0, H_V) = v - r$ when $v \geq r$ and zero otherwise.

When a bidder faces m *deterministic* competitors (realized nominators) and a Poisson(Λ) number of *stochastic* competitors (non-nominating early arrivals and late arrivals), the expected surplus integrates over the Poisson distribution:

$$\Pi(v;\, m,\, \Lambda,\, H_V) \;=\; \sum_{n=0}^{\infty} \frac{e^{-\Lambda}\Lambda^n}{n!}\, \tilde{\Pi}(v;\, m + n,\, H_V). \tag{4}$$

In this set up, the total expected stochastic competition is the sum of non-nominating bidders with values above the reserve coming from three different groups: (1) later-learners from Stage 1 ($l = 2$); (2) early learners from Stage 1 ($l = 1$) who did not nominate but participate in the auction; and (3) Stage 2 arriving learners ($d = 1$).

4.3.2 Stage 1: Learning and Nomination

Bidders arriving at Stage 1 choose when to learn their values. Payoffs of the three strategies ($l_i \in \{1, 2, 3\}$) depend on three factors: 1. timing when learning cost c_i is paid; 2. difference in number of nominations required to satisfy $M \geq K$; and 3. difference in beliefs about yet-to-be-learned value under positive dependency. In addition, because I make the restricting assumption that nomination decision can be made only after learning in Stage 1, only those who chose $l_i = 1$ can decide whether to nominate ($n_i = 1$) upon paying nomination fee ϕ.

$l_i = 1$ **(Learning Early) Payoff.** For a bidder with learning cost c_i, the value of exploring immediately and nominating is

$$U_{l_i=1}(c_i) \;=\; -c_i + \int_{\underline{v}}^{\bar{v}} \mathbb{I}\{n(v)=1\} \left[-\phi \;+\; \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) \right] f_V(v)\, dv, \tag{5}$$

$$\begin{aligned} U_{l_i=1}(c_i) \;=\; -c_i \;+\; & \int_{\underline{v}}^{\bar{v}} \mathbb{I}\{n(v)=1\} \cdot \left[-\phi \;+\; \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) \right] f_V(v)\, dv \\ & + \int_{\underline{v}}^{\bar{v}} \mathbb{I}\{n(v)=0\} \cdot \sum_{m=K}^{\infty} \Pr(M_{-i}=m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v)\, dv, \end{aligned} \tag{6}$$

where $-c_i$ is the learning cost paid immediately upon choosing $l_i = 1$. Because learning is required for entry and, conditional on having learned, a bidder enters and bids whenever $v \geq r$, an early learner's continuation value has two branches. The first (bracketed) term applies when she nominates ($n(v) = 1$, i.e. $v \geq v^*$):

$$-\phi \;+\; \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) \tag{7}$$

is the payoff from nominating given the realized value v: ϕ is the nomination fee paid when $n(v) = 1$, and $\Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)$ is the expected auction surplus for each realized M_{-i}, the

number of nominations from other bidders. The sum starts at $m = K-1$ so that the bidder's own nomination brings total nominations to $m+1 \geq K$, triggering the auction.

The second term applies when the bidder has learned but does *not* nominate ($n(v) = 0$): they pay no nomination fee, and because they did not trigger the auction auction surplus is collected only when others' nominations reach the threshold ($m \geq K$).

$l_i = 2$ **(Learning Later) Payoff.** A buyer with cost c_i who chooses $l_i = 2$ pays nothing in Stage 1. If the auction triggers ($M \geq K$), she updates her beliefs about her own value using the posterior implied by H_V upon observing the auction was triggered ($\mathbb{I}\{M \geq K\}$, but not M itself), pays c_i to learn v_i, then enters if $v_i \geq r$. The expected payoff is

$$U_{l_i=2}(c_i) \;=\; \sum_{m=K}^{\infty} \Pr(M=m)\left[-c_i \;+\; \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv\right], \tag{8}$$

where M counts only *other* bidders' nominations. The competition argument is Λ_{nn} because conditioning on your own arrival does not change the distribution of the *other* arrivals.[13] $f_V(v \mid \mathcal{E}_m)$ is the posterior over her own value after observing $\mathcal{E}_m$, the event that m other bidders nominated (equation (1)), derived from H_V. By the stochastic dominance result in equation (2), this posterior shifts the non-nominating bidder's value distribution weakly upward relative to the unconditional $f_V(v)$. Section 4.4 shows that the equilibrium nomination rule takes the threshold form $n(v) = \mathbb{I}\{v \geq v^*\}$. Under this rule, $\mathcal{E}_m$ specializes to $\{V_1 \geq v^*, \ldots, V_m \geq v^*\}$, the form used from Section 7.3 onward.

$l_i = 3$ **(Exit) Payoff.** Bidders who choose to exit never return to the game and earn zero payoff.

$d_i = 1$ **(Late Arrivals' Learning) Payoff.** A Stage-2 post-arrival who observes the auction triggered ($M \geq K$) faces a similar decision as the $l_i = 2$ strategy: pay learning cost c, learn v, and bid if $v \geq r$. The two problems differ only in *when* the learning cost is sunk.

Remark 1 (Post-arrivals share the later-learner cutoff: $c^*_{\text{post}} = c^*_H$). Conditional on exactly m nominations, a post-arrival and an $l = 2$ later learner update their value beliefs to the same posterior $f_V(v \mid \mathcal{E}_m)$ (from H_V) and face the same Poisson pool of other non-nominator rivals. They differ only in the timing of paying the learning cost: a post-arrival exists only in a triggered auction and therefore pays c_i *for sure*, whereas an $l = 2$ bidder pays c_i only if

[13] This follows from the environmental equivalence property of Poisson games. See Myerson [1998].

the auction triggers. The post-arrival's expected payoff is thus

$$U_{\text{post}}(c_i) \;=\; -c_i \;+\; \sum_{m=K}^{\infty} \Pr(M=m \mid M \geq K) \int_{r}^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv. \tag{9}$$

Multiplying through by $\Pr(M \geq K)$ recovers the later-learner value, $U_{l=2}(c_i) = \Pr(M \geq K)\, U_{\text{post}}(c_i)$: the two are proportional and therefore share the same zero. Since c_H^* solves $U_{l=2}(c_H^*) = 0$, the post-arrival's cutoff coincides with it, $c_{\text{post}}^* = c_H^*$.

4.4 Equilibrium

I characterize a symmetric Perfect Bayesian Equilibrium (PBE) in which all arriving Stage 1 bidders follow the same cost-cutoff strategy $l_i = l(c_{it}) \in \{1, 2, 3\}$.[14] Bidders form rational expectations about the distribution of competitors, consistent with equilibrium play and derived from the Poisson arrival process and the equilibrium probabilities $(\pi_{l=1}, \pi_{l=2})$. Equilibrium requires these to be a fixed point: given $(\pi_{l=1}, \pi_{l=2})$, optimal behavior for each arriving bidder regenerates the same pair.

I solve the model by backward induction. In **Stage 2**, truthful bidding is the weakly dominant strategy and constitutes the unique Nash equilibrium of the Stage 2 subgame for bidders with bids exceeding reserve price ($b(v_{it}) = v_{it} \cdot \mathbb{I}\{v_{it} \geq r\}$). I characterize expected auction surplus under positively dependent private values (Section 4.3.1).

For **Stage 1**, I employ a threshold strategy. First, the nomination threshold v^* follows from the indifference condition of the marginal nominator who have learned v, where v^{**} is derived by setting (7) equal to zero. Given v^* and expected competition, the three payoffs $U(c; l = k)$, $\forall k \in \{1, 2, 3\}$ yield two cost cutoffs: bidders with $c < c_L^*$ choose $l = 1$ and those with $c_L^* \leq c \leq c_H^*$ choose $l = 2$. Late arrivals' threshold strategy is defined similarly, where those with $c < c_{\text{post}}^*$ choose to learn ($d = 1$).

Since payoffs discussed in Section 4.3.2 are increasing linearly in their respective arguments v and c (higher valuation increases 7, or $n = 1$ payoff; lower cost increases $l = 1$ and $l = 2$ payoffs in different speed), the optimal strategies are weakly increasing in those arguments. Therefore, for each realized (and learned) v and c, cutoff strategies are determined using indifference condition.

[14] With incomplete information over private costs and values, the appropriate solution concept is a symmetric perfect Bayesian equilibrium (PBE). Because Stage 2 has a unique Nash equilibrium (truthful bidding is weakly dominant), backward induction uniquely pins down Stage 2 behavior, and Stage 1 strategies are best responses given correct beliefs. The resulting equilibrium is therefore PBE.

Nomination Threshold v^*. The nomination threshold is

$$v^* = \max\{r,\, v^{**}\}, \tag{10}$$

where v^{**} satisfies

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \Pi(v^{**};\, m,\, \Lambda_{\text{nn}},\, H_V) = \phi. \tag{11}$$

Intuitively, v^{**} is the value at which a bidder is exactly indifferent between nominating and not nominating given expected competition and the per-nomination fee ϕ. When $v < v^{**}$, the fee exceeds the expected surplus from participation. The reserve r imposes an institutional floor: any bidder with $v < r$ cannot profitably participate even unopposed.

This specializes the general nomination-event definition in equation (1) to the threshold form $\mathcal{E}_m = \{V_1 \geq v^*, \ldots, V_m \geq v^*\}$, with $n(v) = \mathbb{I}\{v \geq v^*\}$. This form is used in the payoffs for $l_i = 2$ and $d_i = 1$ calculations.[15]

The rule that a lone nominator who faces no competing bidders is obligated to complete the purchase at r highlights both the benefit and cost of nominating that later-learners never receive when $K = 1$. Nominators face a higher ex-post surplus since they face a higher probability of triggering the auction, yielding a nonnegative surplus ($v - r \geq 0$) if they are the sole bidder. However, they face a lower ex-ante surplus since $v < v^*$ means they won't nominate but their learning cost c is sunk. Late-learners, by contrast, enter only conditional on $M \geq K \geq 1$, so they can never be the sole bidder at auction.[16] Nomination is discouraged only by the learning cost c_i and the fee ϕ, both of which enter the payoff unconditionally; the fallback purchase rule does not.

Plugging v^* into (6), since $\mathbb{I}\{n(v) = 1\} = 1$ when $v \geq v^*$ and zero otherwise, (6) can be rewritten as $-c_i + \int_{v^*}^{\bar{v}} U_{n_i=1}(v) f_V(v)\, dv$ and nomination fee is paid with probability $1 - F_V(v^*)$:

$$U_{l_i=1}(c_i) \;=\; -c_i - \phi \cdot \big[1 - F_V(v^*)\big] \;+\; \int_{v^*}^{\bar{v}} \sum_{m=K-1}^{\infty} \big[\Pr(M_{-i}{=}m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\big] f_V(v)\, dv \tag{12}$$

$l = 1$, $l = 2$ Thresholds c_L^* and c_H^*. Equilibrium strategy should characterize intervals of cost type c such that one strategy dominates the other. I begin by studying an important

[15] The pure-strategy threshold rule is robust to noisy/mixed nomination strategies. See Appendix D.2 for the formal argument.

[16] The $j = 0$ term inside $\Pi(v; K-1, \Lambda_{\text{nn}}, H_V)$ captures exactly this: a strictly positive contribution to nominators' expected surplus that is structurally unavailable to non-nominators.

property of the two payoff functions $U(c_i; l=1)$ and $U(c_i); l=2$ that determines equilibrium structure.

Proposition 1 (Single Crossing)**.** *The payoff functions cross at most once and there exists at most one c_L^* such that $U_{l=1}(c_L^*) = U_{l=2}(c_L^*)$.*

Proof. See Appendix D.7. □

Let $A = \int_{v^*}^{\bar{v}} \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) f_V(v)\, dv$ from (12) be the expected gross auction surplus for a $l=1$ player and
$B = \sum_{m=K}^{\infty} \left[\Pr(M=m) \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv\right]$ from (8) be the expected gross auction surplus for $l=2$ player.

Since $\Pr(M \geq K) < 1$,

$$|dU_{l=1}/dc_i| = 1 > \Pr(M \geq K) = |dU_{l=2}/dc_i| \tag{13}$$

Two lines with different slopes can intersect at most once. If they intersect at c_L^*, then

$$\begin{cases} U_{l=1}(c_i) > U_{l=2}(c_i) & (l=1) \text{ if } c_i < c_L^* \\ U_{l=1}(c_i) < U_{l=2}(c_i) & (l=2) \text{ if } c_i > c_L^* \end{cases}$$

Depending on the intercepts and slopes, exactly one of the three equilibrium cases holds.

Proposition 2 (Equilibrium Structure)**.** *Define*

$$\Delta_0 := U_{l=1}(0) - U_{l=2}(0) = A - \phi\big[1 - F_V(v^*)\big] - B \quad \textit{(payoff difference at zero cost)} \tag{14}$$

$$\Delta_{l=2} := U_{l=1}(c_H^*) - U_{l=2}(c_H^*) = U_{l=1}(c_H^*) \quad \textit{(payoff from choosing } l=1 \textit{ at } c_H^*\textit{)} \tag{15}$$

Then exactly one of the following holds. The empirically relevant case is Case 1 (interior solution), described below. The two corner solutions in Case 2 (coordination failure: everyone waits and learns later) and Case 3 (everyone learns early) are presented in Appendix D.4.

Case 1: Interior Solution (Single Crossing) *If $\Delta_0 \geq 0$ and $\Delta_{l=2} < 0$, then there exists an unique $c_L^* \in (0, c_H^*)$ where $U_{l=1}(c_L^*) = U_{l=2}(c_L^*)$ such that*

$$\begin{cases} c_i < c_L^* : l = 1 \\ c_L^* \leq c_i \leq c_H^* : l = 2 \\ c_i > c_H^* : l = 3 \end{cases}$$

Figure 4 illustrates all three equilibrium cases. Case 1 (left panel, interior solution) is the empirically relevant scenario: low-cost bidders ($c_i < c_L^*$) learn early ($l=1$), intermediate-

cost bidders ($c_L^* \leq c_i \leq c_H^*$) learn later ($l = 2$), and high-cost bidders exit ($l = 3$). This matches the ND data, where both nominators ($l = 1$) and non-nominators ($l = 2$) co-exist in equilibrium. Cases 2 and 3 are corner solutions, arising when the nomination fee is prohibitively high (coordination failure, no nominations) or negligible (all $l = 1$). Which case prevails depends on the auction parameters K and ϕ. See Appendix D.4.

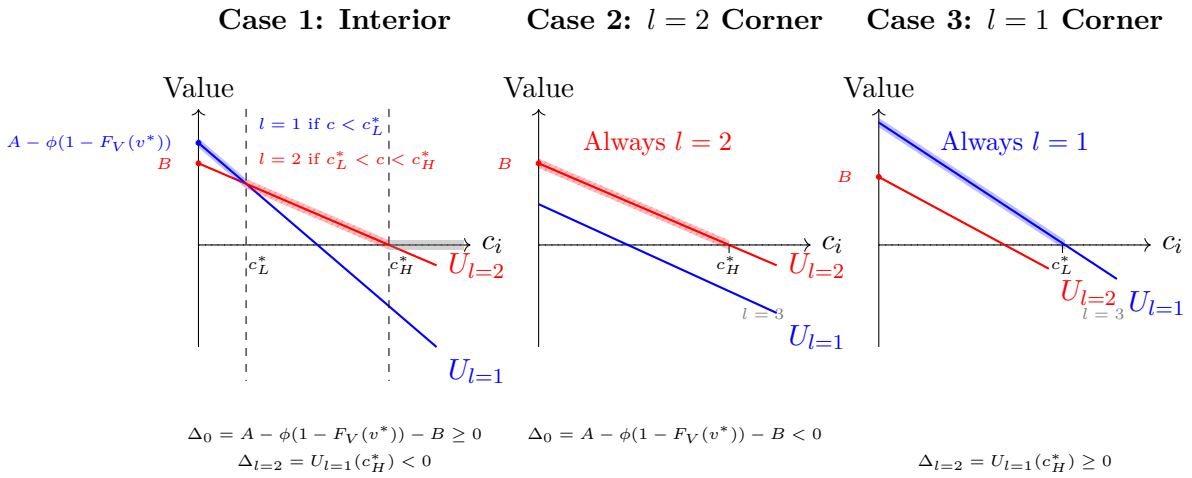


Figure 4: Equilibrium cases based on value function intercepts and slopes.

Notes: Three equilibrium cases based on the value of early learning ($U_{l=1}$) and learning later ($U_{l=2}$) as functions of learning cost c. The intersection points define cutoff costs c_L^* and c_H^*. The single crossing property ensures at most one intersection.

In Case 1 (interior solution), the equilibrium cutoff c_L^* satisfies

$$U_{l=1}(c_L^*) = U_{l=2}(c_L^*) \tag{16}$$

Expanding and solving for c_L^*

$$c_L^* = \frac{A - B - \phi\big[1 - F_V(v^*)\big]}{\Pr(M < K)} \tag{17}$$

Then, the probability of a randomly arriving bidder to choose $l = 1$ is

$$\pi_0^{l=1} = \Pr[c < c_L^*] = F_C(c_L^*) \tag{18}$$

By independence of c and v, the unconditional nomination probability is the product of the two marginals:

$$\Pr[c < c_L^*, v > v^*] = \pi_0^{l=1} \cdot [1 - F_V(v^*)]$$

Similarly, the cutoff c_H^* satisfies $U(c_H^*; l = 2) = 0$, i.e. the learning cost at which the expected auction surplus of choosing $l = 2$ exactly covers the cost of exploring in Stage 2:

$$c_H^* = \frac{\sum_{m=K}^{\infty} \Pr(M = m) \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv}{\Pr(M \geq K)}. \tag{19}$$

Then probability of a random bidder choosing $l = 2$ is

$$\pi_0^{l=2} = \Pr[c_L^* < c < c_H^*] = F_C(c_H^*) - F_C(c_L^*) \tag{20}$$

4.4.1 Competition Structure

The probabilities $\pi_0^{l=1}$ and $\pi_0^{l=2}$ determine the distribution of competitors that any individual bidder expects to face.

Competition from Other Nominators Given nomination probability $\pi_0^{l=1}$, the number of other nominators M_{-i} follows from the Poisson arrival process.

Lemma 1 (Poisson Arrivals of Nominators)**.** *If bidders arrive according to Poisson(λ_{pre}) and each independently choose $l = 1$ and then $n = 1$, meaning the probability of such event happening is $\pi_0^{l=1} \cdot [1 - F_V(v^*)]$, then the number of competing nominators is Poisson with the* nominator rate Λ_{nom}*:*

$$M_{-i} \sim \mathit{Poisson}\left(\Lambda_{\text{nom}}\right), \qquad \Lambda_{\text{nom}} := \lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - F_V(v^*)] \tag{21}$$

Proof. See Appendix D.6. □

Competition from Non-nominators The expected competition from Stage 1 arrivals choosing $l = 2$ and Stage 2 arrivals choosing $d = 1$ is given by the following two components:

$$\Lambda^{l=2} = \lambda_{pre} \cdot \pi_0^{l=2} \cdot [1 - F_{\text{post}}(r)], \tag{22}$$

$$\Lambda^{d=1} = \lambda_{post} \cdot (\pi_0^{l=1} + \pi_0^{l=2}) \cdot [1 - F_{\text{post}}(r)], \tag{23}$$

$$\Lambda_{\text{nn}} = \Lambda^{l=2} + \Lambda^{d=1}, \tag{24}$$

where Λ_{nn} is the *non-nominator competition rate* (Stage-1 later learners plus Stage-2 post-nomination arrivals), and

$$F_{\text{post}}(r) \;=\; \sum_{m=K}^{\infty} F_{V_i|\mathcal{E}_m}(r)\,\Pr(M=m \mid M \geq K) \tag{25}$$

is the reserve-clearing CDF averaged over the posterior $F_{V_i|\mathcal{E}_m}$ of (2).

All non-nominators enter the auction when $v \geq r$. Because they act only after observing the auction triggered ($M \geq K$), correlation shifts their own-value belief upward, so their reserve-clearing probability is the posterior $[1 - F_{\text{post}}(r)]$ of (25) rather than the prior $[1 - F_V(r)]$. Post-nomination arrivals face the same cost cutoff c_H^* (Remark 1), so the fraction that finds exploration profitable is $F_C(c_H^*) = \pi_0^{l=1} + \pi_0^{l=2}$.

4.4.2 Fixed Point Equilibrium Characterization

Consistency Requirement The payoff functions $U_{l=1}(c)$ and $U_{l=2}(c)$ depend on the expected competition $(M_{-i}, \Lambda_{\text{nn}})$, which depends on the probabilities $(\pi_0^{l=1}, \pi_0^{l=2})$, which depend on the payoff functions through equations (17), (18) and (19), (20). An equilibrium requires these objects to be mutually consistent.

Definition 1 (Equilibrium Consistency)**.** *Probabilities* $(\pi_0^{l=1}, \pi_0^{l=2})$ *are* ***consistent*** *if*

1. *Given* $(\pi_0^{l=1}, \pi_0^{l=2})$*, expected competition is characterized by* $M_{-i} \sim$ *Poisson(*Λ_{nom}*) and* Λ_{nn}*, both of which can be computed.*

2. *Given expected competition,* $U_{l=1}(c)$*,* $U_{l=2}(c)$ *and* $U_{post}(c)$ *can be rewritten via equations* (12)*,* (8) *and* (9)

3. *Given payoff functions, the optimal threshold strategy cutoff points defining* l, n, d *can be derived.*

4. *The optimal strategies generate probabilities* $(\pi_0^{l=1}, \pi_0^{l=2})$ *via equations* (18) *and* (20) *that equal the original probabilities*

Putting everything together, a symmetric equilibrium is defined as the following.

Definition 2 (Symmetric PBE)**.** *A symmetric equilibrium is a tuple* $(v^*,\, c_L^*,\, c_H^*,\, \pi_{l=1},\, \pi_{l=2})$ *such that*

1. ***Nomination threshold.*** $v^* = \max\{v^{**}, r\}$*, where* v^{**} *is the unique solution to*

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\,\tilde{\Pi}(v^{**};\, m,\, \Lambda_{\text{nn}},\, H_V) \;=\; \phi. \tag{26}$$

2. ***Cost cutoffs.*** *c_L^* solves $U(c_L^*; l=1) = U(c_L^*; l=2)$, and c_H^* solves $U(c_H^*; l=2) = 0$.*

3. ***Strategy probabilities.***

$$\pi_{l=1} = F_C(c_L^*), \qquad \pi_{l=2} = F_C(c_H^*) - F_C(c_L^*;\, \theta_c). \tag{27}$$

4. ***Competition rates (consistency).***

$$M_{-i} \sim Poisson(\lambda_{pre}\, \pi_{l=1}\, [1 - F_V(v^*)])\,, \tag{28}$$

$$\Lambda_{\text{nn}} = \lambda_{pre}\, \pi_{l=2}\, [1 - F_{\text{post}}(r)] \;+\; \lambda_{post}\, (\pi_{l=1} + \pi_{l=2})\, [1 - F_{\text{post}}(r)]. \tag{29}$$

The equilibrium is a fixed point: $\pi_{l=1}$ and $\pi_{l=2}$ determine the competition rates via (28)–(29), which determine the value functions $U_{l=1}(c), U_{l=2}(c)$, determining the cutoffs c_L^*, c_H^* and recovering $\pi_{l=1}, \pi_{l=2}$ via (27).

4.5 Expansion to Heterogeneous Game

Sections 4 characterize equilibrium behavior for a single, implicitly homogeneous tract. In the data, however, no two tracts are alike: tracts differ in size, royalty rate, proximity to producing wells, and the timing of nearby drilling activity ("spud date"). These differences are not incidental and are exactly the variables ND's own tract listings advertise to every potential bidder before nomination even begins, visibly correlate with how much bidders are willing to pay (Table 11). Before turning to estimation, I extend the model to admit this heterogeneity, and in doing so motivate the multiplicative value decomposition used throughout the empirical analysis.

Let X_t denote the vector of tract characteristics observable to every bidder upon arrival (size, royalty rate, distance to the nearest existing well, and that well's spud date). One might think that, because X_t is public information, bidders already know their value for the tract and the costly-learning structure of Section 4.2 is unnecessary. However, what a given X_t *implies* for a specific firm's payoff depends on firm-specific drilling technology, available machinery, and the firm's existing lease portfolio. [17] Translating X_t into a usable valuation is precisely the costly step that Section 4.2 models via the learning cost c_i.[18] Observability

[17] Phone interviews with bidders indicate that X_t is necessary, but far from sufficient, to determine a firm's value for a tract: most areas "have been explored" in the sense that their basic geological and locational characteristics are known, but translating X_t into a firm-specific dollar value still requires private, costly evaluation.

[18] For instance, large firms generally maintain in-house land men and geologists for this purpose, while small firms rely on third parties whose availability is itself uncertain.

of X_t and costly learning of V_i are therefore not in tension: X_t is the public starting point, and c_i is the cost of figuring out what that starting point is worth to a specific firm.

Because X_t is public and known to shift values in a common direction (a tract close to a recently productive well is known by every bidder to be more promising than a distant one, before anyone pays learning cost) the location of the value distribution differs systematically across tracts. Treating two tracts with very different X_t as though they generate values on the same scale is not innocuous: it would conflate genuine cross-bidder, private heterogeneity (the object of interest) with heterogeneity that is mechanically driven by observable, public differences across tracts.

This is compounded by an asymmetry in units. The policy instruments governing the auction – the nomination fee ϕ and the reserve price r – are fixed in dollar terms by the state, and a bidder's learning cost c_i (land-men's time, third-party consulting fees) is similarly denominated in dollars and does not automatically rise or fall with how lucrative a particular tract happens to be.[19] Values (V_t), by contrast, scale directly with X_t: a tract twice as promising is (by definition) worth roughly twice as much to every bidder who values it. Because V scales with X_t while ϕ, r, and the distribution of c_i do not, the equilibrium derived in Section 4.4, which holds the units of V fixed relative to the units of ϕ, r, c_i, does not simply rescale tract by tract. The game itself takes a different shape as X_t varies, and the model needs to be rewritten to make this dependence explicit before it can be taken to data.

The observable component s_t. I summarize the public, tract-level effect of X_t on values with a single scalar, $s_t \equiv s(X_t)$, which I call the tract's *observable component*. Each bidder's value is then

$$V_{it} = s_t \cdot \tilde{V}_{it}, \tag{30}$$

where $\tilde{V}_{it}$ is bidder i's residual, private value for tract t, distributed according to a tract-invariant $H_{\tilde{V}}$ as in Section 4, again allowing for correlation.[20] [21]

I deliberately call s_t the tract's *observable component* rather than a *common value*: unlike in Milgrom and Weber [1982] where a common value is an unknown payoff state that all

[19] A flat \$10 nomination fee or reserve prices starting from \$2 per-acre is a rounding error on a highly promising tract but can be a meaningful deterrent on a low-value one.

[20] Formally, H_V in Section 4 is the distribution of $s_t \cdot (\tilde{V}_{1t}, \ldots, \tilde{V}_{Nt})$ induced by $H_{\tilde{V}}$. Nothing in the equilibrium characterization of Section 4.4 relied on H_V being unscaled, so that characterization carries over unchanged once restated in terms of $\tilde{V}$.

[21] Phone interview evidence directly supports treating residual values as private and correlated, rather than common: "value depends on firm-specific tech, machinery, [and] portfolio constraints," consistent with firms drawing correlated information about a tract's prospects from unobservable characteristics while placing a firm-specific, private value on that information.

bidders learn about through noisy signals, s_t is known with certainty to every bidder the moment they arrive and no exploration is required to learn it. I refer to $\tilde{V}_{it}$ as bidder i's private *value*, not a *signal*: once bidder i pays c_{it} and learns $\tilde{V}_{it}$, that realization is her private payoff-relevant valuation, independent to any other bidder, and is never revised after the fact.[22]

Implications for the game. Because ϕ, r, and the (unrealized) distribution of c_i are fixed in dollar terms while V scales with s_t, the equilibrium objects of Section 4.4 (v^*, c_L^*, c_H^*) must in general be solved tract by tract: two tracts with different s_t but otherwise identical primitives face a different relative balance between the fixed-dollar instruments and the value of participating. Section 5.1 operationalizes s_t empirically via a hedonic regression on X_t, and Proposition 3 formalizes the rescaling implied by equation (30): solving the model in *residual* units $\tilde{V} = V/s_t$ is equivalent to solving the dollar-denominated model tract by tract, provided the fixed-dollar instruments are rescaled in tandem ($\tilde{\phi}_t = \phi/s_t$, $\tilde{r}_t = r/s_t$, $\tilde{c} = c/s_t$). This is the sense in which s_t is the object that matters for equilibrium: any two tracts with the same s_t (even with different underlying X_t) generate the same residual-space equilibrium.

5 Estimation

The model is estimated in two steps. I parameterize the joint value distribution as

$$H_V(v_1, \ldots, v_N) \;=\; C(F_V(v_1), \ldots, F_V(v_N);\, \rho)\,,$$

where $C(\cdot; \rho)$ is a Gaussian equicorrelation copula with parameter $\rho \in [0, 1)$. **Step 1** recovers the private-value distribution $\hat{F}_{\tilde{V}}$ and the copula parameter $\hat{\rho}$ from observed auction bids. **Step 2** estimates the nomination-stage structural parameters (Poisson arrival rates $(\lambda_{\text{pre}}, \lambda_{\text{post}})$ and the cost distribution $F_c(\cdot; \xi_c, \omega_c, \alpha_c)$) using GMM with equilibrium conditions imposed as explicit constraints (MPEC). Full estimation details are in Appendix B.

5.1 Step 1: Bid Homogenization and Value Distribution Recovery

Raw bids b_{istc} vary multiplicatively with observable tract characteristics (size, royalty rate, well proximity). I estimate a log-linear hedonic regression to isolate the private-value component, recovering auction-specific scaling factors $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$ and homogenized

[22] The only updating that occurs anywhere in the model is the belief revision of an unexplored bidder about her own not-yet-realized $\tilde{V}_{it}$, upon observing others' nomination behavior (equations (1)– (2)) – updating a prior about one's own draw, which is the defining feature of a correlated *private* values model as opposed to a common or interdependent values model

bids $\tilde{b}_{istc} = b_{istc}/s_t$. An equilibrium translational invariance result (Proposition 3) justifies working in residual space: a uniform shift in log-heterogeneity Z_t shifts all equilibrium actions proportionally, leaving residual-space strategies unchanged. Regression estimates are in Table 11 (reproduced in Appendix B.1); residual diagnostics supporting the log-linear specification are in Appendix B.11.

Note that by Proposition 3, all equilibrium objects discussed in Section 4 can be redefined in residual space. Notably, the model primitives (ϕ, r, F_C) is replaced with $(\tilde{\phi}_t, \tilde{r}_t, F_{\tilde{C}_t}) = (10/s_t, r/s_t, F_{C/s_t})$. This is because the nominal \$10 fee, nomination fee and learning cost c does not scale the same way as v does (e.g. v decreases with distance to nearest well while the nominal fees and learning cost c does not). For computational simplicity, I group the 728 estimation auctions into 19 equal-count decile bins and assign each bin b a representative fee $\tilde{x}_b = \text{median}_{t\in b}(x/s_t)$ for $x \in \{\phi, r, c\}$.

Given the homogenized residual-unit bids and cross-auction variation in bidder counts N_r, I jointly estimate a monotone B-spline $\hat{F}_V$ and the equicorrelation Gaussian copula parameter $\hat{\rho}$, matching the theoretical order-statistic CDF (Theorem 1) to empirical Weibull ranks on a 200-point grid over 100-fold cross-validated train/test splits. The Gaussian copula achieves the lowest hold-out RMSE (Table 13; $\hat{\rho} = 0.8605$) and is also the theoretically appropriate choice for a correlated private-values setting (see Appendix B.3 for the economic rationale). A Lognormal tail is fitted to the B-spline and then refined by a truncation-corrected maximum-likelihood step at the lower bid-support boundary v^* (Appendix B.3) to extrapolate F_V below the observed bid support. The non-nominating bidder's conditional value Details are in Appendix B.3.

5.2 Step 2: Nomination-Stage Parameters

GMM moments. I match four moment conditions per bin to model-implied values: (1) the auction trigger rate $\Pr(M_b \geq K)$; (2) mean nominations per triggered tract (zero-truncated Poisson mean); (3) the variance of nominations per triggered tract (zero-truncated Poisson variance); and (4) mean total bidders per triggered auction $\mathbb{E}[N \mid M \geq K]$. Moment (1) is matched in log scale. Full definitions are in Appendix B.5.

Binning. After homogenization, the flat \$10 nomination fee becomes a bin-specific residual-space fee $\phi_b = 10/s_b$ where $s_b = \text{median}_{t\in b}(s_t)$. Because this varies with s_b, the equilibrium indifference conditions, and therefore the cost cutoffs (v^*, c_L^*, c_H^*), differ across bins. Treating all auctions as having the same equilibrium would be incorrect. I address this by estimating bin-specific equilibrium objects $(v_b^*, c_{L;b}^*, c_{H;b}^*)$ while the cost distribution F_C remains shared across all bins. Bin construction details are in Appendix B.6.

MPEC implementation. Following Su and Judd [2012], I replace the nested fixed point [Rust, 1987] with MPEC: the 62-element decision vector

$$\boldsymbol{z} = \left(\lambda_{\text{pre}},\ \lambda_{\text{post}},\ \xi_c,\ \omega_c,\ \alpha_c,\ v_1^*, \ldots, v_{19}^*,\ c_{L,1}^*, \ldots, c_{L,19}^*,\ c_{H,1}^*, \ldots, c_{H,19}^*\right)$$

is optimized subject to 57 equality constraints (three per bin for 19 bins, namely nomination threshold $h_{1,b}$, early learning cutoff $h_{2,b}$, and non-nominator cutoff $h_{3,b}$). Full MPEC implementation details are in Appendix B.8.

6 Estimation Results

Table 2 reports the five shared structural parameters estimated by the binned MPEC procedure.[23] The skew-normal cost parameters are estimated directly in dollars and shared across bins, so the mean exploration cost implied by these estimates is $\approx$ \$1,161, with implied standard deviation $\approx$ \$416. The negative shape parameter $\hat{\alpha}_c$ makes the cost distribution left-skewed. [24] The equilibrium cost cutoffs are estimated in residual units and convert to dollars through the auction-specific scaler $s_t = \exp(\hat{y}_t)$ as $c^* \cdot s_t$. In every bin the dollar cutoff $\hat{c}_b^{*H} \cdot s_b$ lies well above the cost mean $\hat{\mu}_c$, so the cutoffs are non-binding: essentially all arriving bidders would find exploration worthwhile conditional on arriving. The mean learning cost of \$1,161 is modest relative to the average total bid of \$11,803 across all bidders (and approximately \$13,900 for winning bidders). The barrier to nomination therefore operates through the low pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$ rather than through cost screening.

The per-bin equilibrium objects $(\hat{v}_b^*, \hat{c}_b^{*L}, \hat{c}_b^{*H})$ are reported in Appendix Table 15. The nomination threshold $\hat{v}_b^*$ rises with the residual-space fee ϕ_b across bins, while the early learner and non-nominating bidder cutoffs ($\hat{c}_b^{*L} \in [1.51, 2.21]$; $\hat{c}_b^{*H} \in [1.87, 3.05]$, in residual units) lie in the right tail of the shared cost distribution, so that conditional on arriving almost all bidders find exploration worthwhile. The model's fit to the four GMM moments is reported in Appendix Table 16. The estimated value functions $U_{l=1}(c)$ and $U_{l=2}(c)$ and the implied cost-distribution mass shares are presented in Appendix B.10 (Figure 10).

The average cost of learning differs systematically across bidder types, reflecting cost-based selection into $l = 1$ versus $l = 2$. Early learners ($l = 1$) pay the least, an average of \$979; late learners ($l = 2$) pay the most, \$1,374; late (post-nomination) arrivals ($d = 1$) pay \$1,021; and the unconditional population mean is \$1,161. That nominators being the lowest-cost type while later-learners the higher type is precisely the sorting the model predicts:

[23] These estimates use the Joint GMM Strategy copula parameter ($\hat{\rho} = 0.861$, Table 14), estimated using the full 200-point integration grid (grid200 method).

[24] Mean exploration cost is calculated as $\hat{\mu}_c = \hat{\xi}_c + \hat{\omega}_c\, \delta \sqrt{2/\pi}$ where $\delta = \hat{\alpha}_c / \sqrt{1 + \hat{\alpha}_c^2}$

Table 2: MPEC Structural Parameter Estimates

Parameter	Description	Estimate
$\hat{\lambda}_{\text{pre}}$	Pre-auction Poisson arrival rate	0.005
$\hat{\lambda}_{\text{post}}$	Post-auction Poisson arrival rate	2.894
$\hat{\xi}_c$	Skew-Normal location (cost dist., \$)	1,712
$\hat{\omega}_c$	Skew-Normal scale (cost dist., \$)	690
$\hat{\alpha}_c$	Skew-Normal shape (cost dist.)	−126.9
	Implied SN mean (\$)	1,161
	Implied SN std. dev. (\$)	416

Notes: Binned MPEC estimation, $B = 19$ bins of s_t, break-points derived from the full available-tract universe (2017Q1–2018Q4). The skew-normal cost parameters $(\hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c)$ are estimated directly in dollars and shared across bins.

cost-efficient firms find early exploration worthwhile, whereas higher-cost firms defer and wait on the information a nomination reveals.

Evaluating the cost distribution at the dollar cutoffs $\hat{c}_b^* \cdot s_b$ shows how much exploration cost screens participation. In the high-value bins (bins 1–11) the dollar cutoff sits in the far right tail of the cost distribution, so the early learning ($l = 1$) share $\Pr(c < \hat{c}_b^{*L}\, s_b)$ is 100% and cost imposes no screen. In the high-value, high-s bins the dollar cutoff falls into the body of the distribution: the early learning ($l = 1$) share declines to 3.5% in the lowest-value, lowest-s bin, and the share of arrivals who would *not* nominate on cost grounds, $\Pr(c > \hat{c}_b^{*L}\, s_b)$, rises to 96% in the lowest=s bin. The population-weighted average across all bin becomes $\approx 50\%$. Bidders do exit: the exit share $\Pr(c > \hat{c}_b^{*H}\, s_b)$ is 0% in the high-value bins but reaches 95% in the lowest-s. Population-weighted overall average is $\approx 43\%$, so the model does not imply that no one exits. Cost screening is thus concentrated in low-value tracts due to a cost-relative-to-value effect.

However, the learning cost cannot explain the low nomination rate conditional on learning: even in bins 1–11, where cost imposes no screen and everyone learns upon arriving, the trigger rate is only $\approx 0.5\%$, because the pre-period arrival rate is so low ($\hat{\lambda}_{\text{pre}} \approx 0.005$). The low arrival rate is therefore the binding constraint on the overall nomination rate, while cost screening governs which tracts are nominated.

Two nomination-stage frictions besides exploration cost are non-binding: a counterfactual removing the \$10 fee raises the population-weighted trigger rate by only 1.6%, and removing the purchase commitment (setting nomination threshold $v^* = v^{**}$ rather than $\max(v^{**}, r)$) leaves the trigger rate unchanged, since v^{**} already exceeds the reserve in most bin.

As a robustness check on the nomination-count specification, I re-estimate the model re-

placing the mean and variance of nominations (Moments 2–3) with the conditional multiple-nomination share $\Pr(M \geq 2 \mid M \geq 1)$. The results, and their implications for the single-rate Poisson arrival assumption, are reported in Appendix B.7.

7 Counterfactual Analysis

Using the estimated structural model (Tables 2–16), I conduct two sets of counterfactual experiments that compares a world with no nomination requirement ($K = 0$) against the current policy ($K = 1$), first to decompose the value of nomination and second to quantify the externality due to existence of non-nominators. Throughout, I fix the estimated structural parameters $(\hat{\lambda}_{pre}, \hat{\lambda}_{post}, \hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c)$ and re-solve for the equilibrium objects $(v_b^*, c_{Lb}^*, c_{Hb}^*)$ under each counterfactual scenario.[25] Appendix E.1 presents a simplified version without value uncertainty or learning later to build geometric intuition.

7.1 Decomposing the Value of Nomination: Thickening versus Screening

The nomination system does two things at once. First, it **thickens** the market by inviting more bidders: because values are correlated, a nomination reveals that at least one bidder found $v \geq v^*$, encouraging not-yet-learned bidders to enter by updating their beliefs. Second, it *screens* tracts along two margins. It screens on *value*: because a firm nominates only when it has found $v \geq v^*$, nominated tracts are genuinely more valuable than a tract chosen to be sold arbitrarily. It also screens on *demand*: a nomination guarantees that at least one bidder bids, whereas an arbitrarily offered tract frequently attracts no one.

To separately quantify these channels, I hold each sold tract's characteristics fixed and switch off, one at a time, the distinct things a nomination does. Comparing five counterfactual scenarios isolates four forces. In each world bidders' values for a given auction are correlated because they share a common component (an auction-specific draw W_t); the appendix gives the exact simulation.

[25] I also varied the per-nomination fee ϕ (holding $K = 1$), sweeping it from the current nominal level upward and re-solving the equilibrium for a representative small, median, and large tract. The fee has little leverage because the auction trigger rate is already very low—about 0.5% at the estimate, driven by the thin early-arrival rate $\hat{\lambda}_{pre} \approx 0.005$. Raising ϕ lifts the nomination-value threshold v^{**} and so *screens* the nomination margin, but this only shrinks an already-thin extensive margin: expected revenue per offered tract is maximized at $\phi = 0$ and declines thereafter, so the current fee is essentially revenue-optimal. A subsidy does not help either, since participation is already saturated on the tracts that generate revenue. Appendix E.11 gives details, including the offsetting selection effect that *raises* revenue conditional on a sale even as the number of sales falls.

World $\mathcal{W}_1$ is the estimated $K{=}1$ economy: nominated auctions (at least one nominator with $v \geq v^*$) in which non-nominating entrants expect to draw their value from the *posterior* correlated value. World $\mathcal{W}_2$ takes the same nominated auctions but switches the signal off, where late entrants' expected value is drawn from the raw prior F_V. This difference affects late entrants' learning decision, thereby capturing the **thickening** effect of nomination.

Worlds $\mathcal{W}_3$, $\mathcal{W}_{4a}$ and $\mathcal{W}_{4b}$ removes **screening** in three steps. In all three words, an *arbitrary* tract is auctioned, so there is no $v \geq v^*$ value screening. $\mathcal{W}_3$ assumes the early arriving, potentially nominating bidder, to be always learning to keep the 'guaranteed demand from nominator' effect. $\mathcal{W}_{4a}$ and $\mathcal{W}_{4b}$ remove this: $\mathcal{W}_{4a}$ lets this bidder to make its learning decision, keeping the 'guaranteed arrival', while $\mathcal{W}_{4b}$ removes this and lets bidders arrive at the natural rate ($\lambda_{pre} + \lambda_{post}$). World $\mathcal{W}_{4b}$ mirrors the world without nomination where the auctioneer arbitrarily chooses what to sell. Thus, the four successive differences isolate one force each:

$$\underbrace{(\mathcal{W}_{4a} - \mathcal{W}_{4b})}_{\substack{\text{Screening on demand} \\ \text{(mechanical, guaranteed arrival)}}} + \underbrace{(\mathcal{W}_3 - \mathcal{W}_{4a})}_{\substack{\text{Mechanical} \\ \text{(nominator)}}} + \underbrace{(\mathcal{W}_2 - \mathcal{W}_3)}_{\substack{\text{Screening} \\ \text{on value}}} + \underbrace{(\mathcal{W}_1 - \mathcal{W}_2)}_{\text{Thickening}} = \underbrace{(\mathcal{W}_1 - \mathcal{W}_{4b})}_{\text{Total}}.$$

I report each world *per sold auction* (auctions that draw at least one active bidder), and separately report the sale rate. This is because most of the effect comes from whether the auction happens rather than on the outcome conditional on happening. Tables 19–22 give the decomposition for a low, middle, and high bin (1, 10, 19) and pooled across all tracts; the full 19-bin tables are in Appendix C.4 and the simulation details in Appendix B.13.

Pooled across bins, nomination is worth about **$3,090** per sold auction, which is about 32% above the no-nomination benchmark ($9,666) conditional on a sale, and about 46% per offered tract once the higher sale rate is credited (a nominated tract sells with probability one versus 0.90 without nomination). About **80%** ($2,453) is *screening on demand*. A separate mechanical *nominator* piece is essentially zero ($5, +0.01 bidders), because at the estimates learning is almost always worthwhile. *Screening on value* adds about $631 in the right direction (nominated tracts are genuinely better) but is *noisy* (s.e. $728) and barely changes the number of bidders (+0.09 per auction, s.e. 0.03). The correlation signal (**thickening**) is essentially zero on average. This is because the nomination threshold sits close to the minimum price ($v^* \approx r$; Section 6), so a nomination is barely informative. In the lowest-value bins (Bins 15–19), however, v^* is high enough that a nomination is meaningfully selective and thickening becomes non-negligible (Figure 5): from Table 24, the thickening component is exactly zero in Bins 1–14 and rises to $44.5, $38.7, $21.8, $5.4, and $82.7 per

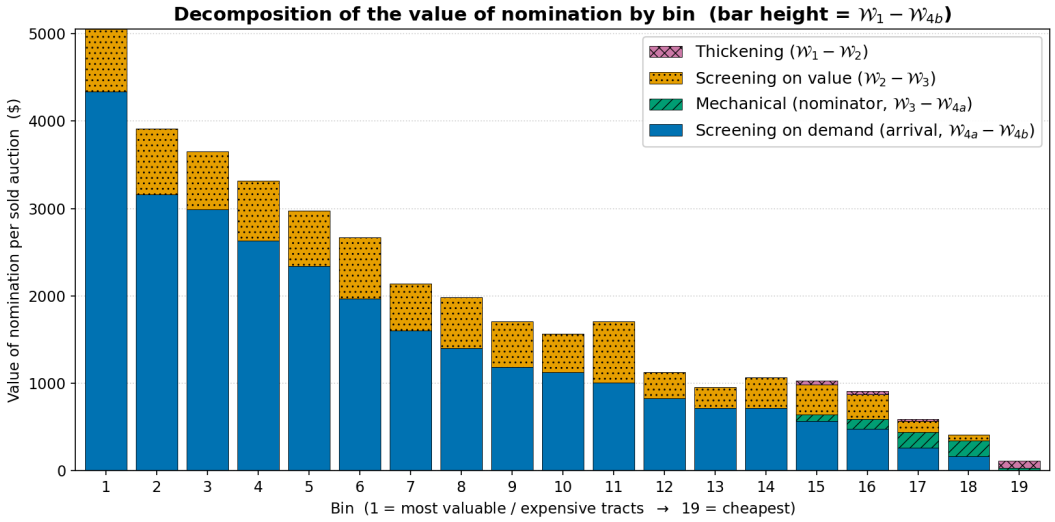
sold auction in Bins 15 through 19, respectively. In the cheapest bin (Bin 19), thickening is in fact the *single largest* channel—$82.7 of that bin's $116.5 total, about 71%—because screening on demand and screening on value nearly vanish for tracts that so rarely clear the reserve, leaving the correlation signal as the main remaining source of value. In level terms, however, even this largest per-bin thickening effect ($82.7 per sold auction) is small relative to the pooled $3,090 that nomination is worth overall (see Figure 5a).

The value of nomination also comes from the extensive margin on whether the offered tract gets sold. Table 22 summarizes the sale-rate for each counterfactual world and Table 23 decomposes this effect by removing each channel. Nominated tracts ($\mathcal{W}_1, \mathcal{W}_2$) always sell, and the pattern then splits by tract value. In the expensive bins almost every auction sells even without nomination (the bin-1 no-sale rate is only 7.3% under $\mathcal{W}_{4b}$), so nomination's value there is intensive (a higher price given a sale). In the cheapest bin, on the other hand, a randomly offered tract draws no bidder 97.3% of the time ($\mathcal{W}_{4b}$), and nomination lifts its sale rate to 100%. Decomposing that 97-point gap, the pure arrival ($\mathcal{W}_{4a} \to \mathcal{W}_{4b}$) contributes under one point while *screening on value* ($\mathcal{W}_2 \to \mathcal{W}_3$) contributes 70 points. This is because bidders arriving at cheaper tracts are more likely to draw values that does not clear the reserve ($v < r$). Another 26 points are the mechanical guarantee that this bidder actually pays to learn ($\mathcal{W}_3 \to \mathcal{W}_{4a}$). This binds in these cheapest tracts because learning is rarely worthwhile there. The value of nomination for cheaper tracts is that the nominator's high value clears the reserve and therefore screening on value.[26]

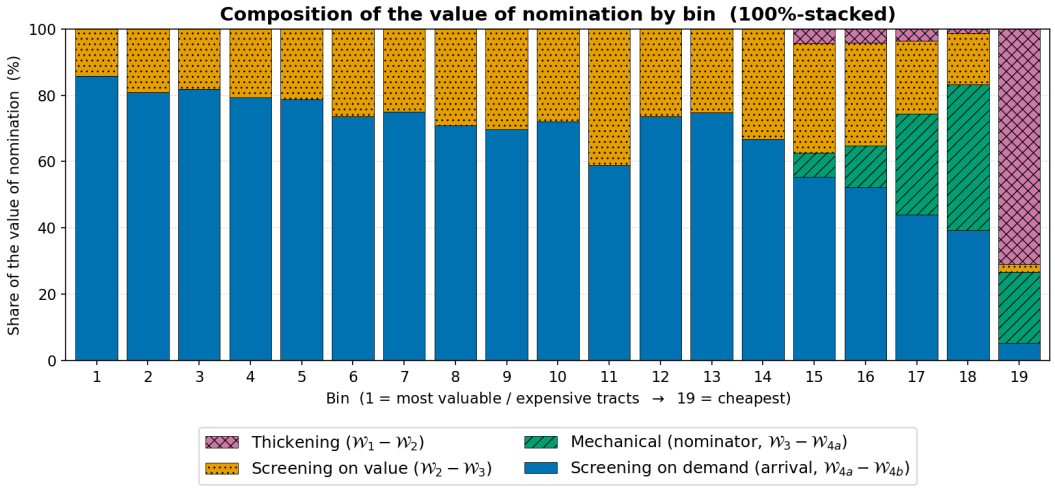
Note that the single-nomination rule ($K = 1$) is found to be the *only* viable trigger threshold: raising it to $K \geq 2$ produces complete coordination failure, eliminating nominations and auctions altogether.[27]

[26] This is the quantitative counterpart to the Alaska example in Section 1: when the seller cannot tell ex ante which tracts have demand, offering a low-value tract directly yields an auction with no bidders. Nomination avoids this because a committed buyer's willingness to nominate reveals that a bidder with $v \geq v^* \geq r$ exists, so tracts that would draw no qualifying bid are never offered.

[27] Raising the trigger threshold to $K \geq 2$ collapses the trigger rate to essentially zero. The cause is *thin early arrival*, not late learners. Early learners arrive at $\hat{\lambda}_{pre} \approx 0.005$ per tract, so the number of nominators is $M \sim \text{Poisson}(\Lambda_{nom}) \approx 4 \times 10^{-3}$. Hence two or more coincident nominations essentially never occur, so a rule demanding them discourage early arrivals from nominating, shutting the auctions off entirely. However, a credible $K \geq 2$ counterfactual cannot hold the arrival process fixed because raising the threshold could itself induce more firms to arrive and learn early to secure a nomination. Without assumptions on how arrival rates respond to K, little can be said about the resulting equilibrium. The simplified fixed-N coordination model in Appendix E.1, which assumes a fixed pool of $N = 4$ potential nominators rather than thin Poisson arrival instead, fails only at $K = 3$, supporting the claim that arrival structure is behind the coordination failure equilibrium. I therefore leave a full treatment of $K \geq 2$ for future work.



(a) Levels: bar height $= \mathcal{W}_1 - \mathcal{W}_{4b}$, in dollars per sold auction.



(b) Composition: the same decomposition as within-bin shares (100%-stacked).

Figure 5: **Decomposition of the value of nomination by bin.** Each bar splits the total value of nomination per sold auction, $\mathcal{W}_1 - \mathcal{W}_{4b}$, into four channels: screening on demand (guaranteed arrival, $\mathcal{W}_{4a} - \mathcal{W}_{4b}$), the mechanical nominator/learning guarantee ($\mathcal{W}_3 - \mathcal{W}_{4a}$), screening on value ($\mathcal{W}_2 - \mathcal{W}_3$), and thickening ($\mathcal{W}_1 - \mathcal{W}_2$). Bins run from the most valuable/expensive tracts (Bin 1) to the cheapest (Bin 19). Panel (a) shows levels—so the bar height equals $\mathcal{W}_1 - \mathcal{W}_{4b}$. Panel (b) shows within-bin shares.

7.2 Non-Nominator Externality, Reserve Prices, and Nominator Welfare

Turning to the welfare of the nominators, the current nomination system imposes two costs on a nominator: the explicit nomination fee ϕ, and the implicit cost of revealing information to non-nominators and post-nomination arrivals, who then enter and compete away part of the nominator's auction surplus. A natural question is whether this competitive externality harms nominators.

The answer is not uniformly yes. The presence of non-nominator competition in the auction raises the seller's expected conditional revenue at any given reserve price. This in turn *lowers* the seller's optimal reserve, from $r^*_{\text{no}} = \$142.97$/acre under nominators-only to $r^*_{\text{cur}} = \$65.03$/acre under the current mechanism.

Because a lower reserve directly expands the set of values at which nomination is profitable, the reserve-price channel partially offsets the competitive harm to nominators under the current mechanism. [28] The competition harm dominates the reserve-price benefit only for the most expensive tracts (bin 1).

I compare three scenarios, each evaluated at its own constrained-optimal reserve price:

(i) **Current $K = 1$, $r = \$2$/acre (baseline):** The current mechanism that allows non-nominators and late-arrivals, at the nominal data reserve of \$2/acre.

(ii) **Current $K = 1$, $r = r^*_{\text{cur}}$:** The current mechanism re-optimized over the reserve price. For each reserve r, the full equilibrium $(v^*_b, c^*_{Hb}, c^*_{Lb})$ is re-solved because Λ depends on $[1 - F_V(\tilde{r}_b)]$ and therefore changes with r (Section 7.2). Seller revenue is maximized over the reserve grid to obtain $r^*_{\text{cur}} = \$65.03$/acre.

(iii) **Nominators-only $K = 1$, $r = r^*_{\text{no}}$:** A counterfactual in which only nominators who paid reserve price as nomination fee bid. The reserve is optimized over a grid to obtain $r^*_{\text{no}} = \$142.97$/acre.[29]

Table 17 reports the trigger rate per tract under each scenario across all 19 cost bins.

The current mechanism sustains a trigger rate about nine times as high as the nominators-only counterfactual: averaged across bins, $\Pr(M \geq 1) \approx 41.0 \times 10^{-4}$ under the current

[28] The reserve here screens at the auction stage and imposes no upfront tax on nomination, setting a floor on the winning payment instead. Nomination fee, on the other hand taxes the decision to learn early and nominate, raising the nomination threshold v^{**}. Because these two instruments screen at different levels, they behave very differently: once the nomination fee exceeds the maximum attainable auction surplus $\max_v \tilde{\Pi}(v)$ no firm nominates and the market collapses, while a reserve preserves this extensive margin. A full comparison of the fee and the reserve as screening instruments is left for future work.

[29] The seller faces fewer total bidders and therefore the optimal reserve rises relative to the current mechanism.

mechanism (scenarios (i)–(ii)) versus only 4.5×10^{-4} under nominators-only (scenario (iii)). This drop is a *reserve* effect: without later-learners the auction is thin (typically a single bidder), so the seller's optimal reserve is much higher ($r^*_{\text{no}} = \$142.97$ vs. $r^*_{\text{cur}} = \$65.03$/acre). The reserve enters as a *lower bound* on the nomination threshold because the winner pays the reserve on sale. This raises the nomination threshold v^* ($= \max(v^{**}, r)$) and suppresses the trigger.

Nominator welfare: the reserve-price channel. The nominators-only mechanism removes the late learners who enter after the nomination and compete without having paid to nominate. Their competition is precisely what lets the reserve-optimizing seller hold the reserve low and without them the reserve increases from $r^*_{\text{cur}} \approx \$65$ to $r^*_{\text{no}} \approx \$143$. This higher reserve is what affects potential nominators.

I summarize welfare by the net surplus of the *average early learner*: a nominator drawn at random from the pool that nominates in the current equilibrium ($v > v^*_{\text{cur}}$), net of the survey/learning cost and the nomination fee. The higher reserve harms this nominator through payments, not participation: she still triggers the auction whenever she wishes, but an early learner who nominates and sinks the survey cost yet draws a value below the raised reserve now wins nothing, and every winner pays a higher floor.

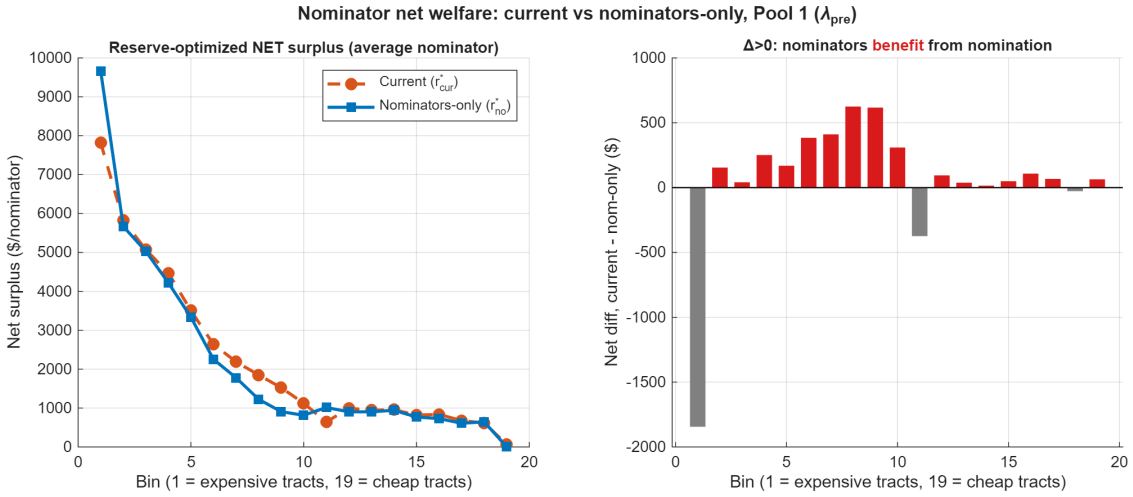


Figure 6: Net surplus of the average early learner ($v > v^*_{\text{cur}}$, net of learning cost and the nomination fee) under the current vs. nominators-only K=1 mechanism, each at its own optimal reserve (baseline λ_{pre} pool). Left: net surplus by tract bin. Right: difference (current − nominators-only); $\Delta > 0$ (red bars) marks bins where the nominator benefits from nomination. Bins run from expensive tracts (1) to cheap tracts (19).

Figure 6 plots this net surplus by tract bin, current versus nominators-only, together with their difference. Once the seller's reserve response is netted against the removed competition,

the welfare effect is *small and mostly adverse*: the average early learner is worse off in 16 of 19 bins, by about \$61 per nominator on average (Table 18). The loss is concentrated in the cheap tracts, where the market is thin and the reserve channel dominates. For the most expensive tracts (bin 1) the sign flips: removing late-learner competition raises the early learner's auction payoff by enough to outweigh the higher reserve, so there she is modestly *better off* under nominators-only (a net gain of \$1,844). The large gross payoff that late-learner competition costs the early learner is thus almost entirely returned to her through the seller's lower reserve (Appendix C.2).

This comparison treats the early arrivals (λ_{pre}) as the potential nominators, letting the non-nominators and late arrivals drop out. As a robustness check, I widen the the potential nominator pool to include late arrivals ($\lambda_{pre} + \lambda_{post}$). Appendix C.2 shows the welfare conclusion is unchanged under that alternative.

Writing $\Delta S(b)$ for the current-minus-nominators-only surplus difference, the effect decomposes into a competition effect and a reserve effect:

$$\Delta S(b) \;=\; \underbrace{\Pi_{\text{cur}}(v,\, r^*_{\text{cur}},\, \Lambda) - \Pi_{\text{no}}(v,\, r^*_{\text{no}},\, 0)}_{\text{competition effect } (-) \,+\, \text{reserve effect } (+)} \;-\; \underbrace{\left[E[c]_{\text{cur}} - E[c]_{\text{no}}\right]}_{\text{cost composition (small)}}, \tag{31}$$

where $\Pi_{\text{no}}(v, r, 0) = \max(0, v - r)$ because a sole nominator wins and pays $\max(r, 0) = r$. Under the current mechanism the outside competition lowers the nominator's payoff (the competition effect is negative), but the lower reserve that this competition sustains raises it (the reserve effect is positive). For cheap tracts, values are low and competition is already thin, leaving the nominator slightly worse off under the nominators-only mechanism due to high reserve. Only for the most expensive tracts (bin 1), where nominators have $v \gg r^*_{\text{cur}}$ and face real competition, does the competition effect dominate and the nominator prefer nominators-only mechanism. The externality is thus double-edged: the same outside entry that competes away auction surplus is what lets the seller hold the reserve low, acting as an indirect subsidy to the nominator, financed by the information her nomination revealed.

The difference column of Table 18 shows that, once the seller's reserve response is taken into account, the average nominator is only slightly affected by the nominators-only counterfactual: she is worse off in 16 of 19 bins, by about \$60 on average. The competition effect dominates the reserve effect in these bins, and the nominator would be *better off* without the outside bidders only for the most expensive tracts (bin 1, −\$1,844).

The mechanism is the reserve-price channel. Without non-nominator and late-arrival competition, the seller has no outside bidders to raise expected revenue at a given reserve, so to extract surplus from this thin market the seller must post a high reserve, $r^*_{\text{no}} = \$142.97$/acre versus $r^*_{\text{cur}} = \$65.03$/acre. This high reserve screens out moderate-

value nominators: a firm with value just above v^*_{cur} but below $\tilde{r}_{\text{no}}$ can nominate and win at the current reserve, yet in the nominators-only world would pay the nomination fee and could fail to meet the reserve. Formally, the surplus difference decomposes as

$$\Delta S(b) \;=\; \underbrace{\Pi_{\text{cur}}(v,\, r^*_{\text{cur}},\, \Lambda) - \Pi_{\text{no}}(v,\, r^*_{\text{no}},\, 0)}_{\text{competition effect } (-) \,+\, \text{reserve effect } (+)} \;-\; \underbrace{[E[c]_{\text{cur}} - E[c]_{\text{no}}]}_{\text{cost composition (small)}}, \tag{32}$$

where $\Pi_{\text{no}}(v, r, 0) = \max(0, v - r)$ because a sole nominator wins and pays $\max(r, 0) = r$. The competition effect dominates for the most expensive tracts, where nominators have $v \gg r^*_{\text{cur}}$ and face meaningful competition, so removing the outside bidders raises surplus by more than the higher reserve costs. For cheaper tracts the reserve effect dominates: values are low and competition is already thin, so the nominators-only mechanism's main bite is its high reserve, which the low current reserve beats.

The externality is therefore double-edged. The entry of non-nominators and late arrivals, which superficially harms the nominator by competing away auction surplus, is also what lets the seller post a low reserve; that low reserve is an indirect subsidy to the nominator, financed by the information her nomination revealed. On net the two nearly cancel for all but the most expensive tracts.

7.3 Non-Nominator Value Decomposition: How Nominators Affect the Non-Nominator

This section analyzes how do the m observed nominators affect a *non-nominator*'s surplus through two channels. First, each nominator who enters the auction is an additional competitor whose value is positively correlated with the non-nominator, compressing her expected winning surplus (competition channel). Second, observing m numbers of bidders crossing the v^* threshold shifts her posterior about her own value upward through positive correlation (own-value information channel).

To separate the two, I hold the correlated competition structure fixed and vary only whether the non-nominator reads the nomination information. Given the prior value density $f_V(v)$ be the prior value density, let $f_{\text{post}}(v \mid m) := f_V(v \mid V_1 \geq v^*, \ldots, V_m \geq v^*)$ be the posterior after observing m nominators (Section 4.3.1). Let $\tilde{\Pi}_{\hat{\rho}}(v; n)$ be the expected auction surplus for a bidder of value v facing n competitors whose values are correlated at the estimated $\hat{\rho}$. With $N \sim \text{Poisson}(\hat{\Lambda})$ existing non-nominator competitors (late-learners

and late arrivals present regardless of the nomination count), define

$$S_0(m) = \mathbb{E}_N\left[\int_{v^*}^{\infty} \tilde{\Pi}_{\hat{\rho}}(v;\, m+N)\, f_V(v)\, dv\right] \tag{33}$$

$$S_{l=2}(m) = \mathbb{E}_N\left[\int_{v^*}^{\infty} \tilde{\Pi}_{\hat{\rho}}(v;\, m+N)\, f_{\text{post}}(v \mid m)\, dv\right] \tag{34}$$

$S_0(m)$ is the surplus of a non-nominator who faces the m correlated nominators as competitors but does *not* read the nomination information (she keeps her prior beliefs). $S_{l=2}(m)$ is her actual surplus, which also incorporates the posterior update. The two channels then read off the figure directly:

Competition. Each additional nominator enters as one more correlated auction competitor, compressing $\tilde{\Pi}_{\hat{\rho}}$ and pushing both curves down as m rises. This channel operates whether or not she reads the information.

Nomination information. The gap $S_{l=2}(m) - S_0(m) \geq 0$ is the value of the information to the non-nominator, the own-value learning made possible by the copula. It is zero at $m = 0$ (with no nominators there is nothing to update, so $f_{\text{post}}(\cdot \mid 0) = f_V$ and $S_{l=2}(0) = S_0(0)$. The gap widens with m in proportion to the threshold's selectivity $u^* = F_V(v^*)$.

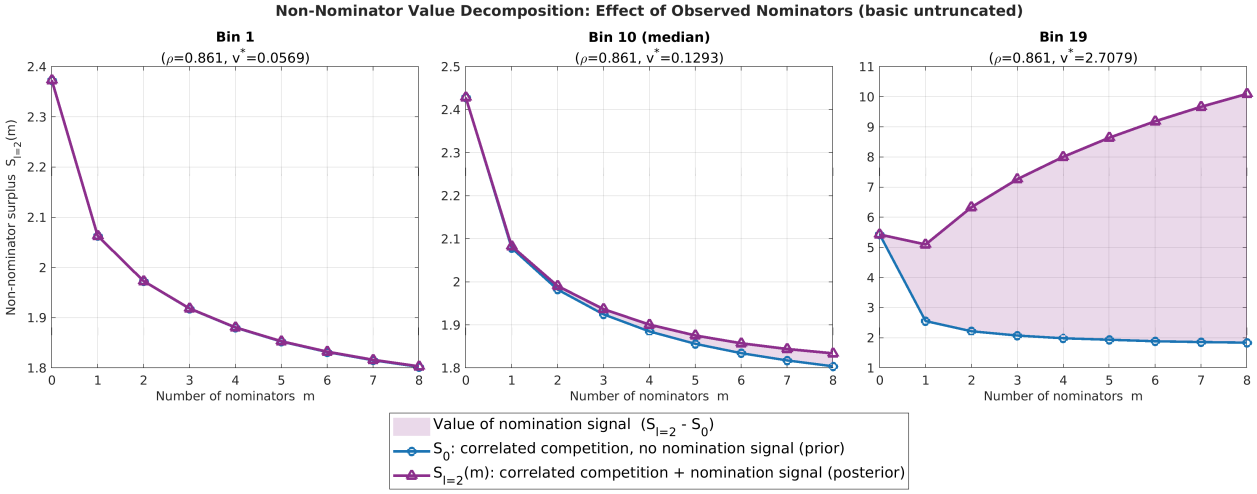


Figure 7: Non-Nominator Value Decomposition: Competition versus Nomination Signal

Notes: Non-nominator surplus $S_{l=2}(m)$ as a function of the observed nominator count m, at $\hat{\rho} = 0.861$, for Bin 1 (most expensive), Bin 10 (median), and Bin 19 (cheapest). **Blue** (S_0, eq. (33)): use prior density f_V (no updates from nomination). **Purple** ($S_{l=2}$, eq. (34)): use posterior density $f_{\text{post}}(v \mid m)$; adds the own-value information channel. The shaded band $S_{l=2} - S_0$ is the value of the nomination signal (zero at $m = 0$). Vertical scales differ across panels.

Figure 7 shows the decomposition at $\hat{\rho} = 0.861$ for three bins. In the most expensive (Bin 1; $Pr(v > v^*) = 0.04$ where $v^* = 0.0568$) and median (Bin 10; $Pr(v > v^*) = 0.10$ where $v^* = 0.1293$) bins the threshold v^* is mostly uninformative, so the signal band is negligible and $S_{l=2}$ and S_0 nearly coincide: the nominators affect the non-nominator almost entirely through competition so her surplus simply erodes as m rises. In the cheapest bin (Bin 19; $Pr(v > v^*) = 0.72$ where $v^* = 2.7079$), the signal becomes more valuable: the band $S_{l=2} - S_0$ is wide and grows with m, and because own-value learning outpaces the added competition, $S_{l=2}$ rises with m.[30]

8 Conclusion

This paper provides the first structural analysis of pre-auction nomination mechanisms, examining their dual role as demand screening devices for sellers and quality screening mechanisms for buyers. Using detailed data from ND state mineral rights auctions during 2017–2018, I develop and estimate a two-stage game-theoretic model where bidders decide whether to learn early and nominate tracts or wait on others' nominations and learn later.

The empirical analysis reveals three key findings. First, I document that most bidders participate without nominating: while auctions attract multiple bidders, the vast majority are triggered by a single nominator. Second, using a Gaussian equicorrelation copula I find moderate positive dependence in bidders' valuations ($\hat{\rho} = 0.861$), indicating that common geological and market uncertainties create correlated values across competitors. Among four copula specifications (Gaussian, Gumbel, Clayton, and Independence) evaluated by held-out RMSE, the Gaussian achieves the best out-of-sample fit (Table 14). Third, I

estimate a left-skewed learning cost distribution (Skew-Normal with $\hat{\alpha}_c = -126.9$) with a modest mean of approximately \$1,161 and standard deviation \$416 (Table 2). At these estimates exploration cost screens participation only in the low-value bins: the early learning share is 100% in the high-value bins but falls to $\approx 3.5\%$ in the lowest value bins, so cost governs which tracts are nominated rather than the overall rate. The low observed nomination rate is instead a consequence of scarcity of early pre-nomination arrivals: the estimated pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$ is orders of magnitude below the post-arrival rate, so few firms arrive early enough to nominate even though learning early and nominat-

[30] The near-flatness of $S_{l=2}(m)$ between $m = 1$ and $m = 2$ indicates that raising the nomination threshold to $K \geq 2$ would do little to thicken the market in most bins. A second nominator adds almost only competition, leaving per-auction non-nominator surplus unchanged or slightly lower. Moreover, because early arrival is thin ($\hat{\lambda}_{pre} \approx 0.005$), two or more coincident nominations essentially never occur, so a $K \geq 2$ rule would deter almost all tracts from being auctioned. Raising K thus would offer little thickening while destroying the extensive margin.

ing is individually worthwhile. The mean exploration cost of \$1,161 is small relative to the average winning bid (\$11,803 over all bids, approximately \$13,900 for winning bids alone), underscoring that the binding friction is thin early arrival rather than the cost of learning.

The structural model characterizes equilibrium nomination and learning strategies under correlated values and costly learning, focusing on the single-nomination rule ($K = 1$) that ND actually uses.

Comparing the current single-nomination policy to a no-nomination benchmark, the value of nomination rests mainly on guaranteeing participation rather than on the information its signal conveys. Nomination raises revenue by roughly \$3,090 per sold auction, about 32% above the no-nomination benchmark conditional on a sale and about 46% once its higher sale rate is included. About 80% of this gain comes from screening on demand and the rest from screening on value, while the correlation signal is negligible except for the lowest-value tracts.

A welfare analysis of the non-nominator externality shows it to be largely self-correcting. Although non-nominators and late arrivals compete away part of the nominator's auction surplus, their presence raises the seller's expected revenue and supports a substantially lower reserve (\$65 versus \$143/acre without them). Netting the two, the average nominator is worse off under a nominators-only mechanism in all but the most expensive tracts, where the competition effect dominate.

Under the estimated primitives, the thin early-arrival rate (≈ 0.005 per tract) makes single-nomination rule the only threshold consistent with positive auction activity: two or more coincident nominations essentially never occur, so almost no tract would ever be auctioned. Whether this conclusion survives once firms' arrival and early-learning decisions are allowed to respond to the threshold is a question I leave for future work.

Two design questions that this framework can also address but left for future work concerns the level and the form of the nomination charge. The first is whether the seller should raise the nomination fee to screen tract quality: a higher fee raises the nomination threshold v^{**} and selects better tracts, but taxes the decision to learn early and nominate, affecting auction trigger rate. The second is the choice between a nomination fee and a reserve price. The two instruments screen at different levels, leading to different outcomes. Nomination fee screens in the earlier stage before auction, but once the fee exceeds maximum attainable auction surplus, no firm nominates and the market collapses. The reserve preserves this extensive margin because it screens bidder quality at the auction stage, but becomes less reliable when market is thin and seller's cost of running the auction is high. A systematic fee-versus-reserve comparison is left for future work.

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A Data Summary Statistics

Tables 3 and 4 report summary statistics for per-acre and total bid values. The bid values exhibit large variation and are harmonized via the homogenization regression described in Section B.1.

Table 3: Bid Value (Per Acre) Summary

Variable	Average	Std Dev	Max	Min
Overall ($)	157	752	18,001	2
Highest Bid ($)	170	829	18,001	2
Excl. Winner ($)	151	712	18,000	2

Notes: Summary statistics for per-acre bid values in dollars. The highest bid represents the highest bid observed in each auction, which is what the winner has to pay. "Excl. Winner" excludes the winning bid, whose recorded value reflects the ascending price proxy bidding mechanism rather than the winner's true valuation.

Table 4: Bid Value (Total) Summary

Variable	Average	Std Dev	Max	Min
Overall ($)	12,346	44,200	1,120,320	0.02
Highest Bid ($)	13,874	48,372	1,120,320	0.02
Excl. Winner ($)	11,626	42,084	1,120,160	0.12

Notes: Summary statistics for total bid values (per-acre bid × tract size) in dollars. "Excl. Winner" excludes the winning bid, whose recorded value is mechanically determined by the ascending price proxy bidding system rather than reflecting the winner's true valuation.

Tables 5 and 6 report parcel sizes (acres) and royalty rates by season. Royalty rates are determined by the county in which the tract is located.

Tables 7–9 report information on the nearest previously drilled well: distance to the tract centroid, start of drilling date, or 'spud date', in calendar terms, and spud date relative to the auction date.

Table 10 summarizes starting bids (minimum bids set by nominators) by season. Most starting bids are $2; all auctions are modeled with $r = \$2$.

Table 5: Auction Size (acres) by Season

Season	# Auctions	Avg Tract Size (acres)	Min	Max
2017-Q1	173	151	35.5	160
2017-Q2	234	137	3.3	163
2017-Q3	166	137	0.16	311
2017-Q4	67	127	25.8	161
2018-Q1	52	141	28.6	160
2018-Q2	58	131	0.06	203
2018-Q3	29	152	80	160
2018-Q4	99	80.6	0.01	161

Notes: Average, minimum, and maximum tract sizes in acres by auction season. Tract sizes are determined by state regulations and geographic characteristics but cannot exceed 160 acres.

Table 6: Count of Royalty Rates by Season

Season	# Auctions	Royalty Rate 1/16	Royalty Rate 3/16
2017-Q1	173	51	122
2017-Q2	234	58	176
2017-Q3	166	52	114
2017-Q4	67	1	66
2018-Q1	52	25	27
2018-Q2	58	17	41
2018-Q3	29	8	21
2018-Q4	99	2	97

Notes: Distribution of royalty rates (1/16 or 3/16) by season. Royalty rates vary by county location and determine the percentage of production revenue paid to the state.

Table 7: Nearest Well Distance (miles) by Season

Season	Avgerage (miles)	Max	Min
2017-Q1	1.55	9.12	0.0881
2017-Q2	1.15	4.08	0.0105
2017-Q3	1.18	2.63	0.129
2017-Q4	0.912	2.13	0.0475
2018-Q1	0.821	2.41	0.0355
2018-Q2	0.672	1.94	0.0665
2018-Q3	0.722	2.22	0.107
2018-Q4	2.46	4.32	0.0452

Notes: Average, maximum, and minimum distance in miles from each tract to the nearest existing well, by season. Proximity to existing wells provides information about geological characteristics and production potential.

Table 8: Nearest Well Spud Date

Season	Earliest	Oldest	Missing
2017-Q1	2012-10-14	1953-12-15	30
2017-Q2	2014-08-30	1954-05-13	23
2017-Q3	2014-01-28	1955-10-06	23
2017-Q4	2013-07-03	1963-06-17	1
2018-Q1	2014-08-07	1956-05-22	10
2018-Q2	2014-06-03	1954-02-01	5
2018-Q3	2012-03-17	1960-12-02	3
2018-Q4	2014-06-12	1955-04-01	13

Notes: Date range of spud dates (when drilling began) for nearest wells by season. Some tracts have missing spud date information for their nearest well.

Table 9: Nearest Well Spud Date (How Close)

Season	Avg (days)	Max (days)	Min (days)
2017-Q1	11,949.392	23,065	1,577
2017-Q2	11,274.384	23,000	976
2017-Q3	11,988.860	22,580	1,281
2017-Q4	14,514.848	19,867	1,588
2018-Q1	13,457.952	22,540	1,279
2018-Q2	14,075.811	23,465	1,428
2018-Q3	12,373.654	21,067	2,334
2018-Q4	7,561.453	23,230	1,608

Notes: Average, maximum, and minimum time (in days) between the nearest well's spud date and the auction date. Longer time periods indicate less recent drilling activity in the area.

Table 10: Starting Bids by Season

Season	Starting at $2	$3	$4	$21	Total Auctions
2017-Q1	171	2	0	0	173
2017-Q2	210	16	6	2	234
2017-Q3	161	5	0	0	166
2017-Q4	67	0	0	0	67
2018-Q1	52	0	0	0	52
2018-Q2	58	0	0	0	58
2018-Q3	29	0	0	0	29
2018-Q4	98	1	0	0	99
Total	**846**	**24**	**6**	**2**	**878**

Notes: Distribution of starting bids (minimum bids) specified by nominators. The vast majority (846 out of 878) specified a minimum bid of $2, indicating nominators set the minimum at the lowest allowed level.

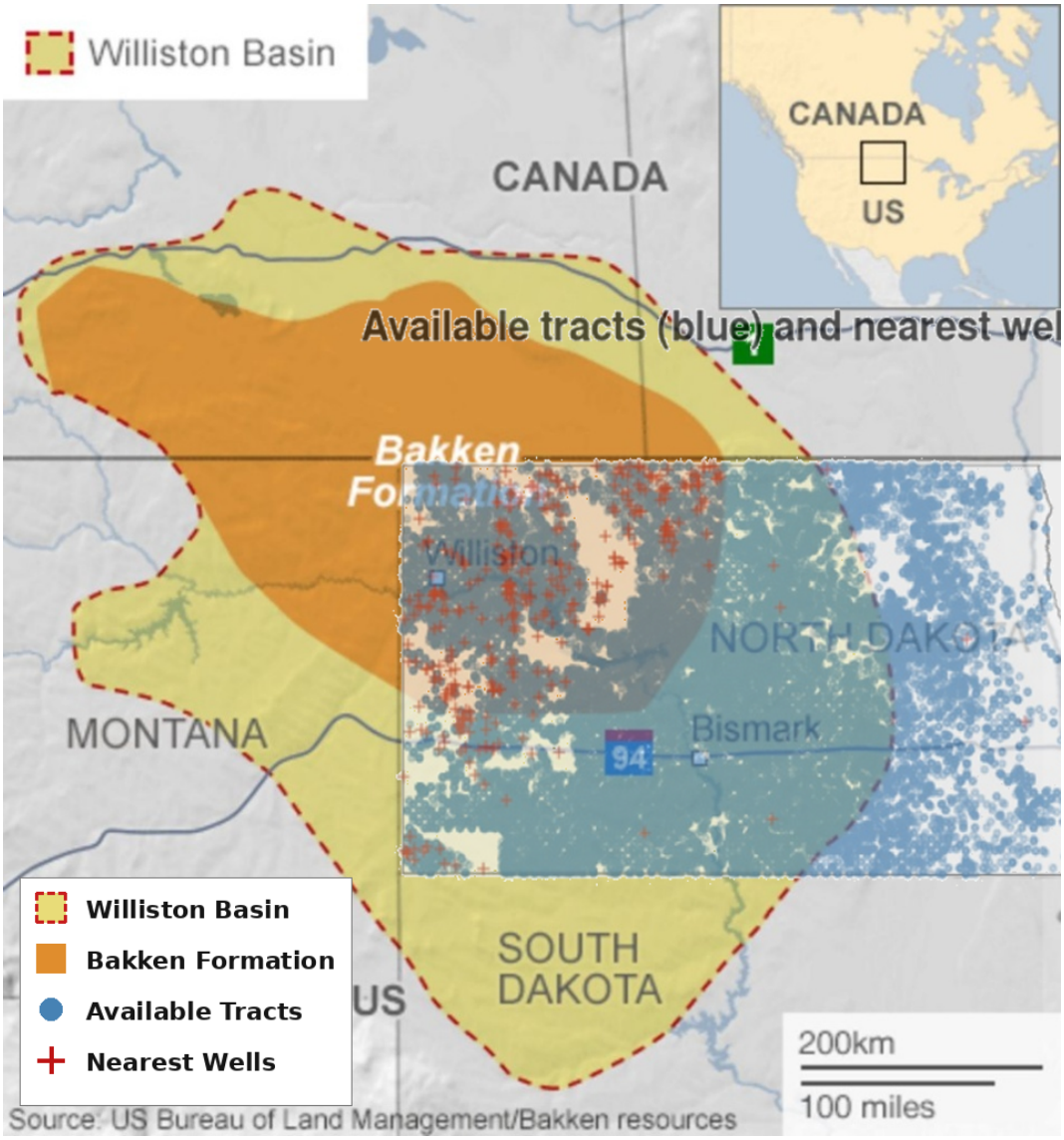


Figure 8: Available Tracts and Nearest Wells in the Williston Basin

Notes: The map overlays the boundary of the Williston Basin and the Bakken Formation with the tracts available for nomination in the estimation sample (blue) and the locations of the nearest producing wells (red crosses). Available tracts vary substantially in their proximity to an existing producing well, motivating the inclusion of distance to the nearest well and its spud date as a regressor in the bid-homogenization procedure (Section B.1): tracts closer to a demonstrated producer or tracts close to a recently developed well carry less geological uncertainty and are expected to command systematically higher bids, all else equal.

B Estimation Details and Results

This appendix provides full details for the two-step estimation procedure summarised in Section 5.

B.1 Bid Homogenization

Raw bid amounts b_{istc} (bid by bidder i in auction season s for tract t in cluster c) vary multiplicatively with observable tract heterogeneity. To isolate the private-value component, I implement a two-step log-linear homogenization procedure. The key identifying assumption is an *equilibrium translational invariance* property: a uniform shift in the game-level heterogeneity Z shifts all equilibrium actions by the same amount, leaving the mapping between markup values and markup actions unchanged . Formal justification is given in Proposition 3 below, and a proof is in Appendix B.2.

I first verify that a multiplicative (log-linear) specification is appropriate. Appendix B.11 compares residual diagnostic plots for the linear and log-linear specifications. In sum, the linear model exhibits severe heteroskedasticity (residual variance fans out with fitted values) while the log-linear model stabilizes variance across the distribution. The formal model-selection rationale is discussed in the Appendix.

Appendix B.11 presents the residual diagnostic plots confirming the suitability of the log-linear specification.

Homogenization regression. I estimate the following log-linear hedonic regression:

$$\begin{aligned}\log(b_{istc}) =& \alpha_s + \alpha_c \\ &+\beta_1 \,\text{size}_t + \beta_2 \,\mathbb{I}[\text{royalty}_t = 3/16] + \beta_3 \,\text{distance}_t \\ &+\beta_4 \,\text{spudAge}_t + \beta_5 \,\mathbb{I}[\text{spudAge missing}] + \varepsilon_{istc}\end{aligned} \tag{35}$$

where α_s and α_c are season and cluster fixed effects; size_t is tract acreage; royalty_t is the royalty rate (1/6 or 3/16); distance_t is distance to the nearest well in miles; and spudAge_t is the number of years elapsed between the spud date of the nearest well and the auction date (with an indicator for missing spud dates). Table 11 reports the estimates; the average fixed effect (approximate constant) is 7.757, and all slope coefficients carry their expected signs.

The auction-specific scaling factor is

$$s_t \;=\; \exp\big(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t\big) \;=\; \exp(\widehat{\log b_{istc}} - \hat{\varepsilon}_{istc}), \tag{36}$$

i.e. the exponentiated fitted value from regression (35). Homogenized bids are then the

exponentiated residuals:

$$\tilde{b}_{istc} \;=\; \exp(\hat{\varepsilon}_{istc}) \;=\; \frac{b_{istc}}{s_t}. \tag{37}$$

Because every bidder in the same auction faces the same X_t, all bids in auction t are divided by the same s_t and the *relative* ranking of bids within an auction is preserved, as is the within-auction positive dependence structure arising from unobservable common shocks (geological uncertainty, commodity prices).

Descaling the nomination fee and cost. The state's flat \$10 nomination fee, $\phi = \$10$, does *not* scale with s_t. It is charged per tract regardless of its acreage or location. In residual space, this fee becomes

$$\phi \;=\; \frac{10}{s_t}. \tag{38}$$

I assume learning costs for bidder i at tract t are largely *fixed in dollar terms* and do not scale the same way as value does to observables. For example, while value decreases in distance to nearby wells, the cost of geological survey or analysis of a nearby well stays constant or, if anything, increases with distance. The estimation maintains that $C_{it} \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$ is drawn from a *common* distribution across all bins so that the bin-specific residual-space distribution is $\tilde{C}_{it} = C_{it}/s_b \sim \text{SN}(\tilde{\xi}_c, \tilde{\omega}_c, \tilde{\alpha}_c)$ where $s_b > 0$.

The per-bin equilibrium structure (Section B.6) partially captures the resulting cross-bin heterogeneity in effective cost barriers through the bin-specific fees $\tilde{\phi}_b = \phi/s_b$.

Dollar-space *values* scale linearly with s_t, meaning auctions over a more valuable tract has proportionally larger bids, competing maxima, and surplus, so $\Pi = s_t \cdot \tilde{\Pi}$. On the other hand, the flat fee ϕ, the reserve r, and the cost distribution F_C are fixed in dollar terms and therefore do not scale. Therefore, when expressed in residual units they become auction-specific objects $\tilde{\phi}_t = \phi/s_t$, $\tilde{r}_t = r/s_t$, and $\tilde{F}_{C,t}(\tilde{c}) = F_C(s_t\tilde{c})$. Under the Skew-Normal specification, $\tilde{F}_{C,t} = \text{SN}(\xi_c/s_t,\, \omega_c/s_t,\, \alpha_c)$. The cross-auction heterogeneity in residual-space equilibria is entirely driven by this asymmetry between scaling values s_t and non-scaling parameters ϕ, r and F_C parameters. The Poisson rates Λ_{nom} and Λ_{nn} are dimensionless counts and are unaffected.

This implies two things for estimation. First, auctions sharing the same residual-space triple $(\tilde{\phi}_t,\, \tilde{F}_{C,t},\, \tilde{r}_t)$ share the same residual-space equilibrium and may be pooled. Section B.6 groups auctions into deciles of s_t and solves one equilibrium per bin. Second, the dollar primitives $(\phi,\, r,\, \xi_c,\, \omega_c,\, \alpha_c)$ and the residual value distribution $F_{\tilde{V}}$, along with other structural primitives (the Poisson rates $(\lambda_{\text{pre}},\, \lambda_{\text{post}})$ and the copula parameter ρ) are estimated as common to all 728 auctions. Their bin-specific residual-space counterparts are recovered by dividing by s_b. The formal proof is in Appendix B.2.

Table 11 reports the regression results used for bid homogenization. The dependent variable is log total bid $\log(b_{istc})$. The reported constant is the average of the absorbed cluster fixed effects. The auction-specific scaling factor is $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$. Size enters as a quadratic (size + size^2) to allow a concave bid-size relationship.

All slope coefficients carry their expected signs: the higher royalty rate (3/16 > 1/16) commands a premium; bids decline with distance to the nearest well and with the age of that well's spud date (older wells are less informative about current reservoir conditions); larger tracts attract higher total bids up to the size at which the quadratic term dominates. The spud-date missing indicator is not significant.

Table 11: Log-Bid Homogenization Regression Result

	Dependent variable:
	log(Total Bid)
Constant	7.757***
	(0.205)
Royalty Rate (3/16)	0.324***
	(0.099)
Distance to Nearest Well (miles)	−0.310***
	(0.044)
Size (acres)	0.013***
	(0.003)
Size^2 (acres^2)	−0.0000313**
	(0.0000141)
Spud Date Age (years)	−0.026***
	(0.002)
Observations	1,875

Notes: Bid homogenization regression (equation 35). Dependent variable is log(per-acre bid × acreage), i.e. log total bid in dollars. Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 12 summarizes the homogenization result. The first row reports overall log bids before homogenization ($\log b_{istc}$); subsequent rows report exponentiated residuals $\tilde{b}_{istc} = \exp(\hat{\varepsilon}_{istc}) = b_{istc}/s_t$ after homogenization, separately for the estimation sample (auctions

where the winner's bid is censored) and the verification sample (auctions where the winner's bid violates the proxy-bid rule and is therefore observed). Homogenization reduces the standard deviation from 1.77 to 1.58, reflecting the removal of observable heterogeneity. The remaining dispersion captures private-value variation and unobservable common shocks.

Table 12: Homogenized Bid (Residuals)

Variable	Average	Std Dev	Max	Min
Per Acre Bid (Log Before Homogenization)	3.25	1.77	9.80	0.693
After Homogenization (Overall)	0.02095452	1.583303	5.493611	-3.973427
For Estimation	0.0305	1.59	5.49	-3.97
For Verification	-0.434	1.39	3.35	-3.74

Notes: Summary statistics of homogenized bids (residuals from the regression in Table 11). Homogenization reduces the standard deviation from 1.77 to 1.58, isolating the private value component from observable heterogeneity.

B.2 Proposition: Residual-Space Reduction

Setup. Each auction t has scaling $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$ recovered from the regression in Appendix B.1. Under private values, dollar quantities scale as $v_{jt} = s_t \tilde{v}_{jt}$ and $V_{(n:n),t} = s_t \tilde{V}_{(n:n),t}$. The fee $\phi = \$10$, reserve $\$r$, and the dollar-cost distribution $F_C \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$ are common to all auctions (none scales with s_t). Define

$$\tilde{\phi}_t := \phi / s_t, \qquad \tilde{r}_t := r / s_t, \qquad \tilde{F}_{C,t}(\tilde{c}) := F_C(s_t \tilde{c}).$$

Under the Skew-Normal, $\tilde{F}_{C,t} = \text{SN}(\xi_c/s_t,\, \omega_c/s_t,\, \alpha_c)$: the location and scale parameters depend on s_t, while the shape α_c is invariant. This follows from the following Lemma.

Lemma 2 (Skew-normal closure under scaling)**.** *Let $C \sim \text{SN}(\xi, \omega, \alpha)$ following the Azzalini parameterization [Azzalini, 2013, eq. 2.3], with density*

$$F_C(c) = \frac{2}{\omega}\, \varphi\left(\frac{c-\xi}{\omega}\right) \Phi\left(\alpha\, \frac{c-\xi}{\omega}\right),$$

where φ and Φ are the standard normal density and CDF. For any $s > 0$, the rescaled variable $\tilde{C} := C/s$ satisfies

$$\tilde{C} \sim \text{SN}(\xi/s,\, \omega/s,\, \alpha)\,.$$

The location and scale parameters rescale by $1/s$ while the shape parameter α is invariant.

Proof. By the standard change-of-variables formula, $\tilde{C} = C/s$ has density

$$f_{\tilde{C}}(\tilde{c}) \;=\; s \cdot F_C(s\tilde{c}) \;=\; \frac{2s}{\omega}\,\varphi\!\left(\frac{s\tilde{c}-\xi}{\omega}\right)\Phi\!\left(\alpha\,\frac{s\tilde{c}-\xi}{\omega}\right).$$

This can be rewritten as

$$f_{\tilde{C}}(\tilde{c}) \;=\; \frac{2}{\omega/s}\,\varphi\!\left(\frac{\tilde{c}-\xi/s}{\omega/s}\right)\Phi\!\left(\alpha\,\frac{\tilde{c}-\xi/s}{\omega/s}\right),$$

which is the density of $\mathrm{SN}(\xi/s,\, \omega/s,\, \alpha)$ by [Azzalini, 2013, eq. 2.3]. □

Surplus scaling. The Gaussian equicorrelation copula depends only on ranks, so the conditional density of the maximum competing value satisfies $f_{V_{(n:n)}|v}(s_t\tilde{y}) = s_t^{-1} f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})$. Substituting $y = s_t\tilde{y}$ into the dollar-space surplus integral yields

$$\Pi(s_t\tilde{v};\, n,\, \rho) = \int_{\tilde{r}_t}^{\tilde{v}} \underbrace{s_t(\tilde{v}-\tilde{y})}_{v-y}\;\underbrace{s_t^{-1} f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})}_{f_{V_{(n:n)}|v}(s_t\tilde{y})}\;\underbrace{s_t\,d\tilde{y}}_{dy} = s_t \cdot \tilde{\Pi}(\tilde{v};\, n,\, \rho), \tag{39}$$

where the residual-space surplus is $\tilde{\Pi}(\tilde{v};\, n,\, \rho) := \int_{\tilde{r}_t}^{\tilde{v}} (\tilde{v}-\tilde{y})\, f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})\, d\tilde{y}$. The Jacobian from dy cancels the Jacobian from the density, leaving a single factor of s_t from the integrand.

Nomination threshold scaling. Before turning to the value functions, I establish how the nomination threshold v^* transforms under the change of variables. Recall from (10) that $v^* = \min\{r,\, v^{**}\}$, where v^{**} satisfies the pointwise indifference condition

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \Pi(v^{**};\, m,\, \Lambda_{\mathrm{nn}},\, \rho) = \phi. \tag{40}$$

By the surplus scaling (39), $\Pi(s_t\tilde{v};\, m,\, \cdot) = s_t \cdot \tilde{\Pi}(\tilde{v};\, m,\, \cdot)$. Substituting $v^{**} = s_t\tilde{v}_0$ into (40) and dividing both sides by s_t:

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \tilde{\Pi}(\tilde{v}_0;\, m,\, \Lambda_{\mathrm{nn}},\, \rho) = \tilde{\phi}_t. \tag{41}$$

By definition, $\tilde{v}^{**} = \tilde{v}_0$, that is, $\tilde{v}_0 = v^{**}/s_t$ satisfies the residual-space pointwise indifference condition with $\tilde{\phi}_t = \phi/s_t$. Therefore, $\tilde{v}^{**} = v^{**}/s_t$.

Next, since $\tilde{r}_t = r/s_t$,

$$\tilde{v}_t^* := \min\{\tilde{r}_t,\, \tilde{v}^{**}\} = \min\{r/s_t,\, v^{**}/s_t\} = (1/s_t)\min\{r, v^{**}\} = v_t^*/s_t.$$

The nomination threshold scales as $\tilde{v}_t^* = v_t^*/s_t$.

Value-function scaling. With the threshold scaling established, the value functions transform as follows. The dollar-space early learner value is $U_{l=1}(c) = -c - \phi[1 - F_V(v^*)] + \sum_m \Pr(M_{-i} = m) S_{l=1}(m)$, where $S_{l=1}(m) = \int_{v^*}^{\bar{v}} \Pi(v;\, m + \Lambda,\, \rho)\, f_V(v)\, dv$. Substituting $v = s_t\tilde{v}$, $v^* = s_t\tilde{v}^*$, $f_V(v) = s_t^{-1} f_{\tilde{V}}(\tilde{v})$, and using the surplus scaling (39):

$$\begin{aligned} S_{l=1}(m) &= \int_{\tilde{v}^*}^{\bar{v}/s_t} \underbrace{s_t\, \tilde{\Pi}(\tilde{v};\, \cdot)}_{\Pi}\, \underbrace{s_t^{-1} f_{\tilde{V}}(\tilde{v})}_{f_V}\, \underbrace{s_t\, d\tilde{v}}_{dv} \\ &= s_t \int_{\tilde{v}^*}^{\bar{v}/s_t} \tilde{\Pi}(\tilde{v};\, \cdot)\, f_{\tilde{V}}(\tilde{v})\, d\tilde{v} = s_t \cdot \tilde{S}_{l=1}(m). \end{aligned} \tag{42}$$

Therefore

$$U_{l=1}(c) = s_t \left[-\underbrace{\tfrac{c}{s_t}}_{\tilde{c}_t} - \underbrace{\tfrac{\phi}{s_t}}_{\tilde{\phi}_t} [1 - F_{\tilde{V}}(\tilde{v}^*)] + \sum_m \Pr(M_{-i} = m)\, \tilde{S}_{l=1}(m) \right],$$

and dividing through by s_t defines the residual-space value function

$$\tilde{U}_{l=1}(\tilde{c}_t) \;:=\; \frac{U_{l=1}(c)}{s_t} \;=\; -\tilde{c}_t - \tilde{\phi}_t[1 - F_{\tilde{V}}(\tilde{v}^*)] + \sum_m \Pr(M_{-i} = m)\, \tilde{S}_{l=1}(m). \tag{43}$$

The identical calculation for the late-learning value yields

$$\tilde{U}_{l=2}(\tilde{c}_t) \;:=\; \frac{U_{l=2}(c)}{s_t} \;=\; \Pr(M \geq K)\Big[\mathbb{E}[\tilde{S}_{l=2}(m) \mid M \geq K] - \tilde{c}_t\Big]. \tag{44}$$

In sum,

$$U_l(s_t\tilde{c};\, \cdot) = s_t \cdot \tilde{U}_l(\tilde{c};\, \cdot), \qquad l \in \{1, 2\}. \tag{45}$$

Proposition 3 (Residual-Space Reduction)**.** *Fix auction t with scaling $s_t > 0$. The dollar-space equilibrium tuple $(v^*, c_L^*, c_H^*, \Lambda_{\text{nom}}, \Lambda_{\text{nn}})$ satisfies the equilibrium system $\mathcal{S}_t$ if and only if the candidate*

$$\tilde{v}^* := v^*/s_t, \quad \tilde{c}_L^* := c_L^*/s_t, \quad \tilde{c}_H^* := c_H^*/s_t,$$

$$\tilde{\Lambda}_{\text{nom}} := \Lambda_{\text{nom}}, \quad \tilde{\Lambda}_{\text{nn}} := \Lambda_{\text{nn}}$$

satisfies the residual-space system $\tilde{\mathcal{S}}_t$. In residual space, auctions differ only through the triple $(\tilde{\phi}_t, \tilde{F}_{C,t}, \tilde{r}_t)$; all other primitives ($\lambda_{\text{pre}}$, λ_{post}, $F_{\tilde{V}}$, ρ, α_c, T, K) are common across auctions.

Proof. $\mathcal{S}_t$ comprises the nomination threshold condition (1), the cost-cutoff indifference conditions (2)–(3), and the Poisson-rate consistency conditions (4)–(5):

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m;\, \Lambda_{\text{nom}})\, \Pi(v^{**};\, m,\, \Lambda_{\text{nn}},\, \rho) = \phi, \quad v^* = \max\{v^{**}, r\} \tag{1}$$

$$U_{l=1}(c_L^*) = U_{l=2}(c_L^*); \quad U_{l=2}(c_H^*) = 0 \tag{2,3}$$

$$\Lambda_{\text{nom}} = \lambda_{\text{pre}}\, F_C(c_L^*)[1 - F_V(v^*)];$$

$$\Lambda = \lambda_{\text{pre}} \left[F_C(c_H^*) - F_C(c_L^*)\right] [1 - F_V(r)] \;+\; \lambda_{\text{post}}\, F_C(c_H^*)\, [1 - F_V(r)] \tag{4,5}$$

The residual-space system $\tilde{\mathcal{S}}_t$ has the same form with all quantities tilded.

Verification of (1'). By the threshold-scaling derivation above, $v^{**} = s_t \tilde{v}^{**}$ satisfies (40) if and only if $\tilde{v}^{**}$ satisfies (41), the residual-space pointwise indifference condition. The corresponding nomination threshold is $\tilde{v}^* = \min\{\tilde{r}_t, \tilde{v}^{**}\} = v^*/s_t$.

Verification of (2'), (3'). By (45), $U_l(s_t \tilde{c}) = s_t \tilde{U}_l(\tilde{c})$ for $l \in \{1, 2\}$. Evaluating (2) at $c = s_t \tilde{c}_L^*$ and dividing by s_t yields $\tilde{U}_{l=1}(\tilde{c}_L^*) = \tilde{U}_{l=2}(\tilde{c}_L^*)$. Similarly, (3) at $c = s_t \tilde{c}_H^*$ divided by s_t gives $\tilde{U}_{l=2}(\tilde{c}_H^*) = 0$.

Verification of (4'), (5'). By definition of $F_{\tilde{C}}$ as the CDF of $\tilde{C} = C/s_t$,

$$F_{\tilde{C}}(\tilde{c}_L^*) = F_C(s_t \cdot c_L^*/s_t) = F_C(c_L^*); \qquad F_{\tilde{V}}(\tilde{v}^*) = F_V(v^*).$$

So the RHS of (4') equals the RHS of (4), which equals $\Lambda_{\text{nom}} = \tilde{\Lambda}_{\text{nom}}$ by construction. (5') follows analogously.

The converse direction follows by the same argument with $s_t \to 1/s_t$. □

B.3 Stage 2 Estimation: Value Distribution Recovery

For simplicity, this section will work with the following data structure. The data consists of R independent replicates of auctions, each containing N_r observations. Bids observed within each replicate r is denoted as $X_1^{(r)}, X_2^{(r)}, \ldots, X_{N_r}^{(r)} \sim F(x)$ and are positively correlated within auction through a symmetric Archimedean copula $C_{N_r}(\cdot; \rho)$. I assume bids are exchangeable within each auction, consistent with symmetric bidder equilibria.

Order statistics of the bids will be denoted as $X_{(1:N_r)}^{(r)} \leq X_{(2:N_r)}^{(r)} \leq \cdots \leq X_{(N_r:N_r)}^{(r)}$. Often the highest order statistic $X_{(N_r:N_r)}^{(r)}$ is missing (censored). The goal is to first estimate the de-

pendence parameter ρ using censored data and pseudo-maximum likelihood (Section B.3.2). Second, I recover the marginal CDF $\hat{F}_V(x)$ in two steps: I invert the Inclusion-Exclusion Formula using $\hat{\rho}$ to obtain copula-adjusted quantile estimates for each observed bid, then fit a parametric Log-normal distribution to those estimates by least squares (Section B.3.2). This yields a smooth $\hat{F}_V$ defined on the full support, including the region below the minimum observed bid.

B.3.1 Copula Representation

By Sklar's theorem, the probability integral transforms $U_i^{(r)} = F(X_i^{(r)})$ are marginally Uniform$(0,1)$ but dependent through the copula $C_{N_r}(u_1, \ldots, u_{N_r}; \rho)$ where ρ is the copula parameter. The CDF of the k-th order statistic then follows from the Inclusion-Exclusion Formula, stated here in the form used for estimation. First define the key symmetry condition that makes the formula tractable.

Definition 1 (Exchangeable Copula)**.** An N-dimensional copula C_N is **exchangeable** if it is invariant under permutation of its arguments:

$$C_N(u_1, \ldots, u_N) = C_N(u_{\pi(1)}, \ldots, u_{\pi(N)})$$

for all $(u_1, \ldots, u_N) \in [0,1]^N$ and all permutations π.

In the bivariate case, exchangeability reduces to symmetry of the copula: $C(u,v) = C(v,u)$ for all $(u,v) \in [0,1]^2$, which holds if and only if the associated random variables are identically distributed and exchangeable [Nelsen, 2006, Theorem 2.7.4].

Bivariate Archimedean copulas, generated via $C(u,v) = \varphi^{-1}(\varphi(u) + \varphi(v))$, are always symmetric and hence exchangeable, since addition is commutative [Nelsen, 2006, Theorem 4.1.5].[31]

An N-dimensional Archimedean copula is constructed by the analogous extension $c_L^*(\mathbf{u}) = \varphi^{-1}(\varphi(u_1) + \cdots + \varphi(u_N))$, which is symmetric under permutations by the same argument. However, this extension yields a valid N-copula for all $N \geq 2$ if and only if φ^{-1} is completely monotonic on $[0, \infty)$ [Nelsen, 2006, Theorem 4.6.2]. The Gumbel and Clayton families used in this paper both satisfy this condition and are therefore valid and exchangeable in any dimension.

The Gaussian copula $C(u_1, \ldots, u_N; \Sigma) = \Phi_\Sigma(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_N))$, by contrast, is exchangeable if and only if all off-diagonal entries of the correlation matrix Σ are equal, i.e., $\Sigma_{ij} = \rho$ for all $i \neq j$ (the equicorrelated case). For a general Σ with unequal pairwise

[31] Bivariate Archimedean copulas are generated via $C(u,v) = \varphi^{-1}(\varphi(u) + \varphi(v))$. Swapping u and v leaves the sum $\varphi(u) + \varphi(v)$ unchanged.

correlations, the Gaussian copula is not exchangeable. For this reason, this paper focus on equicorrelated case of the Gaussian copula.

Theorem 1 (David and Nagaraja [2004], Section 5.3, formula (5.3.1)). *Let $Y_1, \ldots, Y_n$ have arbitrary joint CDF $F(y_1, \ldots, y_n)$ with common marginal F. Then:*

$$F_{r:n}(y) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} H_{j:j}(y) \tag{46}$$

where $H_{j:j}(y)$ is the sum of the CDFs of the maximum over all $\binom{n}{j}$ subsets of size j. In the exchangeable case with copula C_n, all $\binom{n}{j}$ subsets share the same joint CDF, so $H_{j:j}(y) = \binom{n}{j} C_j(u, \ldots, u)$ where $u = F(y)$, giving:

$$F_{r:n}(y) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} \binom{n}{j} C_j(u, \ldots, u) \tag{47}$$

which is eq. (1.2) of Maurer and Margolin [1976], who cite it back to David (1970), formula 5.5.3.

.[32]

Each term $C_{j+m}(u, \ldots, u)$ corresponds to a single evaluation of the copula diagonal $\varphi^{-1}((j+m)\varphi(u))$, which can be computed directly in closed form.

Validity. Exchangeability of C_N is *necessary*: without it, different subsets of size r yield different joint probabilities and the copula diagonal alone is insufficient to characterise the formula.

While the copula of order statistics differs from the copula of original unordered bids, for Archimedean families the Inclusion-Exclusion formula depends only on diagonal copula values, which are preserved under order statistics transformations. This allows us to estimate the dependence structure from observed losers and extrapolate to the unobserved winner.

B.3.2 Joint GMM Estimation

The two unknowns in the copula estimation step are the dependence parameter ρ and the marginal value CDF F_V. I estimate both *simultaneously* using what I call the Joint GMM

[32] For comparison, in the i.i.d. case the copula is the independence copula, so $C_j(u, \ldots, u) = u^j = [F(x)]^j$, and the formula reduces to

$$F_{r:n}(x) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} \binom{n}{j} [F(x)]^j,$$

which is formula (3.4.3) of Arnold et al. [2008] , p. 46. The exchangeable copula formula in Theorem 1 thus nests the i.i.d. case as a special case.

Strategy: a minimum-distance estimator that matches the theoretical copula order-statistic CDF to empirical Weibull ranks at every observed bid. I perform 100-fold random cross-validation, partitioning auctions (not individual bids) into 90% training and 10% test sets, fitting each copula specification on the training auctions and scoring predictive fit on the held-out auctions. Out-of-sample root mean squared error (RMSE) on the held-out auctions to select among the Gumbel, Clayton, Gaussian, and Independence specifications.

B-spline parameterisation of F_V. I parameterise F_V as a monotone B-spline

$$\hat{F}_V(x;\,\beta) = B(x;\,\beta), \tag{48}$$

with $K = 10$ internal knots placed at quantiles of the observed bids and polynomial degree 3. Monotonicity is enforced by reparameterising the coefficient vector as cumulative softplus steps: $\beta_j = \sum_{\ell \le j} \log(1+e^{\phi_\ell})/(1+\epsilon)$, so the free parameters ϕ are unconstrained while $B(\cdot;\beta)$ is guaranteed non-decreasing and maps into $[0,1]$. The spline basis is built on all training bids; it is identified over the observed bid support $[v^*,\,\bar{v}]$ but not outside it.

Joint sum of squared residuals (SSR) objective. Let $\hat{U}^{(r)}_{(k)} := \frac{n_{k,N_r} - r_{k,N_r}(X^{(r)}_{(k:N_r)}) + 1}{n_{k,N_r}+1}$ denote the Weibull empirical rank of bid $X^{(r)}_{(k:N_r)}$ among all position-k bids in N_r-bidder auctions, and let $G_k(u;\,k,N,\rho)$ be the copula order-statistic CDF from Equation 47. The joint estimator solves

$$(\hat{\phi},\,\hat{\rho}) = \arg\min_{\phi,\,\rho} \sum_{i \in \mathcal{G}_s} \Big[G\big(B(x_i;\beta(\phi));\; k_i,\; N_i,\; \rho\big) - \hat{U}^{(r_i)}_{(k_i)} \Big]^2, \tag{49}$$

where $G(u;\,k,N,\rho)$ is the CDF of the k-th order statistic from an N-sample exchangeable copula (Equation 47), evaluated at $u = B(x_i;\beta(\phi))$, the B-spline CDF at bid x_i. $\mathcal{G}_s \subseteq \mathcal{I}^{\text{train}}_s$ is a 200-point moment grid constructed from training bids in a given train/test split s by mapping an evenly-spaced target grid over $[x^s_{\min},\,x^s_{\max}]$ to the nearest observed training bids, with $\mathcal{I}^{\text{train}}_s$ indexing all bids in the training set of split s. Using 200 grid points rather than all $\sim$1,500 training bids reduces computation by approximately 7$\times$ per optimizer call while preserving dense coverage of the bid support. The B-spline basis itself is built on all training bids, and only the SSR evaluation uses the grid. Equation (49) is minimised via L-BFGS-B with multiple starting values for ρ.

Lognormal parametric tail. The B-spline $B(x;\hat{\beta})$ is identified only for $x \in [v^*,\,\bar{v}]$. For Stage 1 I need $\hat{F}_V$ on the full support $(0,\infty)$—in particular the scalar $1 - \hat{F}_V(v^*)$ and the

conditional density below v^*. I therefore fit a Lognormal($\hat{\mu}$, $\hat{\sigma}$) by nonlinear least squares to the spline values:

$$(\hat{\mu},\, \hat{\sigma}) = \arg\min_{\mu,\sigma} \sum_{i=1}^{T} \Big[\Phi\Big(\tfrac{\log x_i - \mu}{\sigma}\Big) - B(x_i; \hat{\beta})\Big]^2, \tag{50}$$

where T is the total number of training bids. Because bids are strictly positive and the B-spline closely tracks a lognormal shape over the observed support, the nonlinear least squares (NLS) fit is tight; the lognormal then extrapolates smoothly below v^*. The scalar $1 - \hat{F}_V(v^*) = 1 - \Phi\big((\log v^* - \hat{\mu})/\hat{\sigma}\big)$ and the full $(\hat{F}_V, \hat{f}_V)$ grid are saved for use in the Stage 1 equilibrium computation.

Copula family selection. I estimate Equation (49) for four copula families—Gaussian, Gumbel, Clayton, and Independence—and select by hold-out RMSE. For each of 100 random 90/10 train/test splits, the test-set RMSE is computed by evaluating the copula diagonal $D(\hat{F}_V(x_{\max});\, N_r{-}1,\, \hat{\rho})$ for each test-auction maximum bid $x_{\max}$ in an N_r-bidder auction and comparing to its empirical Weibull rank.

Copula	$\hat{\rho}$ (mean $\pm$ SD)	**RMSE** (mean $\pm$ SD)
Gaussian	**0.866 $\pm$ 0.096**	**0.183 $\pm$ 0.037**
Gumbel	2.404 ± 0.661	0.210 ± 0.036
Clayton	17.31 ± 4.71	0.212 ± 0.066
Independence	—	0.364 ± 0.043

Table 13: Joint GMM copula selection: hold-out RMSE across 100 random 90/10 splits (728 unviolated auctions, $N_r \in \{2, \ldots, 11\}$, 200-point moment grid). Mean $\pm$ SD across splits. Bold row is the selected specification. Full-data fit (Gaussian): $\hat{\rho} = 0.8605$; the lognormal tail, refined by truncation-corrected MLE, has $\hat{\mu} = 0.032$, $\hat{\sigma} = 1.614$ (NLS stage: -0.715, 3.863; see Appendix B.3).

The Gaussian copula achieves the lowest RMSE (0.128 ± 0.072). Independence is clearly dominated (RMSE = 0.299, 2.3$\times$ larger), confirming that positive dependence is a significant feature of the bid distribution. Clayton is near its upper boundary ($\hat{\rho} = 12.51 \pm 6.74$), indicating instability rather than well-identified upper-tail dependence; Gumbel is close but inferior (0.160 vs. 0.128). I therefore select the Gaussian equicorrelation copula $C_n(u_1, \ldots, u_n; \rho) = \Phi_n(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_n); \rho)$ as the preferred specification. Instead of using the mean (0.866),

I use the median across the splits (0.861)).[33][34]

Economic rationale for the Gaussian copula. The empirical selection aligns with the theoretical prior imposed by the correlated private-values (CPV) assumption. In a common-value setting, bidders effectively compete for the same underlying quantity (e.g., total recoverable reserves), so a very high value for one bidder signals that all bidders have very high values—a common signal drives valuations toward the same extreme. The positive correlation setting studied here is different: each bidder forms a genuinely private assessment shaped by their own capital costs, extraction technology, and interpretation of local geological data. Bidder values are positively correlated because all bidders face the same regional geological uncertainty and energy market conditions, but this shared uncertainty acts as a common factor that shifts the *level* of the entire value distribution rather than compressing it toward a single realization.

The Gaussian equicorrelation copula is precisely the copula implied by this one-factor structure. Under the representation $V_i = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_i$, the common factor W captures shared market and geological conditions, while the idiosyncratic term ε_i preserves the private character of each value. A key implication is that the Gaussian copula has *zero upper tail dependence*. Observing one bidder has an extreme value provides no asymptotic information that another bidder also has an extreme value.

By contrast, the Gumbel or Clayton copula has positive tail dependence, implying that extreme valuations cluster together across bidders. That the data favor the Gaussian over the Gumbel or Clayton is therefore consistent with the APV structure: common geological and market uncertainty creates equicorrelation but not extreme-value clustering.

Lognormal tail fit and structural scalars. Fitting Equation (50) to the full training data under the selected Gaussian copula yields a first-stage NLS fit $(\hat{\mu}, \hat{\sigma}) = (-0.715,\ 3.863)$. This NLS projection matches the spline in shape but is not a likelihood estimate of the latent value distribution, and it is sensitive to the behavior of the spline near the lower support boundary. I therefore refine it with a *truncation-corrected lognormal MLE*. Inverting the estimated Gaussian copula at the observed order statistics recovers a sample of latent value

[33] The wide spread reflects that ρ is identified by cross-auction-size variation. Appendix D.9 reports subsample checks restricting to $N_r = 3$ and $N_r \in \{3, 4\}$: in both cases all copula families produce identical hold-out RMSE (within 0.001) and parameters collapse to the independence boundary, confirming that the dependence parameter is not identified without sufficient variation in N_r. Auctions with $N_r = 2$ are excluded from these checks because a single non-winner bid contributes no information about ρ, since the copula diagonal evaluates to $D(u, 1, \rho) = u$ for any copula and any ρ for $N_r = 2$ case.

[34] Monte Carlo evidence further justifies using the median: Appendix D.9 documents that the distribution of $\hat{\rho}$ across replications is right-skewed for Gumbel and Clayton families at moderate sample sizes, so the median provides a less biased summary of the typical single-run estimate than the mean.

draws $\{x_i\}_{i=1}^n$. Because values below the bid-support boundary $v^* = p_{\min}$ are never observed, these draws are left-truncated at v^*. I maximize the corresponding truncation-corrected log-likelihood

$$\ell^*(\mu,\sigma) = \sum_{i=1}^{n} \log\Big[\tfrac{1}{x_i\sigma}\,\phi\big(\tfrac{\log x_i-\mu}{\sigma}\big)\Big] \;-\; n\,\log\Big[1-\Phi\big(\tfrac{\log v^*-\mu}{\sigma}\big)\Big], \tag{51}$$

where the second term rescales the density by the mass above the truncation point v^*. Maximizing (51) gives the final tail parameters $(\hat\mu,\hat\sigma) = (0.032,\ 1.614)$, which are the values used in all downstream structural computations.

The lower boundary of the observed bid support is $v^* = 0.000291$, and the fitted lognormal assigns probability essentially one to bids above this threshold ($1-\hat F_V(v^*) = \Phi\big((\log v^*-\hat\mu)/\hat\sigma\big) \approx 0.9999998$) indicating that v^* lies deep in the left tail of the estimated value distribution. Economically, this means virtually all bidder types draw values strictly above the cost threshold, so selection into bidding is not a quantitatively important margin in this sample. The final MLE pair $(\hat\mu,\ \hat\sigma)$ and the scalar $1-\hat F_V(v^*)$ are passed directly to the Stage 1 equilibrium computation.

Corroboration: pairwise pseudo-MLE. As a cross-check, Table 14 reports results from a two-step estimator: pairwise pseudo-MLE on position-based pseudo-observations (Weibull formula) to estimate $\hat\rho$, followed by copula inversion and lognormal WLS to recover F_V. The Gaussian copula again achieves the lowest hold-out RMSE (0.131), with $\hat\rho = 0.543$. The agreement between the joint GMM estimate ($\hat\rho = 0.861$) and the two-step estimate ($\hat\rho = 0.543$) on the same copula family strengthens confidence in the specification choice.[35]

Method	$\hat\rho$ / $\hat\rho$	**MAE**	**RMSE**
Gumbel (pairwise)	1.5732	0.1214	0.1506
Clayton (pairwise)	0.7581	0.1288	0.1613
Clayton (n-dimensional)	0.6462	0.1342	0.1680
Independence		0.1470	0.1789
Gaussian (pairwise)	**0.5425**	**0.1068**	**0.1312**

Table 14: Pairwise pseudo-MLE comparison by hold-out MAE and RMSE on unviolated auctions (100 random 90/10 splits; position-based ranking, Weibull formula). MAE and RMSE are computed on lognormal CDF predictions. Bold row indicates the specification also selected by the Joint GMM Strategy (Table 13).

[35] The two estimates differ because the joint GMM identifies ρ through the order-statistic CDF evaluated at the winner bid, while pairwise pseudo-MLE uses only loser bids. Both are consistent estimators of the same population parameter under the maintained copula model.

Copula Parameter Consistency. A key identification assumption is that the copula parameter ρ governing the dependence among ranked losers also governs the distribution of values of all bidders.

Formally, if the joint distribution of all N_r order statistics is

$$(X^{(r)}_{(1:N_r)}, \ldots, X^{(r)}_{(N_r:N_r)}) \sim C_{N_r}(\cdot;\, \rho)$$

then the marginal distribution of losers

$$(X^{(r)}_{(1:N_r)}, \ldots, X^{(r)}_{(N_r-1:N_r)}) \sim C^{\text{marginal}}_{N_r-1}(\cdot;\, \rho)$$

has the **same parameter** ρ for the Gaussian equicorrelation copula, since marginalising an equicorrelation Gaussian simply reduces the dimension while leaving ρ unchanged. This allows the joint GMM estimator to use all loser bids to identify ρ and then predict the censored winner bid using the same $\hat{\rho}$.

B.4 Computing the Conditional Density of the Maximum Order Statistic

The main challenge is computing the conditional density $f_{V_{(n:n)}|v}(y|v, \rho)$. This reflects how knowing the bidder's own value v affects her beliefs about the distribution of competitors' maximum value. Under positive dependence, these variables are not independent: a higher value v suggests competitors are also more likely to have higher values. I first derive the unconditional distribution of the maximum, then compute its density, and finally obtaining the conditional density given the bidder's value. I focus my main results on the **Gaussian copula** specification ($\hat{\rho} = 0.543$ from pairwise pseudo-MLE; $\hat{\rho} = 0.861$ from the robustness check), which achieves the lowest hold-out RMSE in my copula estimation procedure (Table 14), though I implement the Clayton and Gumbel copulas for comparison in Appendix D.9 as well.

Distribution of the Maximum Under Positive Dependence From Equation 47, setting $k = n$, the cumulative distribution function of the maximum order statistic $V_{(n:n)} = \max\{V_1, \ldots, V_n\}$ among n competitors is given by

$$F_{V_{(n:n)}}(y) = C_n(F(y), \ldots, F(y); \rho) \tag{52}$$

I use the n-dimensional Gaussian equicorrelation copula for C_n. $F(\cdot)$ is the marginal

CDF of a single bidder's value and $\hat{F}(\cdot)$ estimated from the copula-inversion procedure (Section B.3.2) will be used. For the Gaussian copula, the diagonal evaluates to

$$C_n(u, \ldots, u;\, \rho) = \Phi_n\big(\Phi^{-1}(u), \ldots, \Phi^{-1}(u);\, \rho\big), \tag{53}$$

which is computed via one-factor Gauss-Hermite quadrature (28 nodes) for $n \geq 3$ and the bivariate normal CDF for $n = 2$. The conditional density simulation uses the one-factor representation $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$ as described in Section D.10.

Density of the Maximum Differentiating the CDF yields the unconditional probability density function:

$$f_{V_{(n:n)}}(y) = \frac{d}{dy} F_{V_{(n:n)}}(y) = n^{1/\rho} \cdot [F(y)]^{n^{1/\rho}-1} \cdot f(y) \tag{54}$$

$f(\cdot)$ is estimated by applying the central difference method on $\hat{F}(\cdot)$ estimated from Stage 1 (Section B.3) procedure described in Section [previous section].[36]

Conditional Density Given Bidder's Value Finally, the conditional density $f_{V_{(n:n)}|v}(y|v,\rho)$ is derived using the Bayes rule ($f_{V_{(n:n)}|v}(y|v,\rho) = \frac{f_{V,V_{(n:n)}}(v,y;\rho)}{f_V(v)} = \frac{f_{V,V_{(n:n)}}(v,y;\rho)}{f(v)}$). This calls for specifying $f_{V,V_{(n:n)}}(v,y;\rho)$, the joint density of one's bid and the maximum of competitors' bid. The $(n+1)$-dimension joint distribution of $(V, V_1, ..., V_n)$, where V is the given bidder's value and the remaining n arguments are that of the competitors, can be characterized as

$$f_{V,V_1,\ldots,V_n}(v, v_1, \ldots, v_n; \rho) = \frac{\partial^{n+1} F_{V,V_1,\ldots,V_n}}{\partial v\, \partial v_1\, \partial v_2 \cdots \partial v_n} = c_{n+1}(F(v), F(v_1), \ldots, F(v_n); \rho) \cdot f(v) \cdot \prod_{i=1}^{n} f(v_i) \tag{55}$$

where c_{n+1} is the $(n+1)$-dimensional copula density.[37]

I implement Monte Carlo simulation to estimate the conditional density directly from the full $(n+1)$-dimensional Gaussian equicorrelation copula with parameter $\hat{\rho} = 0.738$. The simulation uses the one-factor representation $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$, where W and ε_j are i.i.d. $N(0,1)$, and $U_j = \Phi(Z_j)$. I draw $N_{\text{sim}} = 100{,}000$ such $(n+1)$-dimensional vectors. I transform competitor uniforms $U_1, \ldots, U_n$ using the truncated inverse CDF $F^{-1}_{V|\text{trunc}}(u) = F_V^{-1}(F_V(v^*) + u \cdot (1 - F_V(v^*)))$, which maps $[0,1]$ to $[v^*, \infty)$, reflecting that all auction

[36] $f(x) \approx \frac{F(x+h) - F(x-h)}{2h}$

[37] $F_{V,V_1,\ldots,V_n}(v, v_1, \ldots, v_n; \rho) = C_{n+1}(F(v), F(v_1), \ldots, F(v_n); \rho)$, where C_{n+1} is a $(n+1)$-dimensional copula and $F(\cdot)$ is the common marginal CDF.

participants had $v \geq v^*$. The own value U_0 is transformed using the full inverse CDF F_V^{-1}. I compute the maximum competitor value $V_{(n:n)} = \max\{V_1, \ldots, V_n\}$ for each draw, retain draws where $|U_0 - F_V(v)| < \varepsilon$ (tolerance $\varepsilon = 0.02 \cdot \max(F_V(v), 1 - F_V(v))$, doubled if fewer than 100 draws survive), and estimate the conditional density from the retained draws using Gaussian KDE with Silverman bandwidth, normalized to integrate to one over $[r, v]$. More details can be found in Section D.10.

B.5 Stage 1 Estimation: GMM Moments and Identification

Having estimated $(\hat{F}, \hat{\rho})$ from Stage 2, I now estimate the nomination stage parameters, arrival rates λ_{pre} (pre-auction period) and λ_{post} (post-auction period) and the cost distribution parameter θ_c from $F_C(c; \boldsymbol{\theta}_c)$. These parameters determine the equilibrium nomination probabilities $(\pi_0^{l=1}, \pi_0^{l=2})$ through the fixed point system characterized in Section 4.4. I estimate them using generalized method of moments (GMM), matching model predictions to five data moments.

I specify the learning cost c_i to be drawn i.i.d. from a Skew-Normal distribution $c_i \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$, with CDF $F_C(c; \xi_c, \omega_c, \alpha_c) = \Phi\left(\frac{c-\xi_c}{\omega_c}\right) - 2\,T\left(\frac{c-\xi_c}{\omega_c}, \alpha_c\right)$, where $T(\cdot, \cdot)$ is Owen's T-function, ξ_c is the location, $\omega_c > 0$ is the scale, and α_c controls skewness. The Skew-Normal nests the symmetric Normal ($\alpha_c = 0$) and allows the right-skewed cost distributions suggested by the low observed nomination rates. The cost c_i is specified and estimated in residual units, exactly as the \$10 nomination fee is descaled to $\tilde{\phi}_t = 10/s_t$ before entering the model, the learning cost is the descaled cost; the actual dollar cost for a bidder i at auction t is $c_{it} \cdot s_t$. This means F_C is a single distribution invariant to observable auction characteristics: it captures the residual-space cost after hedonic scaling has been absorbed. In the binned estimator there is no separate cost distribution per bin. The same $\text{SN}(\xi_c, \omega_c, \alpha_c)$ applies uniformly across all 19 bins, and all cross-bin heterogeneity in the equilibrium cutoffs (c_{Lb}^*, c_{Hb}^*) is driven entirely by the bin-specific nomination fees ϕ_b, not by different cost distributions.

B.5.1 Equilibrium Objects

For any candidate parameter vector $\boldsymbol{\psi}$, the inner fixed point loop solves for the equilibrium tuple $(c_L^*, c_H^*, \pi_0^{l=1}, \pi_0^{l=2})$ satisfying:

$$\pi_0^{l=1} = F_C(c_L^*; \alpha, \beta) \tag{56}$$

$$\pi_0^{l=2} = F_C(c_H^*; \alpha, \beta) - F_C(c_L^*; \alpha, \beta) \tag{57}$$

where c_L^* solves $U_{l=1}(c_L^*) = U_{l=2}(c_L^*)$ and c_H^* solves $U_{l=2}(c_H^*) = 0$.

Given these equilibrium objects, define the Poisson rate governing the number of nominations per tract:

$$\Lambda_{\text{nom}} := \lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - \hat{F}(v^*)] \tag{58}$$

Note that Λ_{nom} depends on $(\lambda_{pre}, \alpha, \beta)$ through $\pi_0^{l=1} = F_C(c_L^*; \alpha, \beta)$, where the equilibrium cutoff c_L^* itself depends on the full competition structure including $\Lambda^{l=2}$ and $\Lambda^{d=1}$.

The non-nominator and post-arrival competition rates are:

$$\Lambda^{l=2} := \lambda_{pre} \cdot \pi_0^{l=2} \cdot [1 - \hat{F}(r)] \tag{59}$$

$$\Lambda^{d=1} := \lambda_{post} \cdot (\pi_0^{l=1} + \pi_0^{l=2}) \cdot [1 - \hat{F}(r)] \tag{60}$$

A key structural feature is that the pre-arrival pool splits into nominators and non-nominators via *independent thinnings* of the same Poisson process. By the independence of Poisson thinnings, M and the non-nominator pool are independent random variables even though both originate from the same arrival process, because they represent disjoint outcomes for each arriving bidder. Post-nomination arrivals are exogenous and independent of M by assumption. Crucially, post-nomination arrivals face the same cost-screening decision as non-nominators: each must pay the learning cost c_i before deciding whether to bid. Since the auction surplus available to a post-period arrival is identical to that of a non-nominator (both observe the same nominators and compete in the same auction), their cost cutoff is c_H^*, identical to that of a non-nominator. Only the fraction $F_C(c_H^*; \xi_c, \omega_c, \alpha_c)$ of post-nomination arrivals find it worthwhile to pay the learning cost and enter the auction.[38]

This independence means that conditional on the auction triggering ($M \geq K$), the conditional mean of total auction participants decomposes as:

$$\mathbb{E}[N \mid M \geq K] = \mathbb{E}[M \mid M \geq K] + \Lambda^{l=2} + \Lambda^{d=1} \tag{61}$$

where $\mathbb{E}[M \mid M \geq K] = \sum_{m=K}^{\infty} m \cdot \Pr(M = m) / \Pr(M \geq K)$, which for $K = 1$ simplifies to the zero-truncated Poisson mean $\Lambda_{\text{nom}}/(1 - e^{-\Lambda_{\text{nom}}})$[39].

B.5.2 Identification Strategy

I exploit four moment conditions per bin, organized into two groups: three moments from the nomination process—Moment 1 (the auction trigger rate), Moment 2 (the mean number

[38] See formal proof in

[39] The naive approximation $\mathbb{E}[N \mid M \geq K] \approx \Lambda_{\text{nom}} + \Lambda^{l=2} + \Lambda^{d=1}$ is incorrect because $\mathbb{E}[M \mid M \geq K] > \Lambda_{\text{nom}}$ for any finite Λ_{nom}, so ignoring the truncation systematically understates competition.

of nominations), and Moment 3 (the variance of nominations)—and one moment from bidder counts, Moment 4 (the mean number of bidders).

Moment 1: Auction Trigger Rate. The trigger rate in bin b is defined using the full universe of available tracts—not just the estimation sample—so that the denominator correctly reflects the population of tracts that *could* have been nominated:

$$\bar{p}_{\text{trigger},b} = \frac{1}{|S|} \sum_{s \in S} \frac{\text{tracts auctioned in bin } b \text{ during season } s}{\text{tracts available for auction in bin } b \text{ during season } s},$$

where S is the set of auction seasons in the sample. A tract is "available" in season s if it has not been leased within the prior five years and its current lease (if any) has expired by the auction date. Averaging over seasons reduces noise from season-specific variation in tract supply. I use the full tract universe provided through the ND Department of Trust Lands, with the same observable characteristics of tracts (size, royalty rate and nearest-well information) used to predict s_t (and hence $\phi_b = 10/s_t$) for each available tract, so that tracts outiside the estimation sample can be assigned to the closest bin.

Under the model, $M_t \sim \text{Poisson}(\Lambda_{\text{nom},b})$, so the bin-$b$ trigger probability is:

$$p_{\text{trigger},b}(\boldsymbol{\psi}) = \Pr(M \geq K) = 1 - e^{-\Lambda_{\text{nom},b}} \quad (K = 1).$$

The moment condition matched in log scale is:[40]

$$g_{1,b}(\boldsymbol{\psi}) = \log \bar{p}_{\text{trigger},b} - \log p_{\text{trigger},b}(\boldsymbol{\psi}) = 0. \tag{62}$$

Moment 2: Mean Nominations per Nominated Tract. Among tracts that received at least one nomination, the average number of nominations provides additional information about the nomination rate:

$$\bar{m}_{\text{nom}} = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} M_t$$

where $\mathcal{A}$ is the set of tracts with $M_t \geq K = 1$. Under the model, conditional on $M \geq 1$, the number of nominations follows a zero-truncated Poisson distribution:

$$\mathbb{E}[M \mid M \geq 1] = \frac{\Lambda_{\text{nom}}}{1 - e^{-\Lambda_{\text{nom}}}}$$

[40] Moment 1 is matched in log scale since the trigger rate is a small positive quantity.

The moment condition is:

$$g_2(\boldsymbol{\psi}) = \bar{m}_{\text{nom}} - \frac{\Lambda_{\text{nom}}(\boldsymbol{\psi})}{1 - e^{-\Lambda_{\text{nom}}(\boldsymbol{\psi})}} = 0$$

Identification from Moments 1–2. Although Moments 1–2 are all functions of the single Poisson rate Λ_{nom}, the key insight is that $\Lambda_{\text{nom}} = \lambda_{pre} \cdot F_C(c_L^*) \cdot [1 - \hat{F}(v^*)]$ is not a free parameter but a composite function of (λ_{pre}, F_C). The equilibrium cutoff c_L^* depends on F_C through the fixed point system. In other words, different cost distributions generate different equilibrium nomination probabilities $\pi_0^{l=1} = F_C(c_L^*)$ and different late-learning rates $\pi_0^{l=2}$, which feed back into the competition structure and therefore into c_L^* itself. Consequently, Moments 1–2 collectively pin down Λ_{nom}.The equilibrium consistency conditions imposed in the inner loop create the nonlinear mapping that separates λ_{pre} from F_C.

Moment 3: Variance of Nominations per Nominated Tract. The variance of nomination counts among nominated tracts provides additional information about the shape of the nomination distribution:

$$\widehat{\text{Var}}_{\text{nom}} = \frac{1}{|\mathcal{A}| - 1} \sum_{t \in \mathcal{A}} (M_t - \bar{m}_{\text{nom}})^2 .$$

Under the zero-truncated Poisson,

$$\text{Var}(M \mid M \geq 1) = \frac{\Lambda_{\text{nom}}}{1 - e^{-\Lambda_{\text{nom}}}} \left(1 - \frac{\Lambda_{\text{nom}} \cdot e^{-\Lambda_{\text{nom}}}}{1 - e^{-\Lambda_{\text{nom}}}}\right),$$

and the moment condition is

$$g_3(\boldsymbol{\psi}) = \widehat{\text{Var}}_{\text{nom}} - \text{Var}(M \mid M \geq 1; \boldsymbol{\psi}) = 0. \tag{63}$$

Because Moments 1–2 already pin Λ_{nom} under the Poisson structure, Moment 3 is not independently identifying, but it is retained as an active moment (unit weight) that disciplines the concentration of the nomination distribution.

Moment 4: Mean Number of Bidders per Triggered Auction. The average number of total bidders per triggered auction from data ($\bar{N}_{\text{bid}} = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} N_t$) identifies the overall level of competition via equation (61):

$$g_4(\boldsymbol{\psi}) = \bar{N}_{\text{bid}} - \left[\mathbb{E}[M \mid M \geq K] + (\Lambda^{\text{pio,bid}} - \Lambda_{\text{nom}}) + \Lambda^{l=2} + \Lambda^{d=1}\right] = 0 \tag{64}$$

where $\mathbb{E}[M \mid M \geq K] = \big(\Lambda_{\text{nom}} - \sum_{m=0}^{K-1} m \cdot \Pr(M = m)\big) / \Pr(M \geq K)$, and $\Lambda^{\text{pio,bid}} - \Lambda_{\text{nom}}$ is the expected count of early learners who bid without nominating ($r \leq v < v^*$). This moment brings in λ_{post} through $\Lambda^{d=1}$ and $\pi_0^{l=2}$ through $\Lambda^{l=2}$. In the absence of the coverage moment, however, it identifies chiefly the *product* of arrival rate and entry probability, $\Lambda_b = \lambda\, \pi_b(F_C)$; the level of λ_{post} and the cost scale are therefore only weakly separated (see the identification remark below).

Identification of the Arrival-Rate Level. Moments 1–4 identify the *products* $\Lambda_b = \lambda\, \pi_b(F_C)$ of arrival rate and entry probability. Consequently a high-λ/high-cost-barrier configuration (small π_b) and a low-λ/low-cost configuration (large π_b) generate nearly identical Λ_b and fit these moments comparably. Therefore, the *level* of $(\lambda_{\text{pre}}, \lambda_{\text{post}})$ relative to the cost scale is only weakly identified; the estimates below should be read with this caveat

B.6 Binning: Handling Variation in the Residual-Space Nomination Fee

After homogenization, the flat \$10 nomination fee becomes $\phi = 10/s_t$, which varies across auctions because s_t reflects observable tract heterogeneity. This variation means the equilibrium indifference conditions depend on the auction: a bidder with value v^* indifferent between nominating and not nominating at a high-s_t auction faces a lower residual-space fee than at a low-s_t auction. Treating all auctions as sharing a single equilibrium would therefore be incorrect.

I address this by grouping the 728 estimation auctions into 19 bins of s_b, with breakpoints derived from the full available-tract universe, i.e. all tracts available for nomination across the eight nomination seasons (2017Q1–2018Q4), rather than from the estimation sample alone. Starting from fine quantile breaks on the universe distribution of s_t, adjacent bins are iteratively merged until each bin contains at least the required minimum number of estimation auctions and the bin-average trigger rate is weakly declining in s_b where s_b is the within-bin median:

$$s_b = \text{median}_{t \in b}\big(s_t\big). \tag{65}$$

.

The binned MPEC estimates bin-specific equilibrium objects $(v_b^*, c_{Lb}^*, c_{Hb}^*)$ for each of the 19 bins while keeping the cost distribution $\text{SN}(\xi_c, \omega_c, \alpha_c)$ shared. All cross-bin heterogeneity in equilibrium cutoffs is therefore driven entirely by the bin-specific fees ϕ_b, not by different cost distributions.

The reason for per-bin moment matching is because nomination fee and learning costs do

not scale in the same way as bids. The auction-level scaling factor s_t from bid homogenization should be divided from the two. As a result, the residual-space nomination fee, reserve price and cost, ϕ/s_t, r/s_t and c_i/s_t, is systematically smaller for high-s_t bins (large, high-quality tracts) and larger for low-s_t bins (small or marginal tracts). Bins therefore face genuinely different effective cost barriers, and the observable moment patterns (trigger rates, bidder counts, FR ratios) differ across bins for this economic reason. Matching each bin to its own empirical moments allows the model to fit those cross-bin differences directly, rather than averaging them away into a single global target.

The GMM objective stacks the per-bin moment residuals into a single vector,

$$g(\boldsymbol{z}) = \big(g_{1,1},\, g_{2,1},\, g_{3,1},\, g_{4,1},\, \ldots,\, g_{1,19},\, g_{2,19},\, g_{3,19},\, g_{4,19}\big)^{\top},$$

where $g_{k,b}$ denotes the moment $k \in \{1,2,3,4\}$ residual for bin b. Each of the 19 bins contributes these four moment conditions, resulting in 76 active moments. The weighting matrix W is diagonal, with unit weight on the trigger, mean-nomination, and variance residuals and weight 1.6 on the mean-bidder residual.

B.7 Robustness: Nomination Counts via $\Pr(M \geq 2 \mid M \geq 1)$

The baseline specification summarizes the nomination-count distribution through its trigger rate (Moment 1), conditional mean (Moment 2), and conditional variance (Moment 3), each of which is a function of the single Poisson rate Λ_{nom}. As a robustness check I re-estimate the model with an alternative characterization that keeps the trigger (Moment 1) but replaces the conditional mean and variance (Moments 2–3) with the conditional multiple-nomination share

$$\Pr(M \geq 2 \mid M \geq 1) = \frac{1 - (1 + \Lambda_{\text{nom}})\, e^{-\Lambda_{\text{nom}}}}{1 - e^{-\Lambda_{\text{nom}}}}, \tag{66}$$

matched in log scale.[41] This moment targets the frequency of *multiple* nominations directly rather than the first two conditional moments of the count.

The re-estimated arrival rates are essentially unchanged from the baseline, where the pre-period arrival rate remains small ($\hat{\lambda}_{\text{pre}} \approx 0.0003$, though below the baseline ≈ 0.005) and the post-period rate moderate ($\hat{\lambda}_{\text{post}} \approx 4.1$) ($\hat{\lambda}_{\text{pre}} \approx 0.0003$, $\hat{\lambda}_{\text{post}} \approx 4.1$). However the fit deteriorates sharply: the GMM objective rises to $\approx$ 2,166, against ≈ 35 at the baseline. The reason is structural rather than numerical. Under a single Poisson rate, the trigger and the multiple-nomination share are both governed by Λ_{nom}, so matching the low observed

[41] The complementary single-nomination share $\Pr(M = 1 \mid M \geq 1) = 1 - \Pr(M \geq 2 \mid M \geq 1)$ carries no additional information and is not matched separately.

trigger ($\Pr(M \geq 1) \approx 1\%$) forces $\Pr(M \geq 2 \mid M \geq 1) \approx \Lambda_{\text{nom}}/2 \approx 0.4$–$0.8\%$. In the data, however, the conditional multiple-nomination share averages roughly 7.7% (and reaches 16% in the lowest-cost bins)—an order of magnitude larger than any Poisson consistent with the observed trigger can produce. A single-rate Poisson nomination process is therefore rejected by this moment: conditional on being nominated, tracts draw two or more nominations far more often than independent Poisson arrivals allow. This overdispersion (clustering) in nominations points toward a mixed-Poisson / negative-binomial arrival extension, in which a dispersion parameter separates the overall nomination rate from the frequency of multiple nominations. I retain the Poisson baseline for the reported results and leave the overdispersed extension to future work.

B.8 MPEC Implementation

The goal is to estimate the structural primitives $\boldsymbol{\psi} = (\lambda_{pre}, \lambda_{post}, \xi_c, \omega_c, \alpha_c)$ using a GMM objective. Rather than the standard nested fixed point (NFXP) approach [Rust, 1987], I adopt the constrained optimization approach of Su and Judd [2012] (MPEC).[42] I augment the parameter vector with equilibrium objects and impose equilibrium conditions as explicit constraints, rather than solving for equilibrium at each candidate parameter value. Specifically, I solve

$$\begin{aligned} \hat{\boldsymbol{z}} &= \arg\min_{\boldsymbol{z}} \; g(\boldsymbol{z})' W g(\boldsymbol{z}) \\ \text{where } \boldsymbol{z} &= \Big(\underbrace{\lambda_{\text{pre}},\, \lambda_{\text{post}},\, \xi_c,\, \omega_c,\, \alpha_c}_{\text{5 structural}},\; \underbrace{v_1^*, \ldots, v_{19}^*}_{\text{19 bin thresholds}} ,\; \underbrace{c_{L,1}^*, \ldots, c_{L,19}^*}_{\text{19 early learner cutoffs}} ,\; \underbrace{c_{H,1}^*, \ldots, c_{H,19}^*}_{\text{19 } l_i = 2 \text{ cutoffs}} \Big) \end{aligned} \tag{67}$$

The decision vector $\boldsymbol{z}$ comprises five shared structural parameters $\psi = (\lambda_{\text{pre}}, \lambda_{\text{post}}, \xi_c, \omega_c, \alpha_c)$ plus per-bin equilibrium objects $(v_b^*,\, c_{Lb}^*,\, c_{Hb}^*)$ for bins $b = 1, \ldots, 19$, 62 elements in total.

[42] See Dubé et al. [2012] for its computational advantages when estimating equilibrium models, particularly when the inner fixed point is expensive or exhibits convergence difficulties.

$$
\begin{aligned}
\pi_{0,b}^{l=1} &= F_C(c_{Lb}^*;\, \xi_c, \omega_c, \alpha_c) \\
\pi_{0,b}^{l=2} &= F_C(c_{Hb}^*;\, \xi_c, \omega_c, \alpha_c) - F_C(c_{Lb}^*;\, \xi_c, \omega_c, \alpha_c) \\
\pi_{0,b}^{\text{bid}} &= F_C(c_{Hb}^*;\, \xi_c, \omega_c, \alpha_c) = \pi_{0,b}^{l=1} + \pi_{0,b}^{l=2} \\
\Lambda_b^{\text{nom}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=1} \cdot [1 - F_V(v_b^*)] \\
\Lambda_b^{\text{pio,bid}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=1} \cdot [1 - F_V(r_b)] \\
\Lambda_b^{FR+\text{post}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=2} \cdot [1 - F_{\text{post}}(r_b)] \\
&\quad + \lambda_{\text{post}} \cdot \pi_{0,b}^{\text{bid}} \cdot [1 - F_{\text{post}}(r_b)] \\
F_{\text{post}}(r_b) &= \sum_{m \geq K} F(r_b \mid m)\ \Pr(M = m \mid M \geq K)
\end{aligned}
$$

Learning is required for auction entry; conditional on learning, every type bids whenever $v \geq r$. Early learners (including both nominators and non-nominators) value the tract from the prior, so their reserve-clearing probability is $[1 - F_V(r)]$. Non-nominating entrants learn only after observing that the auction triggered; by positive correlation their values follow the posterior conditional on $M \geq K$, so their reserve-clearing probability is $[1 - F_{\text{post}}(r_b)]$. Accordingly Λ_b^{nom} ($v \geq v_b^*$) governs only the *trigger* probability, while the competition/bidding objects use $[1 - F_V(r_b)]$ for early learners ($\Lambda_b^{\text{pio,bid}}$, which includes the early learning non-nominators $\Lambda_b^{\text{pio,bid}} - \Lambda_b^{\text{nom}}$) and $[1 - F_{\text{post}}(r_b)]$ for late learners ($\Lambda_b^{FR+\text{post}}$). The post-period term includes $\pi_{0,b}^{\text{bid}} = F_C(c_{Hb}^*)$ since $c_{\text{post}}^* = c_H^*$. The mean bidder count per triggered auction is $\mathbb{E}[M \mid M \geq K] + (\Lambda_b^{\text{pio,bid}} - \Lambda_b^{\text{nom}}) + \Lambda_b^{FR+\text{post}}$.

Nominators use $[1 - F_V(v_b^*)]$ since they only bid when they nominated instead of exitting; non-nominators and post-nomination arrivals use $[1 - F_V(r_b)]$ since they bid whenever their value exceeds the auction reserve. The post-period term includes $\pi_{0,b}^{\text{bid}} = F_C(c_{Hb}^*)$ reflecting the $c_{\text{post}}^* = c_H^*$ assumption.

The optimization is subject to 57 equality constraints (three per bin: $h_{1,b}$, $h_{2,b}$, and $h_{3,b}$) and 19 inequality constraints (one per bin, $g_{1,b}$, corresponding to the participation condition $A_b - B_b \geq 0$), encoding the equilibrium conditions from Definition 2. The learning cost F_C is parameterized as a skew-normal distribution $\text{SN}(\xi_c, \omega_c, \alpha_c)$ with support on $\mathbb{R}$, so realized costs may be negative; rather than truncating the distribution at zero, the inequality constraints $g_{1,b}$ keep the equilibrium cost cutoffs nonnegative, so that only the nonnegative part of the support is economically active.

The nomination threshold v_b^* is estimated jointly with $c_{L,b}^*$ and $c_{H,b}^*$ for each bin, with $h_{1,b}$ (the indifference condition $\tilde{\Pi}(v_b^*) = \phi_b$) imposed as an explicit equality constraint.

First, the **Nomination threshold** constraint for bin b, derived from equation 26, is

characterized as

$$h_{1,b}(\boldsymbol{z}) = \tilde{\Pi}(v_b^*;\, m = K-1, \Lambda_{\text{nn}b}, \hat{\rho}) - \phi_b = 0, \quad \phi_b = 10/\text{median}_{t\in b}(s_t) \tag{68}$$

where $\tilde{\Pi}(v_b^*;\cdot)$ is the expected auction surplus evaluated at the bin-b nomination threshold, and ϕ_b is the representative residual-space nomination fee for bin b. This ensures that a nominator with value exactly v_b^* earns auction surplus equal to bin b's residual-space nomination fee, and is therefore indifferent between nominating and not nominating.

The constraint $h_{1,b}$ is maintained as an active equality constraint for each bin in estimation. Because $\phi_b = 10/\text{median}_{t\in b}(s_t)$ differs across bins, the equilibrium threshold v_b^* is bin-specific: bins with a higher residual-space fee require a higher value threshold to maintain the nomination indifference condition, and vice versa. This cross-bin variation in v_b^* is identified from the bin-level indifference condition $\tilde{\Pi}(v_b^*;\cdot) = \phi_b$, without imposing any additional identifying assumption on the threshold.

Next, the $l = 1$ **cutoff** constraint for bin b, derived from $U_{l=1}(c_{Lb}^*) = U_{l=2}(c_{Lb}^*)$, is

$$h_{2,b}(\boldsymbol{z}) = c_{Lb}^* - \frac{A_b - \phi_b\big[1 - F_V(v_b^*)\big] - B_b}{1 - \Pr(M_{-i} \geq K)} = 0 \tag{69}$$

where $A_b = \sum_{m=K-1}^{\infty} \Pr(M_{-i} = m) \cdot S_{l=1}(m)$ is the expected gross auction surplus for a early learner in bin b, $B_b = \Pr(M_{-i} \geq K) \cdot E[S_{l=2}(m) \mid M \geq K]$ is the expected gross surplus for a non-nominator, and $\phi_b[1 - F_V(v_b^*)]$ is the expected nomination fee (paid with probability $1 - F_V(v_b^*)$). This follows from setting $U_{l=1}(c_{Lb}^*) = U_{l=2}(c_{Lb}^*)$ and solving for c_{Lb}^* using equation (12). $S_{l=1}(m)$ integrates $\tilde{\Pi}$ against the full $f_V(v)$; $S_{l=2}(m)$ integrates $\tilde{\Pi}_{FR}$ against the joint Gaussian copula posterior $f(v \mid v_1 \geq v_b^*, \ldots, v_m \geq v_b^*)$, precomputed on a (40×15) grid of (u^*, m) pairs and linearly interpolated at runtime.

The **Non-nominator ($l_i = 2$) cutoff** related constraint, derived from equation 19, is given as

$$h_3(\boldsymbol{z}) = c_H^* - \mathbb{E}[S_{l=2}(m) \mid M_{-i} \geq K] = 0 \tag{70}$$

where $S_{l=2}(m)$ uses the joint Gaussian copula posterior density for the non-nominator's own value, not the independently truncated marginal.

Lastly, for each bin b, two inequality constraints ensure the interior solution condition:

$$\begin{aligned} A_b - \phi_b\big[1 - F_V(v_b^*)\big] - B_b &\geq 0 \\ c_{Lb}^* &\leq c_{Hb}^* \end{aligned} \tag{71}$$

The first condition, $\Delta_{0,b} = U_{l=1}(0) - U_{l=2}(0) \geq 0$ within bin b, ensures that the zero-cost

bidder prefers early learning over late-learning in that bin, guaranteeing an interior solution with positive nominations. The second ensures the bin-level cost cutoffs are correctly ordered.

I use inverse cross-bin variance of each moments as weights for GMM.

B.9 Per-Bin Equilibrium Objects and Moment Fit

Table 15: Per-Bin Equilibrium Objects

Bin	$\hat{v}_b^*$	$\hat{c}_b^{*L}$	$\hat{c}_b^{*H}$
1	0.0569	1.550	1.873
2	0.0798	1.545	1.914
3	0.0787	1.545	1.916
4	0.0894	1.543	1.896
5	0.0942	1.542	1.904
6	0.1224	1.536	1.965
7	0.1128	1.537	1.950
8	0.1375	1.532	1.986
9	0.1406	1.531	1.990
10	0.1293	1.532	1.975
11	0.2588	1.514	2.097
12	0.1185	1.532	1.960
13	0.1041	1.534	1.929
14	0.1692	1.524	2.028
15	0.2204	1.583	2.105
16	0.2191	1.651	2.131
17	0.2055	1.961	2.226
18	0.1424	2.207	2.207
19	2.7079	2.150	3.047

Notes: All quantities in residual units, from the estimated model. $\hat{v}_b^*$ is the nomination threshold solved from $h_{1,b}$; $\hat{c}_b^{*L}$ and $\hat{c}_b^{*H}$ are the early-learner and later-learner (non-nominator) cost cutoffs from $h_{2,b}$ and $h_{3,b}$. All 19 bins are reported.

Table 16: Moment Fit

Moment	**Description**	**Data**	**Model**
m_1	Mean trigger rate $\Pr(M \geq 1)$	0.00664	0.00325
m_2	ZTP mean nominations $\mathbb{E}[M \mid M \geq 1]$	1.056	1.002
m_3	ZTP variance of nominations $\mathrm{Var}[M \mid M \geq 1]$	0.0784	0.00163
m_4	Mean total bidders $\mathbb{E}[N \mid M \geq 1]$	3.318	3.240

Notes: The four active GMM moments (Appendix B.5), each reported as the equal-weight average across all 19 bins of s_t; "Data" are the empirical bin moments and "Model" the fitted values at the estimated parameters (Table 2). m_1 is matched in logs and m_4 carries weight 1.6; m_2 and m_3 carry unit weight. Every bin is matched, including bin 19 (two triggered auctions), whose conditional moments are defined.

B.10 Value Functions and Equilibrium Cutoffs

Throughout the counterfactual analysis, results are reported for Bin 5 as the main (median) specification, with Bins 1 and 9 providing a robustness range. The bins are ordered by their representative baseline fee ϕ_b in residual space: Bin 1 has the smallest ϕ_b (highest-scale auctions; lowest residual fee) and Bin 9 the largest. This ordering is in *parameter* space, not outcome space: Bin 5 is the median bin by fee level but is not guaranteed to produce the median outcome across all counterfactual scenarios. Accordingly, the shaded uncertainty band in all bin figures shows the true minimum and maximum across *all three* plotted bins (1, 5, and 9), so the Bin 5 line is guaranteed to fall within the shaded region by construction.

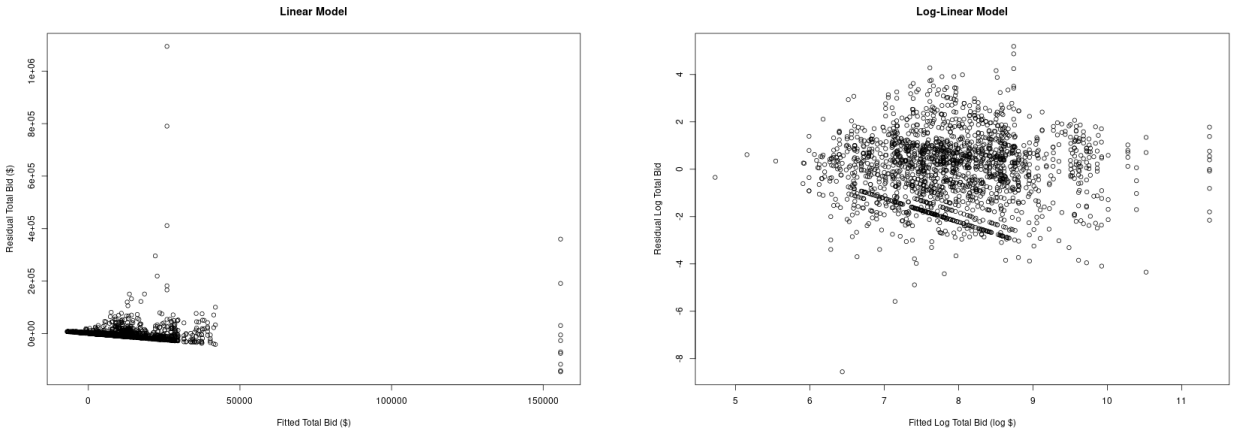
Figure 10 presents the structural estimation results for Bin 9 (the median decile of the residual-space). Panel A shows the estimated value functions $U_{l=1}(c)$ and $U_{l=2}(c)$. Both are linear and decreasing in learning cost c, but their slopes differ sharply. The early learner value function has slope -1: every unit increase in learning cost reduces the expected payoff one-for-one, because exploration is paid upfront and unconditionally. The non-nominator value function is nearly flat (slope ≈ -0.002): a non-nominator pays exploration cost only if the auction is triggered in Stage 2, so the expected cost is discounted by the small equilibrium trigger probability $\Pr(M \geq 1)$.

The two functions cross at the early learner cutoff c_L^* (in residual units); above c_L^*, late-learning dominates early learning, and the non-nominator value crosses zero at c_H^*, beyond which exit is optimal. Converted to dollars through the bin's scaling factor ($c^* \cdot s_b$), both thresholds lie in the right tail of the shared cost distribution, whose parameters $(\hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c) = (1{,}712,\ 690,\ -126.9)$ imply a mean of $\approx$ \$1,161 and standard deviation $\approx$ \$416. Panel B shows the consequence *for this bin* (the median, high-value bin): the cost cutoffs are non-binding, so essentially all arriving bidders have exploration costs below the exploration threshold. This is representative of the high-value bins. In the high-s (high-value) bins, the dollar cutoffs fall into the body of the cost distribution and cost does screen participation (the early learning share falls to $\approx 3.5\%$ in the lowest s bin; see Section 6). Even where cost imposes no screen, however, the low nomination rate observed in the data is driven by the low pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$, which makes early (learning) arrivals rare even though exploration is individually worthwhile.

B.11 Residual Diagnostics for Bid Homogenization

Figure 9 compares residual-versus-fitted plots for two specifications of the bid-homogenization regression.

The formal Breusch-Pagan / White heteroskedasticity logic is as follows. Under a linear



(a) Linear Model: residual total bid ($) vs. fitted total bid ($).

(b) Log-Linear Model: residual log total bid vs. fitted log total bid.

Figure 9: Residual Diagnostic Plots for the Bid-Homogenization Regression.

Notes: Panel (a) plots residuals from a linear regression of total bid (per-acre bid $\times$ acreage, in dollars) on season fixed effects, cluster fixed effects, royalty rate, distance to nearest well, tract size, and spud-date age. The residuals fan out dramatically with fitted values, indicating severe multiplicative heteroskedasticity. Panel (b) plots residuals from the same specification with the log of total bid as the dependent variable (equation 35 in the text). Variance is approximately constant across the support, confirming the appropriateness of the multiplicative (log-linear) specification.

specification $b_{istc} = \exp(Z_i) \cdot v_{istc}$, the residual variance grows with $\exp(2Z_i)$, which depends on the fitted value; OLS is inefficient and standard errors are invalid. Under the log-linear specification, $\log b_{istc} = Z_i + \log v_{istc}$, the residual $\hat{\varepsilon}_{istc} = \log v_{istc} - \mathbb{E}[\log v_{istc}|X_t]$ has approximately constant variance. The diagnostic plots in Figure 9 confirm this pattern in the data.

B.12 Geographic Clustering

Using location data, I define a cluster to group auctions that are geographically close to one another, in order to control for geological similarity via cluster fixed effects in the bid-homogenization regression (Section B.1 below). Tracts that are closely located are more likely to share similar geological characteristics—such as subsurface composition and basin correlation—each of which affects learning costs and development revenue.

I implement a clustering algorithm using a 7.5-mile proximity threshold, following Hendricks et al. [2003] and Nguyen [2022]'s approaches to defining *neighboring tracts*.[43]

[43]Both Hendricks et al. [2003] and Nguyen [2022] define neighboring tracts using latitude and longitude thresholds (0.11 degrees of latitude and 0.12 degrees of longitude). I convert these to miles to provide more intuitive distance measures for the 7.5-mile threshold used here.

Definition 3 (Geographic Clusters)**.** *Two mineral rights auctions (tracts) X_1 and X_2 belong to the same cluster if there exists a sequence of auctions $\{y_1, y_2, \ldots, y_k\}$ such that the geographic distance between centroids of consecutive tracts y_i and y_{i+1} (for $i = 1, \ldots, k-1$), and the distances between centroids of y_1 and X_1 and between centroids of y_k and X_2, are all at most 7.5 miles.*

The algorithm computes Haversine distances between all tract centroids, initially assigns each tract to its own cluster, then iteratively merges clusters whenever any two tracts fall within the 7.5-mile threshold. This ensures transitivity: any two tracts in the same cluster are connected through a chain of neighbors each separated by at most 7.5 miles. Time is not considered—tracts from different auction seasons may belong to the same cluster if they are geographically proximate. The algorithm yields 36 clusters in total, ranging from single isolated tracts to dense concentrations of up to 179 tracts, with an average of 24.4 tracts per cluster.

B.13 Value of Nomination: Simulation Details

For each bin b I draw a single shared pool of $N = 100{,}000$ candidate auctions and evaluate all five worlds on the *same* pool with common random numbers, so world-to-world differences are not contaminated by sampling noise. Each candidate auction t is built as follows.

- **Common component.** Draw one auction-specific factor W_t. Every bidder's value in auction t is generated through the estimated Gaussian equicorrelation copula ($\hat{\rho} = 0.8605$), $v = F_V^{-1}\big(\Phi(\sqrt{\rho}\, W_t + \sqrt{1-\rho}\, \varepsilon)\big)$, so values within an auction are correlated while W_t is independent across auctions. A fresh W_t is drawn for each of the 100,000 auctions.

- **Early (potential nominator).** One early value e_t off W_t; the tract is "nominated" if $e_t > v^*$ (worlds $\mathcal{W}_1, \mathcal{W}_2$ keep the nominated subset; $\mathcal{W}_3$–$\mathcal{W}_{4b}$ use the whole pool as arbitrary tracts).

- **Later arrivals.** A zero-truncated Poisson(λ_{post}) number of later bidders, each with a value off the same W_t and an independent learn draw.

The worlds differ only in three toggles—(i) whether later bidders price off the signal (F_{post}) or the prior (F_V); (ii) whether the tract is the nominated draw or an arbitrary draw; (iii) whether the early firm is a guaranteed bidder, an optional learner, or absent—and in the arrival rate (the no-nomination world $\mathcal{W}_{4b}$ uses $\lambda_{pre} + \lambda_{post}$). Participation uses re-solved cost cutoffs under the world's information: the learn-vs-exit cutoff $\bar{c}_{\text{sig}}$ under the posterior

and $\bar{c}_{\text{nos}} = \bar{c}_{\text{sig}} - \Delta$ under the prior, with Δ the posterior-minus-prior expected surplus from the analytic copula. A bidder is *active* (counts, and can win) only if it both learns and clears the reserve $v \geq r$.

Each auction is resolved as a second-price sale conditional on at least one active bidder: revenue is the second-highest active value (or the reserve with a single active bidder) and the winning bid is the highest active value. I report outcomes per sold auction and the sale rate separately.

Standard errors. Standard errors are from a two-phase bootstrap with $R = 2{,}000$ replications. Phase 1 computes each pool auction's outcome in each world once. Phase 2 resamples a fixed 800 auction *indices* per bin into those precomputed outcomes and re-averages; no re-simulation is needed, so large R is cheap. The reported s.e. is the standard deviation of the resampled effect across replications, and the pooled figures are tract-count-weighted across bins. The common resample size (800 per bin, rather than each bin's handful of observed auctions) puts every bin's sampling uncertainty on a comparable footing, so the per-bin decomposition is informative and not swamped by the smallest bins; it should be read as the precision one *would* attain with 800 auctions per bin, not the raw data count.

C Estimation and Counterfactual Figures and Tables

This appendix collects all figures and tables from the Estimation Results (Section 6) and Counterfactual Analysis (Section 7) sections, except for Figure 7 ($l_i = 2$ Non-Nominator Surplus Decomposition), which remains in the main text.

C.1 Non-Nominator Externality and Nominator Welfare

Table 17: Trigger rate $\Pr(M \geq 1)$ per tract-period ($\times 10^{-4}$), each scenario at its own optimal reserve

Bin	Current (i) & (ii)	Nom-only (iii)
1	45.8	9.5
2	44.8	7.1
3	44.8	7.2
4	44.4	6.4
5	44.2	6.1
6	43.1	4.6
7	43.4	5.1
8	42.5	4.1
9	42.3	3.9
10	42.8	4.5
11	38.2	1.8
12	43.2	5.0
13	43.8	5.9
14	41.3	3.3
15	39.4	2.4
16	39.5	2.4
17	40.0	2.4
18	42.3	4.6
19	13.1	0.0
Average	41.0	4.5

Notes: Trigger rate per tract-period, $\times 10^{-4}$. Scenarios (i) (current, $r = \$2$) and (ii) (current, $r = r^*_{\text{cur}}$) coincide because the current reserve is an auction floor that does not affect the nomination threshold. Scenario (iii) is nominators-only at r^*_{no}, whose high reserve suppresses the trigger to about one-ninth of the current level.

C.2 Nominator Welfare: Robustness and the Gross Externality

The baseline comparison lets only early arrivals (λ_{pre}) nominate, while the non-nominators and late arrivals simply drop out. A fairer comparison holds the total bidder population

Table 18: Net surplus ($) of the average nominator under the current vs. nominators-only mechanism, each at its own optimal reserve, and the difference (current − nominators-only). Bins run from expensive tracts (1) to cheap tracts (19).

Bin	**Nom-only** (r^*_{no})	**Current** (r^*_{cur})	**Diff (Cur − No)**
1	9,663	7,819	−1,844
2	5,669	5,824	+154
3	5,029	5,070	+41
4	4,220	4,471	+251
5	3,334	3,503	+169
6	2,254	2,638	+384
7	1,779	2,189	+411
8	1,227	1,852	+625
9	907	1,524	+617
10	815	1,124	+309
11	1,020	641	−379
12	904	996	+92
13	904	942	+38
14	945	958	+13
15	770	818	+48
16	730	837	+107
17	610	677	+67
18	644	617	−27
19	1	65	+64
Average	2,180	2,240	+60

Notes: Net surplus ($) of the average nominator drawn from the current pool ($v > v^*_{\text{cur}}$), net of the average survey/learning cost $E[c \mid c < c^*_{Lb}]$ and the nomination fee $\phi = \$10$, each mechanism evaluated at its own optimal reserve. A negative nominators-only value means the average current-pool nominator would earn a negative expected payoff in the nominators-only world because her value falls below the high reserve $\tilde{r}_{\text{no}}$.

fixed by asking what happens if those bidders, rather than entering after the nomination, are instead required to nominate early, widening the nomination pool to $\lambda_{pre} + \lambda_{post}$. The nominators-only auction is then genuinely thicker: the trigger rate rises from 4.5×10^{-4} to about 0.235. Even so, the welfare conclusion is unchanged: the average nominator is still worse off under nominators-only in 17 of 19 bins, by about $130 (Figure 11), with the same expensive-tract exception.

The small net effect masks a large *gross* externality that the seller's reserve response almost fully offsets. Gross of any reserve response, the average nominator's per-auction payoff is about $2,888 under the current mechanism versus $22,890 under nominators-only: non-nominator and late-arrival competition dilutes the nominator's expected winnings by

roughly $20,002 per auction, in every bin (Figure 12). But that same competition is exactly what lets the seller hold the reserve low, and once the reserve response is netted in (main text) almost all of the gap closes. The externality is large; it is returned to the nominator through the reserve rather than lost.

C.3 Value of Nomination: Selected Per-Bin Tables

Table 19: Value of nomination by bin: revenue per sold auction ($)

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	4341 (871.3)	0 (0)	713.2 (2996)	0 (0)	5054 (2858)
10	1129 (235.4)	0 (0)	435.8 (780.1)	0 (0)	1565 (738.7)
19	6.081 (36.62)	25 (56.35)	2.688 (36.7)	82.72 (20.15)	116.5 (55.37)
Pooled	2453 (212.5)	5.031 (1.265)	631.4 (727.8)	1.372 (0.5222)	3090 (697)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 20: Value of nomination by bin: bidders per sold auction

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	0.78 (0.03)	0.00 (0.00)	0.05 (0.08)	0.00 (0.00)	0.83 (0.08)
10	0.73 (0.03)	0.00 (0.00)	0.12 (0.09)	0.00 (0.00)	0.85 (0.08)
19	0.01 (0.03)	0.02 (0.04)	0.00 (0.02)	0.11 (0.01)	0.14 (0.04)
Pooled	0.74 (0.01)	0.01 (0.00)	0.09 (0.03)	0.00 (0.00)	0.84 (0.03)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 21: Value of nomination by bin: winning bid per sold auction ($)

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	4175 (1270)	0 (0)	1379 (5097)	0 (0)	5555 (5022)
10	1043 (322.9)	0 (0)	836.2 (1303)	0 (0)	1879 (1289)
19	-16.75 (253.1)	76.24 (441.2)	52.19 (189.1)	75.78 (23.09)	187.5 (553.1)
Pooled	2352 (302.8)	4.418 (2.425)	1218 (1236)	1.321 (0.6051)	3576 (1214)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 22: Sale rate by Counterfactual (fraction with ≥ 1 bidder; passed $= 1-$this)

Bin	$\mathcal{W}_1$	$\mathcal{W}_2$	$\mathcal{W}_3$	$\mathcal{W}_{4a}$	$\mathcal{W}_{4b}$
1	1.000	1.000	0.987	0.987	0.927
10	1.000	1.000	0.962	0.962	0.897
19	1.000	1.000	0.299	0.036	0.027
Pooled	1.000	1.000	0.968	0.964	0.901

Table 23: Sale-rate loss from removing each nomination channel (drop in the fraction of auctions with ≥ 1 bidder)

Bin	Mech (arrival) $\mathcal{W}_{4a}-\mathcal{W}_{4b}$	Mech (nominator) $\mathcal{W}_3-\mathcal{W}_{4a}$	Screening $\mathcal{W}_2-\mathcal{W}_3$	Thickening $\mathcal{W}_1-\mathcal{W}_2$	Total $\mathcal{W}_1-\mathcal{W}_{4b}$
1	0.060	0.000	0.013	0.000	0.073
10	0.065	0.000	0.038	0.000	0.103
19	0.009	0.263	0.701	0.000	0.973
Pooled	0.063	0.004	0.032	0.000	0.099

Each entry is the fall in the sale rate when the indicated nomination channel is switched off, i.e. the successive differences of Table 22; columns sum to the total ($\mathcal{W}_1 - \mathcal{W}_{4b}$). In the expensive bins (1, 10), the small sale-rate loss is led by the arrival guarantee; in the cheapest bin (19) screening on value dominates (0.70), because a nominator's $v \geq v^* \geq r$ is guaranteed to clear the reserve.

C.4 Value of Nomination: Full Per-Bin Tables

Tables below give the four effects (thickening, screening, guaranteed bidder, mechanical existence) and the total for all 19 bins, by revenue, bidders, and winning bid, plus the sale rate by world.

Table 24: Revenue per sold auction ($), all bins

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	4341 (871.3)	0 (0)	713.2 (2996)	0 (0)	5054 (2858)
2	3162 (644.5)	0 (0)	748.1 (2252)	0 (0)	3910 (2155)
3	2988 (625.8)	0 (0)	662.8 (1935)	0 (0)	3651 (1812)
4	2633 (555.6)	0 (0)	685.9 (1800)	0 (0)	3318 (1710)
5	2340 (481.1)	0 (0)	631.5 (1539)	0 (0)	2972 (1473)
6	1966 (411)	0 (0)	705 (1349)	0 (0)	2671 (1286)
7	1607 (335.7)	0 (0)	533.7 (1125)	0 (0)	2141 (1061)
8	1406 (289.2)	0 (0)	578.4 (1017)	0 (0)	1985 (967.5)
9	1190 (245.8)	0 (0)	518.8 (832.8)	0 (0)	1709 (805)
10	1129 (235.4)	0 (0)	435.8 (780.1)	0 (0)	1565 (738.7)
11	1008 (216)	0 (0)	700 (752.1)	0 (0)	1708 (724.3)
12	827.1 (169.5)	0 (0)	296.7 (574.4)	0 (0)	1124 (545.4)
13	714.9 (140.2)	0 (0)	241.2 (496.5)	0 (0)	956.1 (477.5)
14	714.2 (150.7)	0 (0)	356.3 (525.1)	0 (0)	1071 (501)
15	568.9 (131.8)	74.72 (51.35)	340.8 (428.3)	44.51 (30.73)	1029 (405.3)
16	475.5 (121.9)	115 (62.68)	281.6 (384.6)	38.72 (30.31)	910.9 (370.8)
17	261.3 (89.42)	180.2 (79.28)	130.3 (249.4)	21.78 (18.43)	593.5 (244.3)
18	163.2 (66.59)	182.3 (76.23)	64.14 (182)	5.434 (10.22)	415.1 (181.9)
19	6.081 (36.62)	25 (56.35)	2.688 (36.7)	82.72 (20.15)	116.5 (55.37)
Pooled	2453 (212.5)	5.031 (1.265)	631.4 (727.8)	1.372 (0.5222)	3090 (697)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 25: Bidders per sold auction, all bins

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	0.78 (0.03)	0.00 (0.00)	0.05 (0.08)	0.00 (0.00)	0.83 (0.08)
2	0.77 (0.03)	0.00 (0.00)	0.08 (0.09)	0.00 (0.00)	0.85 (0.08)
3	0.77 (0.03)	0.00 (0.00)	0.07 (0.09)	0.00 (0.00)	0.84 (0.09)
4	0.76 (0.03)	0.00 (0.00)	0.08 (0.08)	0.00 (0.00)	0.84 (0.08)
5	0.76 (0.03)	0.00 (0.00)	0.09 (0.09)	0.00 (0.00)	0.84 (0.08)
6	0.74 (0.03)	0.00 (0.00)	0.11 (0.09)	0.00 (0.00)	0.85 (0.09)
7	0.74 (0.03)	0.00 (0.00)	0.11 (0.09)	0.00 (0.00)	0.85 (0.08)
8	0.73 (0.03)	0.00 (0.00)	0.13 (0.08)	0.00 (0.00)	0.85 (0.08)
9	0.72 (0.03)	0.00 (0.00)	0.13 (0.09)	0.00 (0.00)	0.85 (0.08)
10	0.73 (0.03)	0.00 (0.00)	0.12 (0.09)	0.00 (0.00)	0.85 (0.08)
11	0.67 (0.03)	0.00 (0.00)	0.20 (0.09)	0.00 (0.00)	0.87 (0.09)
12	0.74 (0.03)	0.00 (0.00)	0.11 (0.09)	0.00 (0.00)	0.85 (0.08)
13	0.75 (0.03)	0.00 (0.00)	0.10 (0.09)	0.00 (0.00)	0.85 (0.09)
14	0.71 (0.03)	0.00 (0.00)	0.15 (0.09)	0.00 (0.00)	0.85 (0.09)
15	0.57 (0.03)	0.08 (0.01)	0.15 (0.08)	0.05 (0.01)	0.85 (0.08)
16	0.49 (0.03)	0.12 (0.02)	0.13 (0.08)	0.05 (0.01)	0.79 (0.07)
17	0.28 (0.03)	0.20 (0.03)	0.07 (0.06)	0.02 (0.01)	0.57 (0.06)
18	0.20 (0.02)	0.21 (0.03)	0.03 (0.05)	0.01 (0.00)	0.45 (0.05)
19	0.01 (0.03)	0.02 (0.04)	0.00 (0.02)	0.11 (0.01)	0.14 (0.04)
Pooled	0.74 (0.01)	0.01 (0.00)	0.09 (0.03)	0.00 (0.00)	0.84 (0.03)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 26: Winning bid per sold auction ($), all bins

Bin	Mech (arrival)	Mech (nominator)	Screening	Thickening	Total
1	4175 (1270)	0 (0)	1379 (5097)	0 (0)	5555 (5022)
2	3141 (891.3)	0 (0)	1443 (3784)	0 (0)	4584 (3739)
3	2781 (830.1)	0 (0)	1280 (3208)	0 (0)	4061 (3167)
4	2439 (755.7)	0 (0)	1324 (3019)	0 (0)	3762 (2956)
5	2217 (676.2)	0 (0)	1216 (2602)	0 (0)	3433 (2540)
6	1868 (577.9)	0 (0)	1346 (2242)	0 (0)	3213 (2178)
7	1452 (447.2)	0 (0)	1026 (1916)	0 (0)	2477 (1876)
8	1312 (415.4)	0 (0)	1119 (1693)	0 (0)	2431 (1660)
9	1205 (383.8)	0 (0)	1003 (1439)	0 (0)	2208 (1418)
10	1043 (322.9)	0 (0)	836.2 (1303)	0 (0)	1879 (1289)
11	906.9 (306.1)	0 (0)	1344 (1247)	0 (0)	2251 (1243)
12	787 (249.1)	0 (0)	576.2 (985.8)	0 (0)	1363 (969.4)
13	708.6 (221)	0 (0)	466.9 (847.7)	0 (0)	1175 (833.9)
14	666.8 (212.1)	0 (0)	683.7 (886.8)	0 (0)	1350 (881.6)
15	508.9 (185.7)	69.51 (76.5)	676.9 (725.4)	44.34 (36.38)	1300 (714.4)
16	422 (185.6)	101.5 (97.17)	579.9 (686.3)	37.91 (29.67)	1141 (683.9)
17	192.8 (146.7)	151.4 (132.1)	315 (468.6)	21.12 (25.93)	680.3 (497.6)
18	164.7 (141.9)	141.9 (132.6)	180.6 (374.8)	3.361 (5.845)	490.6 (399.3)
19	-16.75 (253.1)	76.24 (441.2)	52.19 (189.1)	75.78 (23.09)	187.5 (553.1)
Pooled	2352 (302.8)	4.418 (2.425)	1218 (1236)	1.321 (0.6051)	3576 (1214)

Entry: point estimate (bootstrap s.e.), per sold auction.

Table 27: Sale rate by world, all bins

Bin	$\mathcal{W}_1$	$\mathcal{W}_2$	$\mathcal{W}_3$	$\mathcal{W}_{4a}$	$\mathcal{W}_{4b}$
1	1.000	1.000	0.987	0.987	0.927
2	1.000	1.000	0.979	0.979	0.920
3	1.000	1.000	0.980	0.980	0.919
4	1.000	1.000	0.976	0.976	0.913
5	1.000	1.000	0.974	0.974	0.912
6	1.000	1.000	0.962	0.962	0.899
7	1.000	1.000	0.966	0.966	0.904
8	1.000	1.000	0.957	0.957	0.893
9	1.000	1.000	0.957	0.957	0.891
10	1.000	1.000	0.962	0.962	0.897
11	1.000	1.000	0.909	0.909	0.841
12	1.000	1.000	0.967	0.967	0.904
13	1.000	1.000	0.973	0.973	0.910
14	1.000	1.000	0.946	0.946	0.878
15	1.000	1.000	0.919	0.907	0.829
16	1.000	1.000	0.916	0.890	0.801
17	1.000	1.000	0.905	0.797	0.685
18	1.000	1.000	0.934	0.741	0.621
19	1.000	1.000	0.299	0.036	0.027
Pooled	1.000	1.000	0.968	0.964	0.901

D Theory and Proofs

D.1 Positive Correlation Details

Formal definition. Under affiliated private values (APV), the joint density f of H_V satisfies

$$f(\mathbf{v} \vee \mathbf{v}')\, f(\mathbf{v} \wedge \mathbf{v}') \;\geq\; f(\mathbf{v})\, f(\mathbf{v}') \qquad \text{for all valuation vectors } \mathbf{v}, \mathbf{v}',$$

where $\vee$ and $\wedge$ denote coordinate-wise maximum and minimum. The parametric form of H_V adopted in Section 5 generates joint distributions satisfying this property [Nelsen, 2006, Müller and Scarsini, 2005], ensuring the empirical findings are consistent with the theoretical framework underlying equilibrium existence and revenue ranking results in auction theory [Castro, 2007].

Posterior stochastic dominance. The stochastic dominance result in equation (2) is a consequence of this affiliation property: for affiliated random variables,

$$\Pr(V_i > v \mid V_j \geq v^*,\, j = 1, \ldots, m) \;\geq\; \Pr(V_i > v) \qquad \text{for all } v$$

[Milgrom and Weber, 1982]. This holds for *any* H_V satisfying **positive correlation**, not only for the Gaussian copula parameterization adopted in Section 5. Therefore, the equilibrium characterization in Section 4.4 does not rely on the specific copula choice.

D.2 Robustness to Noisy Nomination

The main text models nomination as a deterministic threshold rule: $n(v) = \mathbb{I}\{v \geq v^*\}$. Here I show the framework is robust to mixed or noisy nomination strategies.

Suppose a positive fraction $p \in (0, 1)$ of nominations are uninformative: a bidder with realized value below v^* randomizes and nominates with probability p, independent of her realized value (a bidder above v^* still nominates with probability 1, as in the main text). This dilutes the informational content of a nomination. The non-nominator's effective screening probability – the probability that a nominator's value actually clears v^*, given that she nominated – becomes

$$\pi \;\equiv\; \Pr(v \geq v^* \mid \text{nominate}) \;=\; \frac{1 - F_V(v^*)}{1 - F_V(v^*) + p\, F_V(v^*)},$$

rather than $\pi = 1$ under the pure-strategy rule of the main text.

D.3 Posterior Updating of the Non-Nominator's Value Distribution

This section derives the mechanism through which observing m nominators above threshold v^* shifts the non-nominator's own value distribution upward, as claimed in Section 4.3.1. The mechanism operates through the one-factor representation of the Gaussian equicorrelation copula. Each normal score $Z_i = \Phi^{-1}(U_i)$ decomposes as

$$Z_i = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_i, \qquad W, \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where W is a shared common factor and ε_i is idiosyncratic. Conditional on $W = w$, the Z_i are independent with $Z_i \mid W = w \sim \mathcal{N}(\sqrt{\rho}\, w,\, 1-\rho)$, so each nominator's threshold-crossing probability is $P(Z_i > z^* \mid W = w) = 1 - \Phi\left(\frac{z^* - \sqrt{\rho}\, w}{\sqrt{1-\rho}}\right)$, where $z^* = \Phi^{-1}(u^*)$. Observing m threshold-crossings updates the posterior on W via Bayes' theorem:

$$p(w \mid m \text{ crossings}) \;\propto\; \phi(w) \cdot \left[1 - \Phi\left(\frac{z^* - \sqrt{\rho}\, w}{\sqrt{1-\rho}}\right)\right]^m.$$

This posterior shifts rightward as m increases — each additional nominator above z^* is independent evidence that W is high. Since the non-nominator's own normal score satisfies $Z_0 = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_0$, the rightward shift in $p(w \mid m)$ stochastically increases Z_0, and therefore $U_0 = \Phi(Z_0)$ and the non-nominator's value $V_0 = F_V^{-1}(U_0)$.

D.4 Corner Equilibrium Cases

This appendix presents the two corner-solution equilibrium cases that arise in the nomination game when the interior-solution condition fails. Recall the definitions (equations 14–15):

$$\Delta_0 = A - \phi\big[1 - F_V(v^*)\big] - B \quad \text{(value difference at zero cost)}$$
$$\Delta_{l=2} = U_{l=1}(c_H^*) \quad \text{(value of early learning at the non-nominator cutoff)}$$

where $A = \sum_{m=K-1}^{\infty} \Pr(M_{-i}{=}m)\, S_{l=1}(m)$ is the expected gross auction surplus from early learning and $B = \Pr(M \geq K)\, \mathbb{E}[S_{l=2}(m) \mid M \geq K]$ is the expected gross surplus from late-learning.

Case 2: Corner Solution (All Wait and Learn Later)

Condition: $\Delta_0 < 0$ and $\Delta_{l=2} < 0$.

When $\Delta_0 < 0$, the value of early learning at zero cost falls below the value of late-learning: $U_{l=1}(0) < U_{l=2}(0)$. Combined with $\Delta_{l=2} < 0$ (late-learning is still preferable even at the non-nominator cutoff), the function $U_{l=1}(c_i) - U_{l=2}(c_i)$ is strictly negative for all $c_i \in [0, c_H^*]$. There is therefore no interior crossing point, and no bidder finds it optimal to nominate.

Formally, $\pi_0^{l=1} = 0$ and the equilibrium degenerates to a mass of non-nominators (those with $c_i \leq c_H^*$) and exits. This outcome is a *coordination failure*: each potential nominator would prefer to wait for someone else to incur the nomination cost and fee, and in equilibrium no one does.[44]

As shown in the center panel of Figure 4, the $U_{l=1}$ intercept lies below the $U_{l=2}$ intercept ($A - \phi[1 - F_V(v^*)] < B$), so $U_{l=2}(c_i) > U_{l=1}(c_i)$ for all c_i in the relevant range.

Case 3: Corner Solution (All Learn Early)

Condition: $\Delta_{l=2} \geq 0$.

When $\Delta_{l=2} = U_{l=1}(c_H^*) \geq 0$, the value of early learning remains non-negative at the non-nominator cutoff c_H^*. Since $U_{l=1}$ is weakly decreasing in c_i and $U_{l=2}(c_H^*) = 0$ by definition of c_H^*, this means $U_{l=1}(c_i) \geq U_{l=2}(c_i)$ for all $c_i \in [0, c_H^*]$. No bidder prefers late-learning over early learning within the participation range.

In this case the equilibrium threshold is defined by $U_{l=1}(c_L^*) = 0$ (where the early learning value just breaks even, not where it crosses the late-learning value). Any bidder with $c_i < c_L^*$ learns early; any bidder with $c_i > c_L^*$ exits. The late-learning strategy is dominated and $\pi_0^{l=2} = 0$.

The intuition is that either the nomination fee is negligible relative to the expected auction surplus (low ϕ) or the pivotal probability is very high, making the nomination benefit large enough that all participating bidders prefer to bear the cost and nominate outright. As shown in the right panel of Figure 4, the $U_{l=1}$ line lies above $U_{l=2}$ everywhere in the feasible cost range.

Empirical relevance. In the ND data, both nominators and non-nominators are observed in every auction season, consistent with the Case 1 interior solution. Cases 2 and 3 are not observed in the data but are relevant for counterfactual policy analysis: a sufficiently large increase in the nomination fee ϕ can push the equilibrium from Case 1 into Case 2, eliminating nominations entirely.

[44] The pivotal probability $\Pr(M_{-i} = K - 1)$ is the probability that a given bidder's nomination is decisive for triggering the auction. When this probability is low (many other nominators expected) or the fee ϕ is high relative to the nomination benefit, $\Delta_0 < 0$ and the coordination failure arises.

D.5 Proofs for Section 4.4

D.6 Proof of Lemma 1: Poisson Arrivals of Nominators

Proof. Let $N \sim \text{Poisson}(\Lambda)$ denote the total number of arriving bidders, where $\Lambda = \lambda_{pre}\cdot$. Each arriving bidder independently becomes a competing nominator with probability $p \equiv \pi_0^{l=1} \cdot [1 - F_V(v^*)]$, reflecting two independent thinning events: the bidder chooses the early learning ($l = 1$) strategy with probability $\pi_0^{l=1}$, and conditional on that, has a value exceeding the threshold v^* with probability $[1 - F_V(v^*)]$.

I derive the distribution of M_{-i}, the number of competing nominators excluding bidder i, directly. Conditioning on the total number of arrivals $N = j$ gives:

$$\Pr(M_{-i} = m) = \sum_{j=m}^{\infty} \Pr(M_{-i} = m \mid N = j)\ \Pr(N = j) \tag{72}$$

$$= \sum_{j=m}^{\infty} \binom{j}{m} p^m (1-p)^{j-m} \cdot \frac{e^{-\Lambda}\Lambda^j}{j!} \tag{73}$$

where the binomial factor $\binom{j}{m} p^m (1-p)^{j-m}$ is the probability that exactly m of the j arrivals become competing nominators. Rearranging terms,

$$\Pr(M_{-i} = m) = \sum_{j=m}^{\infty} \frac{j!}{m!\,(j-m)!} \cdot p^m (1-p)^{j-m} \cdot \frac{e^{-\Lambda}\Lambda^j}{j!} \tag{74}$$

$$= \frac{e^{-\Lambda} p^m}{m!} \sum_{j=m}^{\infty} \frac{\Lambda^j (1-p)^{j-m}}{(j-m)!} \tag{75}$$

Substituting $k = j - m$,

$$\Pr(M_{-i} = m) = \frac{e^{-\Lambda} p^m \Lambda^m}{m!} \sum_{k=0}^{\infty} \frac{[\Lambda(1-p)]^k}{k!} \tag{76}$$

$$= \frac{e^{-\Lambda}(\Lambda p)^m}{m!} \cdot e^{\Lambda(1-p)} \tag{77}$$

$$= \frac{e^{-\Lambda p}(\Lambda p)^m}{m!} \tag{78}$$

where the Taylor expansion $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$ is used and combines $e^{-\Lambda} \cdot e^{\Lambda(1-p)} = e^{-\Lambda p}$. This is the PMF of Poisson(Λp), so

$$M_{-i} \sim \text{Poisson}(\Lambda p) = \text{Poisson}\big(\lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - F_V(v^*)]\big)\,. \quad \square$$

□

D.7 Proof of Proposition 1: Single Crossing

Proof. Both value functions are linear in c_i:

$$U_{l=1}(c_i) = A - \phi\big[1 - F_V(v^*)\big] - c_i \quad \text{with slope } -1 \tag{79}$$

$$U_{l=2}(c_i) = B - \Pr(M \geq K) \cdot c_i \quad \text{with slope } -\Pr(M \geq K) \tag{80}$$

where $A = \sum_{m=K-1}^{\infty} \Pr(M_{-i} = m) S_{l=1}(m)$ is the expected gross auction surplus for an early learner, $B = \Pr(M \geq K) \cdot \mathbb{E}[S_{l=2}(m) | M \geq K]$ is the expected gross surplus for a non-nominator, and $\phi[1 - F_V(v^*)]$ is the expected nomination fee paid with probability $1 - F_V(v^*)$.

Since $\Pr(M \geq K) < 1$,

$$|dU_{l=1}/dc_i| = 1 > \Pr(M \geq K) = |dU_{l=2}/dc_i|. \tag{81}$$

Two lines with different slopes can intersect at most once. If they intersect at c_L^*, then

$$\begin{cases} U_{l=1}(c_i) > U_{l=2}(c_i) & \text{(learn early) if } c_i < c_L^* \\ U_{l=1}(c_i) < U_{l=2}(c_i) & \text{(learn later) if } c_i > c_L^* \end{cases}$$

□

D.8 Numerical Implementation Details for Section B.4

I implement several numerical techniques to ensure stable computation across the parameter space. First, I compute Poisson probabilities in log space to avoid overflow for large n or λ

$$\log P(N = n) = -\lambda + n \log \lambda - \sum_{i=1}^{n} \log i \tag{82}$$

and then exponentiate only when summing across different values of n. This allows us to handle cases with expected numbers of competitors as large as 20 or more without numerical issues.

Second, when computing the copula density, I handle extreme values of the CDF. I clamp probabilities to the range $[\epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-10}$ before taking logarithms, and I check for numerical overflow in the copula density formula, capping density values at 10^{10} to prevent propagation of numerical errors.

Third, for the numerical integration in equation (2), I use the trapezoidal rule with 500 evaluation points, which I verified provides sufficient accuracy by comparing with higher-resolution grids. I also normalize the conditional density to integrate to one over its support before computing the expected surplus, which ensures that any small errors in the density estimation do not compound during integration.

D.9 Joint GMM Estimation of F_V and ρ

Method. As a robustness check I re-estimate the copula parameter using a joint GMM estimator ('Joint GMM Strategy') that avoids the two-step structure of pairwise pseudo-MLE. The objective is

$$(\hat{\beta},\, \hat{\rho}) = \arg\min_{\beta,\,\rho} \sum_{i \in \mathcal{G}} \Big[G_k\big(B(x_i;\beta);\, k_i,\, N_i,\, \rho\big) - \hat{U}^{(r_i)}_{(k_i)} \Big]^2, \tag{83}$$

where $B(x;\beta)$ is a monotone B-spline (10 internal knots, degree 3) with coefficient vector β constrained to be non-decreasing, G_k is the copula order-statistic CDF (Equation 47), $\hat{U}$ are Weibull empirical ranks, and $\mathcal{G}$ is a 200-point evenly-spaced x-grid selected from the training bids. The B-spline basis is built on all training bids; only the SSR evaluation is restricted to the 200-point grid to reduce computation.

After minimising (83), the lognormal parameters $(\hat{\mu}, \hat{\sigma})$ are initialized by NLS: fitting $\Phi\big((\log x - \mu)/\sigma\big)$ to the spline values $B(x_i;\hat{\beta})$ for all training bids x_i. This provides a smooth, parametric extrapolation of F_V below v^*, but is then refined by the truncation-corrected lognormal MLE described in Appendix B.3 (Eq. 51); the MLE pair $(\hat{\mu}, \hat{\sigma}) = (0.032,\, 1.614)$

is the one used downstream.

Estimation uses 100 random 90/10 train/test splits of the 728 unviolated auctions. The test-set RMSE for each split is computed by evaluating the copula diagonal $D(\hat{F}_V(x_{\max});\, N_r{-}1,\, \hat{\rho})$ for each test-auction maximum bid $x_{\max}$ in an N_r-bidder auction, then comparing to the empirical Weibull rank of $x_{\max}$ among all observed winner bids.

Grid200 results. Table 28 reports the Joint GMM Strategy estimates for the full sample ($N_r \in \{2, \ldots, 11\}$, 728 auctions) that is used as the main estimation method in the main text.

Table 28: Joint GMM Strategy estimates: joint estimation of F_V (monotone B-spline) and ρ by SSR minimisation (200-point moment grid, 100 random 90/10 splits). RMSE is evaluated by applying the copula diagonal $D(\hat{F}_V(x_{\max}),\, N_r{-}1,\, \hat{\rho})$ to test-set maximum bids. Mean $\pm$ SD across 100 splits. Bold row: selected specification (lowest RMSE).

Copula	$\hat{\rho}$ (mean $\pm$ SD)	**RMSE** (mean $\pm$ SD)
Gaussian ($\hat{\rho}$)	**0.866 $\pm$ 0.096**	**0.183 $\pm$ 0.037**
Gumbel	2.404 ± 0.661	0.210 ± 0.036
Clayton	17.31 ± 4.71	0.212 ± 0.066
Independence	—	0.364 ± 0.043

Full-data fit (Gaussian, 728 auctions): $\hat{\rho} = 0.8605$. Lognormal tail: NLS stage $(\hat{\mu}, \hat{\sigma}) = (-0.715,\, 3.863)$, refined by truncation-corrected MLE (Eq. 51) to (0.032, 1.614) (final; used downstream).

The Gaussian copula achieves the lowest RMSE (0.183 ± 0.037). Independence is clearly dominated (RMSE 0.364, 2.0$\times$ larger), confirming that positive dependence is a significant feature of the bid distribution. Clayton is unstable ($\hat{\rho} = 17.31 \pm 4.71$, near the upper bound of 20), which reflects boundary behavior rather than well-identified upper-tail dependence.

Monte Carlo evidence for median aggregation. This section provide formal evidence to why I use median rather than the mean across the 100 train/test splits as main specification of the correlation parameter, I run a Monte Carlo study in which I simulate fresh datasets from the correctly-specified model across all three copula families and apply the same Joint GMM estimator, then compare the mean and median as aggregation strategies across replications. I document that under Gumbel and Clayton dependence structures, the distribution of the aggregated $\hat{\rho}$ is right-skewed shape, providing justification to why median is a better predictor of the true correlation parameter.

Design. For each of six DGP cells (Gumbel $\rho_0 \in \{1.5,\, 4.0\}$, Clayton $\rho_0 \in \{0.5,\, 3.0\}$, Gaussian $\rho_0 \in \{0.3,\, 0.8\}$) crossed with six lognormal marginal variants ($\mu \in \{-0.5,\, 0.0,\, 0.5\}$,

$\sigma \in \{0.3, 0.7\}$), I draw $R = 100$ independent datasets for each auction-count level $S \in \{50, 100, 700\}$. Each dataset consists of S auctions with $N_s \sim \text{Uniform}\{2, \ldots, 11\}$ bidders and one observed bid vector per auction (mimicking the single-observation-per-auction structure of the empirical data). The Gaussian copula objective uses a 20-node Gauss-Hermite quadrature approximation to evaluate the copula diagonal. I apply the same Joint GMM procedure to each dataset and record $\hat{\rho}^{(r)}$ for $r = 1, \ldots, 100$.

The key measure for comparison is the *normalised deviation* $\delta^{(r)} = (\hat{\rho}^{(r)} - \rho_0)/\rho_0$, and I compareOk bu these two different aggregation method across $r = 1, \ldots, 100$ datasets: mean($= \bar{\rho} = \frac{1}{100}\sum_r \hat{\rho}^{(r)}$) and median($= \tilde{\rho} = \text{median}(\hat{\rho}^{(1)}, \ldots, \hat{\rho}^{(100)})$).

Note that the root mean squared error RMSE $= \sqrt{\frac{1}{100}\sum_i (\delta^{(i)})^2}$ characterizes the *full distribution* of individual estimates and is the same regardless of which aggregator is reported.

Results. Figure 13 shows the RMSE of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ as a function of S for each copula family and dependence level. All six DGP cells converge toward zero as S grows, confirming consistency of the Joint GMM estimator. The Gaussian high cell ($\rho_0 = 0.8$) converges fastest, while Clayton low ($\rho_0 = 0.5$) and Gaussian low ($\rho_0 = 0.3$) show the highest RMSE at $S = 700$ (25% and 38% respectively), reflecting the flat GMM objective surface at weak dependence.[45]

Table 29 summarises the bias of each aggregator, pooled over the six lognormal marginal variants (600 replications per row).

Interpretation. Four findings emerge from Table 29 and Figures 13–14.

First, for Gumbel and Clayton families the median aggregator is consistently less biased than the mean at every sample size. At $S = 50$, the mean–median gap reaches 35 percentage points for Gumbel high dependence (+109% vs. +74%) and 18 points for Clayton high (+109% vs. +91%); by $S = 700$ the gap shrinks to $\lesssim 5$ points as the objective surface concentrates and optimizer outliers vanish. This pattern is the hallmark of a right-skewed estimator distribution driven by occasional L-BFGS-B convergence to high local minima, not by systematic model misspecification.

Second, the Gaussian copula behaves differently across the two dependence regimes. For strong dependence ($\rho_0 = 0.8$), the estimator is remarkably well-concentrated even at $S = 50$: mean and median bias are both around +16% with RMSE of 16%. By $S = 700$, bias falls to +3% and RMSE to 4% — the tightest convergence of any copula family. The bounded

[45] For all three copula families, the copula diagonal evaluates to the independence case $C_r(u, \ldots, u) = u^r$ as the dependence parameter approaches its independence limit. Since the moment conditions $G_k(u; N, \rho)$ are built from sums of copula diagonals, it follows that $\partial G_k/\partial\rho \to 0$ as dependence weakens: the moment conditions stop responding to ρ near independence, making the GMM objective surface flat in the ρ direction.

Table 29: Monte Carlo bias of mean and median $\hat{\rho}$ aggregators under the Joint GMM estimator, pooled over six lognormal marginal variants (600 replications per row). Bias is $(\hat{\rho} - \rho_0)/\rho_0 \times 100\%$. RMSE is the root mean squared normalised deviation of the *individual* $\hat{\rho}$ draws — it is the same regardless of which aggregator is reported. Gaussian results use the vectorised 20-node Gauss-Hermite evaluator (`strategy3_sim_validation_fast.py`).

DGP	ρ_0	S	**Mean bias**	**Median bias**	**RMSE**
Gumbel low	1.5	50	+57%	+50%	66%
		100	+28%	+25%	32%
		700	+5%	+5%	8%
Gumbel high	4.0	50	+109%	+74%	148%
		100	+48%	+36%	75%
		700	+14%	+10%	24%
Clayton low	0.5	50	+211%	+208%	229%
		100	+112%	+105%	126%
		700	+17%	+16%	25%
Clayton high	3.0	50	+109%	+91%	134%
		100	+60%	+54%	76%
		700	+11%	+9%	20%
Gaussian low	0.3	50	+110%	+110%	114%
		100	+65%	+67%	71%
		700	+10%	+15%	38%
Gaussian high	0.8	50	+16%	+16%	16%
		100	+9%	+10%	11%
		700	+3%	+2%	4%

parameter space $\rho \in (-1, 1)$ prevents the runaway outliers seen in Gumbel and Clayton, so mean and median agree. For weak dependence ($\rho_0 = 0.3$), the Gaussian objective is flatter: RMSE remains elevated at $S = 700$ (38%), and the mean–median gap is *reversed* (mean +10%, median +15%), consistent with occasional near-boundary estimates at $\hat{\rho} \approx 0$ pulling the mean below the median rather than above it.

Third, the estimator converges to the truth as S grows across all families. By $S = 700$ (the actual sample size in this paper), Gumbel and Clayton cells show median bias of at most 16% and RMSE of at most 25%, and Gaussian high shows median bias of 2% with RMSE of 4%.

Fourth, the empirical application uses the Gaussian copula family with an estimated $\hat{\rho} = 0.861$, placing it in the "Gaussian high" regime ($\rho_0 = 0.8$). In that regime, mean and median are nearly identical at $S = 700$ and both provide low-bias estimates. The residual mean–median gap observed in the 100 train/test splits (mean 0.866, median 0.861, SD 0.096) therefore likely reflects a combination of (i) data heterogeneity across splits and (ii) a modest right-skew from splits with fewer unviolated auctions, where the $\hat{\rho}$ distribution behaves more

like the small-S case. The **median** $\hat{\rho} = 0.861$ is the preferred point estimate as it is more robust to this residual skew.

D.10 Implementation: Handling Mixed Competition

I compute $\tilde{\Pi}(v; m, \Lambda, \rho)$, where m is the deterministic component (known competitors) and Λ is the stochastic component that captures the random variable $n \sim \text{Poisson}(\Lambda)$. First, for m known competitors and n potential competition, I determine 90th percentile support ($n_{\max} = \text{poissinv}(0.90, \Lambda) + 3\sqrt{\Lambda}$) and compute Poisson probabilities as $p_n = \frac{e^{-\Lambda}\Lambda^n}{n!}$ for $n \in \{0, ..., n_{\max}\}$. This is then normalized as $p_n \leftarrow p_n / \sum p_n$. Next, for each n, I compute surplus with $m+n$ total competitors and take expectation as $\tilde{\Pi}(v; m, \Lambda, \rho) = \sum_n p_n \cdot \text{surplus}(v, m+n, \rho)$.

Simulation of Gaussian Equicorrelation Copula

I estimate the conditional density $f_{V_{(n:n)}|V_0}(y \mid v, \hat{\rho})$ via Monte Carlo simulation from the $(n+1)$-dimensional Gaussian equicorrelation copula. The simulation exploits the one-factor ("compound symmetric") structure.

Simulation algorithm.

1. **Generate common factor:** Draw $W \sim N(0,1)$ independently for each simulation replicate. This shared factor induces equicorrelation across all dimensions.

2. **Generate idiosyncratic shocks:** Draw $\varepsilon_j \overset{\text{i.i.d.}}{\sim} N(0,1)$, $j = 1, \ldots, n+1$, independent of W.

3. **Construct correlated normals:** $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$. By construction, $\text{Cov}(Z_j, Z_k) = \hat{\rho}$ for $j \neq k$.

4. **Transform to uniform marginals:** $U_j = \Phi(Z_j)$, where Φ is the standard normal CDF. The vector $(U_1, \ldots, U_{n+1})$ follows the Gaussian equicorrelation copula with parameter $\hat{\rho}$.

5. **Transform to bid scale:** $V_j = F_V^{-1}(U_j)$ for $j = 1$ (own value, full distribution) and $V_j = F_{V|\text{trunc}}^{-1}(U_j)$ for $j = 2, \ldots, n+1$ (competitors, truncated distribution $F_V(\cdot \mid v \geq v^*)$).

6. **Record maximum competitor bid:** $V_{(n:n)} = \max\{V_2, \ldots, V_{n+1}\}$.

7. **Conditional sampling:** Retain draws where the own bid $|U_1 - F_V(v)| < \varepsilon$ (adaptive tolerance $\varepsilon = 0.02$, doubled if fewer than 100 draws survive).

8. **Kernel density estimation:** Estimate $\hat{f}_{V_{(n:n)}|V_0}(y \mid v)$ from retained $\{V_{(n:n)}^{(j)}\}$ via Gaussian KDE with Silverman bandwidth. Normalize to integrate to one over $[v^*, v]$.

E Toy Model and Supplementary Analysis

E.1 Simplified Nomination Model: Coordination Without Late-Learners

This appendix presents a simplified version of the nomination game that abstracts from value uncertainty and strategic late-learning. The purpose is to isolate the pure coordination problem inherent in the nomination threshold and to provide geometric intuition for equilibrium multiplicity. The full structural model (Section 7) adds learning costs, endogenous late-learning, and Poisson arrival rates, which substantially amplify coordination failure relative to the analysis here.

E.2 Simplifications Relative to Full Model

The following simplifications transform the two-stage game with early learning ($l = 1$) to nominate and late-learning into a one-shot coordination game where the only decision is whether to pay cost c_i to nominate a parcel of known value v_i.

Assumption 1: No Value Uncertainty. All bidders know their private values v_i ex ante, eliminating learning costs and the late-learning motive. This allows us to focus purely on the coordination problem inherent in the nomination threshold.

Assumption 2: Fixed Competition. The number of potential bidders N is known and fixed (not Poisson-distributed). This removes uncertainty about the number of competitors and isolates the strategic interaction in nomination decisions.

Assumption 3: Nomination Costs Only. Bidders face only nomination costs $c_i \sim \text{Gamma}(\theta_{c1}, \theta_{c2})$, which rationalize the observed low nomination rates $\frac{M_t}{N_t}$ despite the absence of value uncertainty.

These simplifications transform the model from a complex game with sequential exploration and entry decisions into a cleaner coordination game where the trade-off between auction occurrence (extensive margin) and revenue conditional on occurrence (intensive margin) is transparent.

E.3 Nomination Decision and Equilibrium

A bidder with valuation v_i and nomination cost c_i nominates if and only if

$$h(\pi_0; K, N) \cdot \Delta(v_i; N) \geq c_i \tag{84}$$

where $h(\pi_0; K, N) = \binom{N-1}{K-1}\pi_0^{K-1}(1-\pi_0)^{N-K}$ is the probability of being pivotal and $\Delta(v_i; N) = \int_0^{v_i} F_V(y)^{N-1}dy$ is the expected surplus from a second-price auction with $N-1$ competitors. The symmetric equilibrium nomination probability π_0^* satisfies

$$\pi_0^* = \int_{\underline{v}}^{\overline{v}} F_C\left(h(\pi_0^*; K, N) \cdot \Delta(v; N)\right) f_V(v)\, dv. \tag{85}$$

When $K = 1$, nominations are strategic substitutes: $h(\pi_0; 1, N) = (1-\pi_0)^{N-1}$ is strictly decreasing, ensuring a unique equilibrium. When $K \geq 2$, nominations become strategic complements and multiple equilibria, including a coordination-failure equilibrium at $\pi_0^* = 0$, can arise (Figure 15).

E.4 Estimation of Cost Distribution

I estimate the nomination cost distribution $F_C(c; \theta_c)$ using a Gamma specification

$$F_C(c; \alpha, \beta) = \frac{\gamma(\alpha, c/\beta)}{\Gamma(\alpha)} \tag{86}$$

where $\theta_c = (\alpha, \beta)$ are shape and rate parameters. Using the observed nomination rate $\bar{\pi}_{\text{obs}}$ and the variance of nomination rates, I form two moment conditions and estimate via GMM:

$$\hat{\theta}_c = \arg\min_{\theta_c \in \Theta_c} \mathbf{g}(\theta_c)' \mathbf{W} \mathbf{g}(\theta_c). \tag{87}$$

Table 30 presents the estimates.

E.5 Counterfactual: Nomination Threshold K

Table 31 presents counterfactual results varying $K \in \{1, 2, 3\}$ at mean $N = 4$ bidders. In this simplified model (no late-learning), $K = 2$ produces a reduced but positive trigger rate and doubles conditional revenue through selection. $K = 3$ produces complete coordination failure. The full structural model (Section 7.1) finds complete failure already at $K = 2$, because late-learning compounds the coordination problem.

Table 30: Toy Model: Estimated Parameters

Parameter	**Estimate**	**Std. Error**
Shape (α)	0.82	(0.09)
Scale (β)	34.44	(3.12)
Implied Moments		
Mean cost (\$/acre)	28.09	(3.05)
Std. dev. cost (\$/acre)	31.11	(4.21)

Standard errors in parentheses, computed via bootstrap with 1,000 replications.

Table 31: Toy Model Counterfactual: Effect of Nomination Threshold

Metric	$K = 1$ **(Baseline)**	$K = 2$	$K = 3$
Nomination rate (π_0^*)	19.2%	15.4%	0%
Auction occurrence $\Pr(M \geq K)$	58%	24%	0%
# Bidders (if occurs)	3.56	3.56	—
Revenue | occurs (\$/acre)	120	250	—
Revenue unconditional (\$/acre)	70	60	0
Change vs. baseline:			
Δ Revenue | occurs	—	+108%	—
Δ Revenue unconditional	—	−14%	−100%

All results at mean $N = 4$ bidders. Revenue is per-acre winning bid. "—" indicates auction never occurs. Contrast with the full structural model (Section 7.1), where $K = 2$ already produces zero trigger rate.

E.6 Equilibrium Multiplicity Under $K \geq 2$

The hump-shaped $G(\pi)$ function when $K \geq 2$ reflects strategic complementarities: the pivotal probability initially increases with π (making nomination more attractive), peaks at $\pi \approx \frac{K-1}{N-1}$, then declines. This non-monotonicity generates multiple equilibria, with the coordination-failure equilibrium ($\pi = 0$) always present when $K \geq 2$.

E.7 Proofs for Toy Example

E.8 Proof of Uniqueness when $K = 1$

Proposition 4 (Uniqueness for $K = 1$)**.** *When $K = 1$, there exists at most one equilibrium nomination probability $\pi^* \in [0, 1]$.*

Proof. When $K = 1$, the pivotal probability becomes

$$h(\pi) = (1-\pi)^{N-1} \tag{88}$$

Taking the derivative

$$h'(\pi) = -(N-1)(1-\pi)^{N-2} < 0 \quad \forall \pi \in [0,1) \tag{89}$$

The operator $G(\pi, N)$ is defined as

$$G(\pi, N) = \int_{\underline{v}}^{\bar{v}} F_C\left((1-\pi)^{N-1} \cdot \Delta(v; N)\right) f_V(v)\, dv \tag{90}$$

Since F_C is monotonically non-decreasing and $h(\pi)$ is strictly decreasing, $G(\pi, N)$ is strictly decreasing in π. Therefore, the fixed point equation $G(\pi^*, N) = \pi^*$ has at most one solution.

If $G(0, N) \leq 0$, then $\pi^* = 0$ is the unique equilibrium. If $G(0, N) > 0$, then by the Intermediate Value Theorem and monotonicity, there exists a unique $\pi^* \in (0, 1)$ satisfying $G(\pi^*, N) = \pi^*$. □

E.9 Proof of Multiple Equilibria when $K > 1$

Proposition 5 (Multiple Equilibria for $K > 1$)**.** *When $K > 1$, multiple equilibria may exist, including*

1. $\pi^* = 0$ *(coordination failure equilibrium)*
2. *One or more interior equilibria* $\pi^* \in (0, 1)$

Proof. For $K > 1$, the pivotal probability is

$$h(\pi) = \binom{N-1}{K-1} \pi^{K-1} (1-\pi)^{N-K} \tag{91}$$

Step 1: Properties of $h(\pi)$.

Taking the derivative

$$h'(\pi) = \binom{N-1}{K-1} \left[(K-1)\pi^{K-2}(1-\pi)^{N-K} - (N-K)\pi^{K-1}(1-\pi)^{N-K-1}\right] \tag{92}$$

$$= \binom{N-1}{K-1} \pi^{K-2}(1-\pi)^{N-K-1} \left[(K-1)(1-\pi) - (N-K)\pi\right] \tag{93}$$

Setting $h'(\pi) = 0$

$$(K-1)(1-\pi) = (N-K)\pi \implies \pi_{l=1} = \frac{K-1}{N-1} \tag{94}$$

This is the unique interior critical point. Since $h(0) = h(1) = 0$ and $h(\pi_{l=1}) > 0$, $h(\pi)$ is single-peaked.

Step 2: Properties of $G(\pi, N)$.

By Lemma [from sub-document page 21], the transformation through F_C preserves the single-peaked property

$$G(\pi, N) = \int_{\underline{v}}^{\bar{v}} F_C(h(\pi) \cdot \Delta(v; N)) f_V(v)\, dv \tag{95}$$

is also single-peaked at $\pi_{l=1}$, with $G(0, N) = G(1, N) = F_C(0)$.

Step 3: Equilibrium multiplicity.

The fixed point equation $G(\pi, N) = \pi$ can have multiple solutions depending on the height of the peak $G(\pi_{l=1}, N)$. If $G(\pi_{l=1}, N) < \pi_{l=1}$, only $\pi^* = 0$ exists (coordination failure). If $G(\pi_{l=1}, N) > \pi_{l=1}$, the 45-degree line intersects $G(\cdot)$ at $\pi^* = 0$ and at least one interior point. The number of interior equilibria depends on the curvature of $G(\cdot)$ near $\pi_{l=1}$

□

E.10 Posterior Density Ratio: $f_{\text{post}}(v \mid m)/f_V(v)$

Figure 16 provides direct evidence for the nature of the Bayesian update at the estimated parameters ($\hat{\rho} = 0.861$, $v^* \approx r$, $u^* = F_V(v^*) \approx 0$). The left panel overlays the copula posterior density $f_{\text{post}}(v \mid m)$ at $m \in \{1, 2, 4, 8\}$ against the prior density truncated at v^* (black dashed, the $m = 0$ baseline). The right panel plots the pointwise ratio $f_{\text{post}}(v \mid m)/f_{\text{prior}}(v)$.

Across the mass-bearing region the ratio is indistinguishable from 1: the posterior and truncated-prior densities in the left panel are visually almost identical, so the Bayesian update leaves the conditional distribution unchanged.

This confirms that the near-zero nomination threshold $u^* \approx 0$ renders the conditioning event $\{V_j \geq v^*\}$ nearly certain: it conveys almost no information about the common factor W through the *shape* of the conditional distribution within $[v^*, \infty)$. The operative information channel at the estimated parameters is instead the *entry probability*: the mass $\int_{v^*}^{\infty} f_{\text{post}}(v \mid m)\, dv = \Pr(V \geq v^* \mid m)$ rises above $\Pr(V \geq v^*)$ via the strong copula correlation ($\hat{\rho} = 0.861$), even though the shape of the conditional density is unchanged.

E.11 The Nomination Fee Counterfactual

In this counterfactual, nomination fee is raised from its current level, holding the single-nomination rule ($K = 1$) fixed. A dollar fee ϕ enters each tract's equilibrium as the residual fee $\tilde{\phi} = \phi/s_b$, so a common dollar fee is a larger burden on smaller tracts. I re-solve the equilibrium objects (v^{**}, c_L^*, c_H^*) at each fee for the median value bin, with the most expensive and cheapest bins as bounds, and report revenue both *conditional* on a triggered auction and *unconditional* (per offered tract, i.e. trigger rate $\times$ conditional revenue), each net of the fee transfer collected from nominators.

A higher fee raises the nomination-value threshold v^{**}, so the firms that still nominate are more strongly selected on value. Conditional on an auction occurring, this *raises* the winning bid: at the median value tracts, revenue per triggered auction rises from about \$2,900 at the current fee toward \$9,800 as the fee climbs (and analogously \$10,400 $\rightarrow$ \$36,400 among the most expensive tracts). But the same increase drives the trigger rate $\Pr(M \geq 1) = 1 - e^{-\Lambda_{nom}}$ down monotonically, because $\Lambda_{nom} = \lambda_{pre}\, \pi_0^L\, [1 - F_V(v^{**})]$ falls as v^{**} rises. The two effects work against each other.

Netting the two, expected revenue *per offered tract* is maximized at $\phi = 0$ in every value class and declines as the fee rises. For most expensive tracts the seller's net revenue (auction revenue plus the fee transfer) stays roughly flat over a wide range because the fee it collects offsets the lost auctions. It never exceeds the zero-fee level and the underlying auction revenue (excluding the transfer) falls throughout. There is thus no revenue-improving interior fee: the current nominal fee is already essentially optimal, and the flat range means the seller is close to indifferent over modest fees.

The reason the fee cannot help is that the constraint on market thickness is the scarcity of early, informed arrivals ($\hat{\lambda}_{pre} \approx 0.005$) together with the learning-cost distribution, not the level of the \$10 fee. Raising the fee can only screen the extensive margin further and cannot manufacture the lacking early interest.

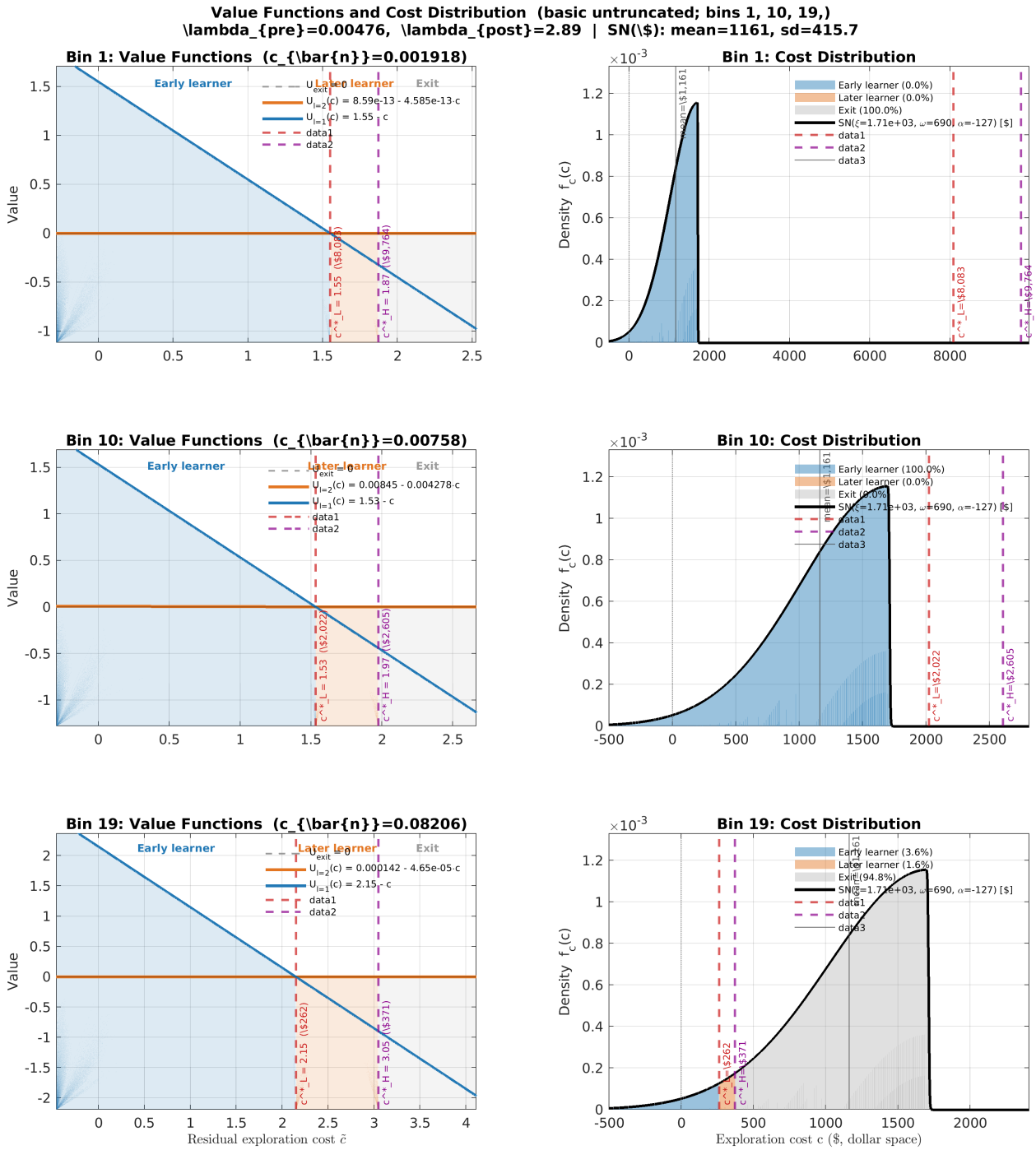


Figure 10: Value Functions and Cost Distribution (Bins 1, 10, and 19)

Notes: Panel A: Early learner value $U_{l=1}(c)$ (blue) and non-nominator value $U_{l=2}(c)$ (orange) as functions of learning cost c, evaluated at Bin 5 estimated parameters. Vertical dashed lines mark c_L^* and c_H^*. Panel B: Estimated Skew-Normal cost density $F_C(c)$ with early learner and $l_i = 2$ non-nominator cutoffs marked. Early learner region: $\approx 0.1\%$; non-nominator region: $\approx 0.2\%$; exit: $\approx 99.8\%$. Costs are in residual units; dollar costs equal $c \times s_t$ where $s_t = \exp(\hat{y}_t)$ is the auction-specific hedonic scaling factor.

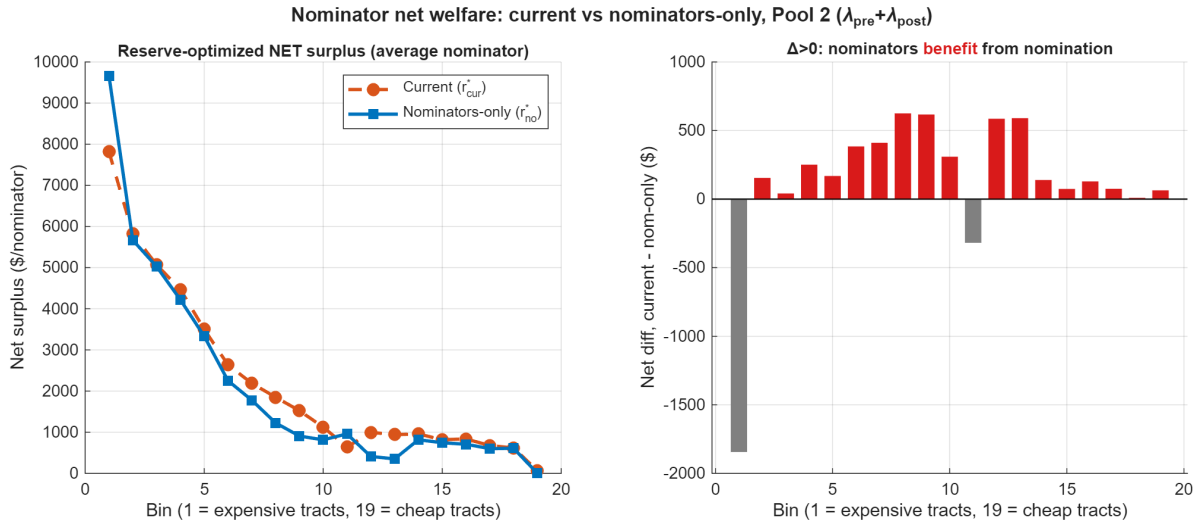


Figure 11: Robustness: net surplus of the average nominator when the non-nominators and late arrivals must instead nominate early, so the pool is $\lambda_{pre} + \lambda_{post}$. Left: net surplus by tract bin; right: difference (current − nominators-only). The conclusion matches the baseline (nominator worse off moving to nominators-only in 17 of 19 bins).

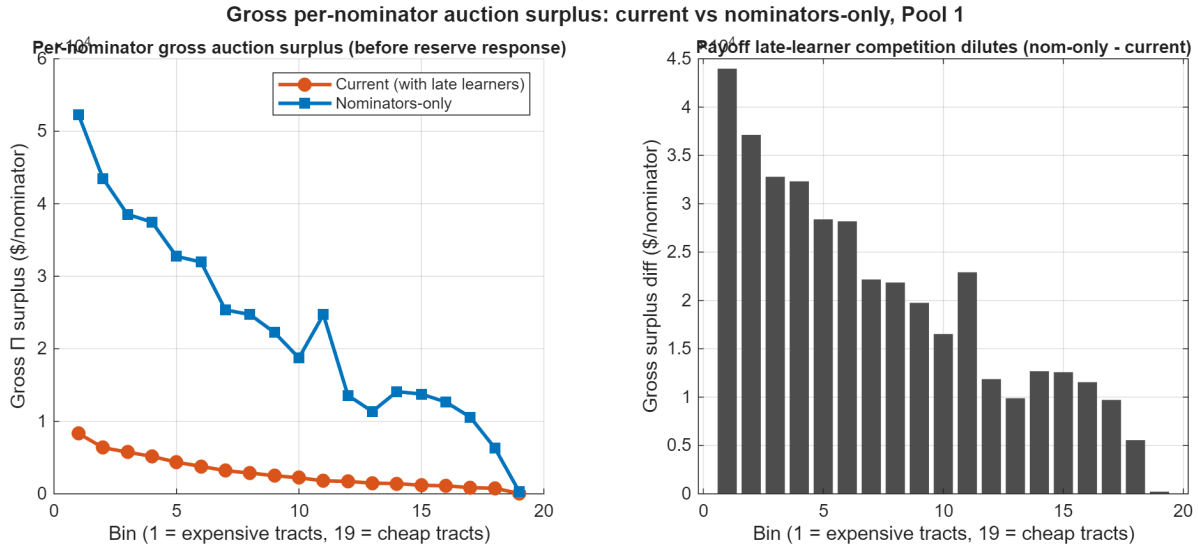


Figure 12: Gross per-nominator auction surplus (before the reserve response), current vs. nominators-only. Left: gross expected surplus $\tilde{\Pi}$ per nominator by tract bin; right: the per-auction payoff that non-nominator and late-arrival competition dilutes (nominators-only − current), about \$20,002 on average.

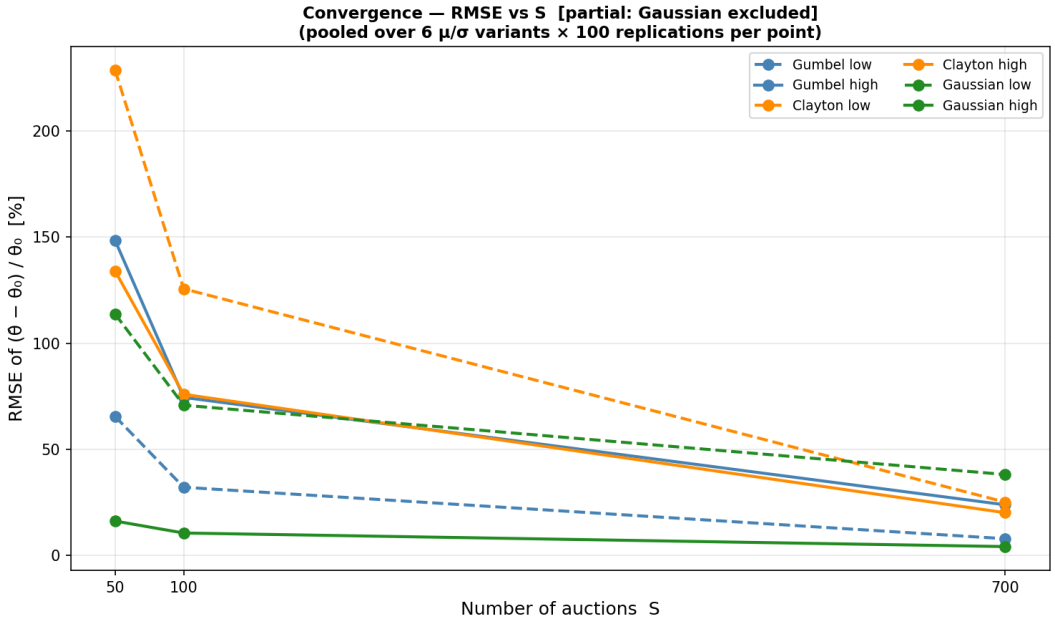


Figure 13: RMSE of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ versus number of auctions S, pooled over six lognormal marginal variants (600 replications per point). Solid lines = high dependence; dashed lines = low dependence. All copula families converge toward zero as S grows.

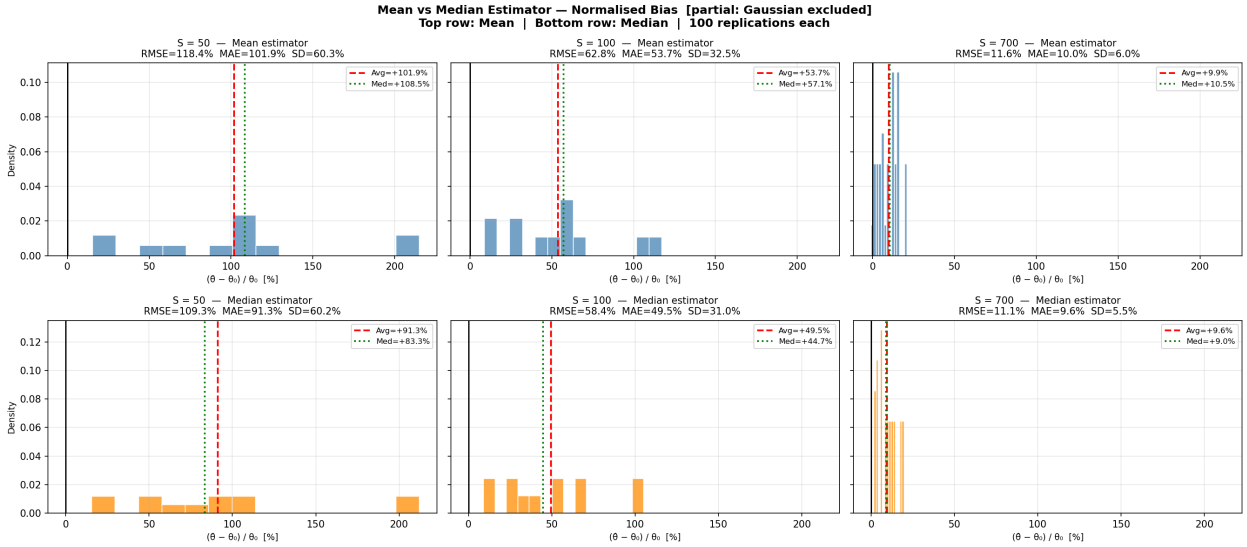
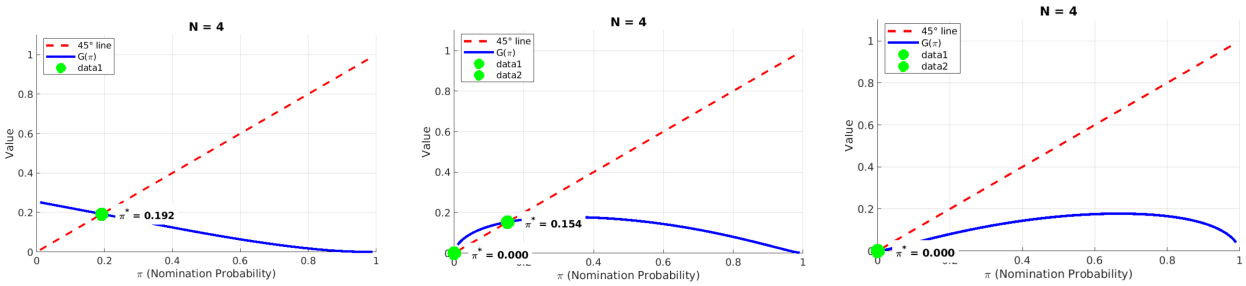


Figure 14: Distribution of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ across all 36 DGP cells, separately for the mean aggregator (top row) and the median aggregator (bottom row), at $S \in \{50, 100, 700\}$ (columns left to right). Red dashed line = average of the displayed distribution; green dotted line = median of the displayed distribution. The mean distribution has a heavier right tail at small S (driven by Gumbel and Clayton cells), which the median aggregator substantially dampens.



(a) $K = 1$: Unique equilibrium (b) $K = 2$: Multiple equilibria (c) $K = 3$: Coordination failure

Figure 15: Fixed Point Problem $G(\pi; K, N) = \pi$ for $N = 4$ Bidders (Simplified Model)

Each panel plots the operator $G(\pi; K, N)$ (blue curve) against the 45-degree line (red dashed). Equilibria correspond to intersections. Panel (a): strategic substitutes under $K = 1$, unique equilibrium. Panel (b): strategic complements under $K = 2$, multiple equilibria including coordination failure. Panel (c): $K = 3$, only coordination failure. This simplified model abstracts from late-learning; the full model (Section 7.1) finds coordination failure already at $K = 2$.

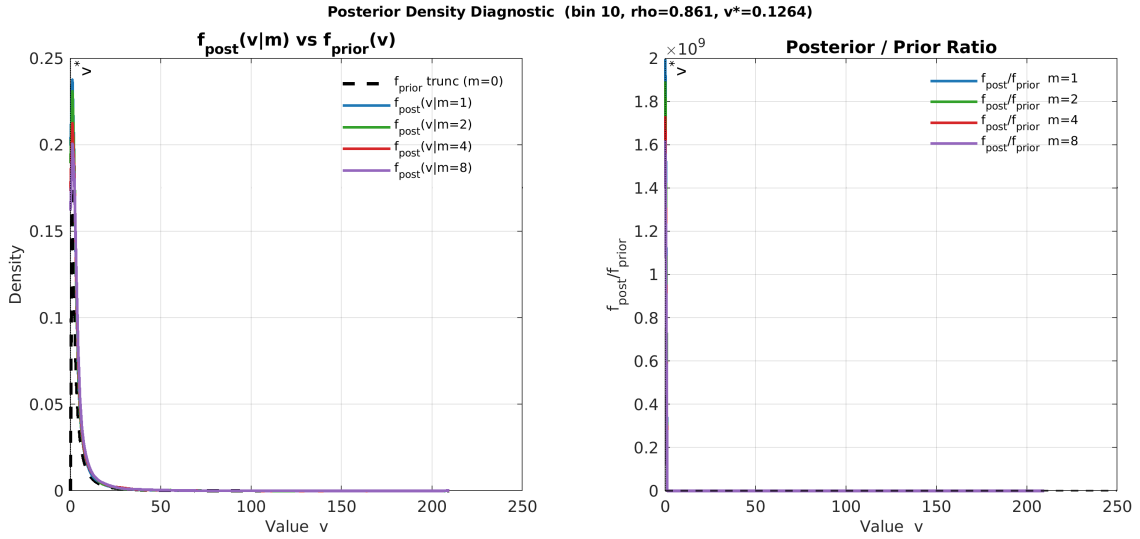


Figure 16: Posterior Density vs. Prior, and Their Ratio (Estimated Parameters)

Notes: Left panel: prior density truncated at v^* (black dashed, the $m = 0$ baseline) and copula posterior $f_{\text{post}}(v \mid m)$ at $m \in \{1, 2, 4, 8\}$ (colored solid). Right panel: point-wise ratio $f_{\text{post}}(v \mid m)/f_{\text{prior}}(v)$. Vertical dotted line marks v^*. Away from the boundary the ratio is indistinguishable from 1, so the conditional density shape is essentially unchanged by the Bayesian update at $u^* \approx 0$; the large spike near the lower boundary $v \approx v^*$ (of order 10^9) is a numerical artifact of dividing by the sharply truncated prior density there and carries negligible probability mass. Reported for Bin 10 (median bin), $\hat{\rho} = 0.861$, $v^* = 0.1264$ ($u^* \approx 0$).